INTRODUCTION TO QUANTUM COMPUTATION • 67596

Final Project:

Part I (Q1–5)

Conceptual, Theoretical, and Algorithmic Questions

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Extended explanations and primers are provided in the separate companion, *Appendices and supplementary material* (Appendices A–D).

Part II, a review of the selected paper, is submitted separately.

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QUESTION 1

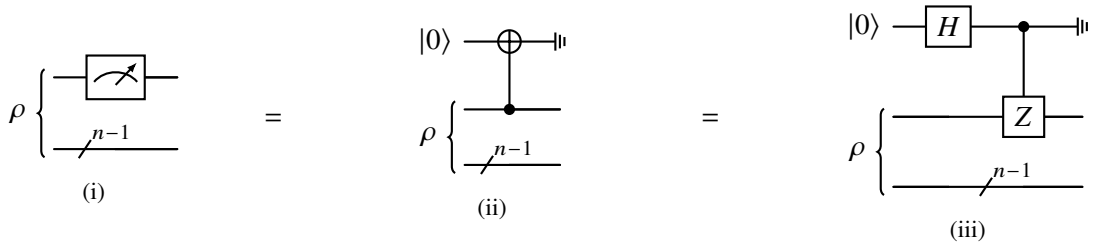
Deferred Measurement

The circuit model allows measurements *during* computation, with later gates chosen by the outcomes. The **Deferred Measurement Principle** says all of them can be postponed to the end of the computation itself; we used this principle extensively in class, while we assumed its correctness, never formally proving it. The standard reference was given by Aharonov–Kitaev–Nisan, §4.1 [1]; we shall prove a simpler, albeit quite general version, of said principle.

1.a Measurement without looking

PROBLEM

Let ρ be a density matrix on n qubits. Let P_0 be the projector onto the span of the standard basis states $|0, x_2, \ldots, x_n\rangle$, and let $P_1 = I - P_0$ be the projector onto the span of the states $|1, x_2, \ldots, x_n\rangle$. Explain what "measurement without looking" does to ρ in terms of P_0, P_1, and prove the equivalence of the three circuits below. Trashing a qubit means tracing over the corresponding two-dimensional Hilbert space.



(i) (ii) (iii)

Figure 1. The three procedures to be compared. The state ρ describes all n original wires. Circuits (ii) and (iii) add one temporary record qubit; only that added qubit is discarded.

All three procedures induce the same transformation on the retained system,

$$\rho \longmapsto \mathcal{D}(\rho) := P_0 \rho P_0 + P_1 \rho P_1. \tag{1.a.1}$$

The three procedures handle the same two alternatives in different physical ways. Circuit (i) **measures** whether the first original qubit is 0 or 1 without retaining the classical answer. Circuit (ii) **records** that answer as $|0\rangle$ or $|1\rangle$ in a fresh ancilla and then **discards** the ancilla. Circuit (iii) records the same answer as $|+\rangle$ or $|-\rangle$ and then discards the ancilla.

Their intermediate states are not identical. The claim is instead an input-output statement: after the classical result or ancillary record is no longer accessible, the density matrix on the original n qubits is the same. The physical record may still exist elsewhere; excluding it from the accessible output does not require destroying it.

We first calculate the output of circuit (i). We then follow the actual gate evolution of circuits (ii) and (iii), discard the record created by each one, and calculate their retained outputs separately. The proof is complete once both calculations reproduce the output of circuit (i).

Let $I_{2^{n-1}}$ be the identity on the last $n-1$ qubits. The two alternatives are selected by

$$P_0 = |0\rangle\langle 0| \otimes I_{2^{n-1}}, \qquad P_1 = |1\rangle\langle 1| \otimes I_{2^{n-1}}. \tag{1.a.2}$$

They satisfy $P_j^2 = P_j$, $P_0 P_1 = P_1 P_0 = 0$, and $P_0 + P_1 = I_{2^n}$.

1.a.i Circuit (i): unread measurement

Unconditioned output. The unread output is a mixture of the conditional states, so we first need the probability of each outcome. Born's rule gives

$$p_0 = \mathrm{Tr}(P_0 \rho), \qquad p_1 = \mathrm{Tr}(P_1 \rho). \tag{1.a.3}$$

The two weights exhaust all possibilities because

$$p_0 + p_1 = \mathrm{Tr}\big((P_0 + P_1)\rho\big) = \mathrm{Tr}(\rho) = 1. \tag{1.a.4}$$

If the answer is 0, the state is restricted to the range of P_0 and then renormalized; similarly for answer 1. When the corresponding denominators are nonzero, the two conditional states are

$$\rho_0 = \frac{P_0\rho P_0}{\mathrm{Tr}(P_0\rho P_0)}, \qquad \rho_1 = \frac{P_1\rho P_1}{\mathrm{Tr}(P_1\rho P_1)}. \tag{1.a.5}$$

By cyclicity of the trace and $P_i^2 = P_i$, we have $\mathrm{Tr}(P_i\rho P_i) = \mathrm{Tr}(P_i\rho) = p_i$ for $i \in \{0, 1\}$. Thus the denominators in (1.a.5) are precisely the corresponding Born probabilities.

The unread output averages the conditional states with their outcome probabilities. For $p_0, p_1 > 0$, the weights cancel the normalizations:

$$\rho_{\mathrm{out}}^{(\mathrm{i})} = p_0\rho_0 + p_1\rho_1 = p_0\frac{P_0\rho P_0}{p_0} + p_1\frac{P_1\rho P_1}{p_1} = P_0\rho P_0 + P_1\rho P_1. \tag{1.a.6}$$

A zero-probability branch is omitted: its positive semidefinite operator $P_j\rho P_j$ has trace zero and is therefore zero.

To see what this changes, compare the output with the input:

$$\rho = (P_0 + P_1)\rho(P_0 + P_1) = P_0\rho P_0 + P_0\rho P_1 + P_1\rho P_0 + P_1\rho P_1. \tag{1.a.7}$$

The output preserves the two within-alternative blocks, including coherence inside each, and removes the cross-alternative blocks $P_0\rho P_1$ and $P_1\rho P_0$. The unread result averages conditional density matrices with their outcome probabilities; it does not add their amplitudes. This is **dephasing in the first-qubit computational basis**. Appendix A.1.i develops the pure-state block picture and its interpretation in detail. Equation (1.a.6) is now the target that circuits (ii) and (iii) must reproduce.

1.a.ii Circuit (ii): computational-basis record

Our target is the output already found for circuit (i) in (1.a.6). Circuit (ii) first writes the same 0-versus-1 information into a fresh ancilla, then discards that ancilla and retains the original n qubits. We therefore follow one complete pure input through the controlled-NOT, display the resulting joint density matrix, explain exactly what discarding the record means, and finally extend the calculation to the arbitrary input density matrix ρ.

Recording the outcome. The first original qubit controls a CNOT whose target is a fresh ancilla A in $|0\rangle_A$; the other original wires are untouched. Writing the original system first and the ancilla last, the two alternatives act as

$$|0\rangle\,|0\rangle_A \longmapsto |0\rangle\,|0\rangle_A\,, \qquad |1\rangle\,|0\rangle_A \longmapsto |1\rangle\,|1\rangle_A\,. \tag{1.a.8}$$

A general normalized pure state of the n original qubits can be grouped by the value of its first qubit as

$$|\psi\rangle = \alpha\,|0\rangle\,|\phi_0\rangle + \beta\,|1\rangle\,|\phi_1\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1, \tag{1.a.9}$$

where $|\phi_0\rangle$ and $|\phi_1\rangle$ are normalized states of the remaining $n-1$ qubits for the branches whose coefficients are nonzero. A branch with zero coefficient is absent and its corresponding normalized vector may be chosen arbitrarily.

Now attach the blank ancilla to this input. Linearity applies the two rules in (1.a.8) separately to the two branches:

$$\big(\alpha\,|0\rangle\,|\phi_0\rangle + \beta\,|1\rangle\,|\phi_1\rangle\big)\,|0\rangle_A \longmapsto \alpha\,|0\rangle\,|\phi_0\rangle\,|0\rangle_A + \beta\,|1\rangle\,|\phi_1\rangle\,|1\rangle_A =: |\Psi_{\mathrm{ii}}\rangle\,. \tag{1.a.10}$$

No measurement has occurred. The controlled-NOT has coherently created a record: the first-qubit-0 branch is accompanied by $|0\rangle_A$, while the first-qubit-1 branch is accompanied by the orthogonal record $|1\rangle_A$.

The joint density matrix of the actual output in (1.a.10) is

$$\begin{aligned}|\Psi_{\mathrm{ii}}\rangle\langle\Psi_{\mathrm{ii}}| = {} & |\alpha|^2\,|0\rangle\,|\phi_0\rangle\,\langle 0|\,\langle\phi_0| \otimes (|0\rangle\langle 0|)_A + \alpha\overline{\beta}\,|0\rangle\,|\phi_0\rangle\,\langle 1|\,\langle\phi_1| \otimes (|0\rangle\langle 1|)_A \\ & + \overline{\alpha}\beta\,|1\rangle\,|\phi_1\rangle\,\langle 0|\,\langle\phi_0| \otimes (|1\rangle\langle 0|)_A + |\beta|^2\,|1\rangle\,|\phi_1\rangle\,\langle 1|\,\langle\phi_1| \otimes (|1\rangle\langle 1|)_A.\end{aligned} \tag{1.a.11}$$

The original system still has four terms. The difference is that each term now also records, on its ancilla side, which first-qubit alternative occurs at its ket and bra endpoints.

The same bookkeeping becomes concrete in ordinary row–column notation. In the ancilla's computational basis,

$$|0\rangle_A = \begin{pmatrix}1\\0\end{pmatrix}, \quad \langle 0|_A = \begin{pmatrix}1 & 0\end{pmatrix}, \qquad |1\rangle_A = \begin{pmatrix}0\\1\end{pmatrix}, \quad \langle 1|_A = \begin{pmatrix}0 & 1\end{pmatrix}. \tag{1.a.12}$$

Consequently, the four ancilla outer products in (1.a.11) are the four matrix units

$$(|0\rangle\langle 0|)_A = \begin{pmatrix}1 & 0\\0 & 0\end{pmatrix}, \quad (|0\rangle\langle 1|)_A = \begin{pmatrix}0 & 1\\0 & 0\end{pmatrix}, \quad (|1\rangle\langle 0|)_A = \begin{pmatrix}0 & 0\\1 & 0\end{pmatrix}, \quad (|1\rangle\langle 1|)_A = \begin{pmatrix}0 & 0\\0 & 1\end{pmatrix}. \tag{1.a.13}$$

Collecting these four terms by their ancilla row and column gives the explicit joint-state block matrix

$$|\Psi_{\text{ii}}\rangle\langle\Psi_{\text{ii}}| = \begin{pmatrix} |\alpha|^2\,|0\rangle\,|\phi_0\rangle\,\langle 0|\,\langle\phi_0| & \alpha\overline{\beta}\,|0\rangle\,|\phi_0\rangle\,\langle 1|\,\langle\phi_1| \\ \overline{\alpha}\beta\,|1\rangle\,|\phi_1\rangle\,\langle 0|\,\langle\phi_0| & |\beta|^2\,|1\rangle\,|\phi_1\rangle\,\langle 1|\,\langle\phi_1| \end{pmatrix}_A. \tag{1.a.14}$$

Every entry is an operator on the original n qubits. The upper-left and lower-right blocks carry matching ancilla labels; the off-diagonal blocks carry different ancilla labels.

Discarding the computational-basis record. The discard symbol says that only the original n qubits belong to the output; the ancilla is no longer retained. We therefore need the density matrix that predicts every measurement on the original qubits alone. It is obtained by the **partial trace over the ancilla**, the quantum counterpart of summing over a variable that is no longer retained.

Definition 1.1 (Partial trace over the record)**.** For any joint operator X on the original system and the one-qubit record A, define

$$\mathrm{Tr}_A(X) := \sum_{a\in\{0,1\}} (I_{2^n}\otimes\langle a|_A)X(I_{2^n}\otimes|a\rangle_A). \tag{1.a.15}$$

The concrete coordinates above show what this definition does to the actual block matrix in (1.a.14). Multiplying on the right by $|0\rangle_A$ selects its first ancilla column, and multiplying on the left by $\langle 0|_A$ then selects the first row; the $a = 0$ summand is therefore the upper-left block. The $a = 1$ summand similarly selects the lower-right block. Thus tracing out the ancilla adds those two diagonal blocks and does not select either off-diagonal block. The corresponding general block and record identities are collected in Appendix A.1.iii.

We now substitute the actual joint state produced by circuit (ii) into the definition:

$$\mathrm{Tr}_A(|\Psi_{\text{ii}}\rangle\langle\Psi_{\text{ii}}|) = (I_{2^n}\otimes\langle 0|_A)\,|\Psi_{\text{ii}}\rangle\langle\Psi_{\text{ii}}|\,(I_{2^n}\otimes|0\rangle_A) + (I_{2^n}\otimes\langle 1|_A)\,|\Psi_{\text{ii}}\rangle\langle\Psi_{\text{ii}}|\,(I_{2^n}\otimes|1\rangle_A). \tag{1.a.16}$$

Applying these two summands to the four concrete terms in (1.a.11) gives

$$\begin{aligned}\mathrm{Tr}_A(|\Psi_{\text{ii}}\rangle\langle\Psi_{\text{ii}}|) &= |\alpha|^2\,|0\rangle\,|\phi_0\rangle\,\langle 0|\,\langle\phi_0| + \alpha\overline{\beta}\;\langle 1|0\rangle\;|0\rangle\,|\phi_0\rangle\,\langle 1|\,\langle\phi_1| \\ &\quad + \overline{\alpha}\beta\;\langle 0|1\rangle\;|1\rangle\,|\phi_1\rangle\,\langle 0|\,\langle\phi_0| + |\beta|^2\,|1\rangle\,|\phi_1\rangle\,\langle 1|\,\langle\phi_1| \\ &= |\alpha|^2\,|0\rangle\,|\phi_0\rangle\,\langle 0|\,\langle\phi_0| + |\beta|^2\,|1\rangle\,|\phi_1\rangle\,\langle 1|\,\langle\phi_1|\,.\end{aligned} \tag{1.a.17}$$

The two within-alternative terms survive. The two coherence terms vanish because their ancilla records disagree and $\langle 1|0\rangle = \langle 0|1\rangle = 0$. This is the precise mathematical effect of discarding a record that had become correlated with the alternatives.

It remains to pass from this pure-state calculation to the arbitrary density matrix in the question. By the density-matrix decomposition theorem, write

$$\rho = \sum_k q_k\,|\psi_k\rangle\langle\psi_k|\,, \qquad q_k \geq 0, \qquad \sum_k q_k = 1, \tag{1.a.18}$$

where the $|\psi_k\rangle$ are normalized pure states. Let U_{ii} denote the controlled-NOT on the original system and the ancilla. We now substitute this decomposition into the circuit input. Ordinary matrix multiplication distributes over the sum, and the definition (1.a.15) shows that the partial trace does as well:

$$\begin{aligned}\rho_{\text{out}}^{(\text{ii})} &:= \mathrm{Tr}_A\Big[U_{\text{ii}}\big(\rho\otimes|0\rangle\langle 0|_A\big)U_{\text{ii}}^\dagger\Big] = \mathrm{Tr}_A\left[U_{\text{ii}}\left(\sum_k q_k\,|\psi_k\rangle\langle\psi_k|\otimes|0\rangle\langle 0|_A\right)U_{\text{ii}}^\dagger\right] \\ &= \sum_k q_k\;\mathrm{Tr}_A\Big[U_{\text{ii}}\big(|\psi_k\rangle\langle\psi_k|\otimes|0\rangle\langle 0|_A\big)U_{\text{ii}}^\dagger\Big] = \sum_k q_k\,(P_0\,|\psi_k\rangle\langle\psi_k|\,P_0 + P_1\,|\psi_k\rangle\langle\psi_k|\,P_1) \\ &= P_0\rho P_0 + P_1\rho P_1 = \rho_{\text{out}}^{(\text{i})}.\end{aligned} \tag{1.a.19}$$

Thus circuit (ii), after its actual computational-basis record is discarded, has exactly the same retained output as circuit (i).

1.a.iii Circuit (iii): phase-basis record

Once again, the target is (1.a.6). The only change from circuit (ii) is the basis used for the temporary record: the ancilla will store the two alternatives as $|+\rangle$ and $|-\rangle$ instead of $|0\rangle$ and $|1\rangle$. We first verify, gate by gate, that the diagram really writes those two records. We then discard the actual phase-basis record and calculate $\rho_{\text{out}}^{(\text{iii})}$.

Phase-basis branch evolution. The ancilla begins in $|0\rangle_A$. The first gate in circuit (iii) is a Hadamard on that ancilla. Thus the state actually entering the controlled-Z is $|+\rangle_A$:

$$|0\rangle_A \xrightarrow{H_A} \underbrace{|+\rangle_A}_{\text{actual ancilla state}} := \frac{|0\rangle_A + |1\rangle_A}{\sqrt{2}}. \tag{1.a.20}$$

In the formulas below, the first original qubit is written first and the ancilla last. When the ancilla component is $|0\rangle_A$, controlled-Z applies the identity to the first original qubit; when it is $|1\rangle_A$, it applies Z. Here $Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$.

If the first original qubit is 0, both ancilla components leave it unchanged:

$$|0\rangle|+\rangle_A = \frac{|0\rangle|0\rangle_A + |0\rangle|1\rangle_A}{\sqrt{2}} \xrightarrow{\text{CZ}} \frac{I|0\rangle|0\rangle_A + Z|0\rangle|1\rangle_A}{\sqrt{2}} = \frac{|0\rangle|0\rangle_A + |0\rangle|1\rangle_A}{\sqrt{2}} = |0\rangle|+\rangle_A . \tag{1.a.21}$$

If the first original qubit is 1, the ancilla-$|1\rangle_A$ component applies the -1 eigenvalue of Z:

$$|1\rangle|+\rangle_A = \frac{|1\rangle|0\rangle_A + |1\rangle|1\rangle_A}{\sqrt{2}} \xrightarrow{\text{CZ}} \frac{I|1\rangle|0\rangle_A + Z|1\rangle|1\rangle_A}{\sqrt{2}} = |1\rangle\left(\frac{|0\rangle_A - |1\rangle_A}{\sqrt{2}}\right) =: |1\rangle|-\rangle_A . \tag{1.a.22}$$

The two record states that have emerged from the calculation are orthogonal:

$$\langle -|+\rangle = \frac{1}{2}(\langle 0|_A - \langle 1|_A)(|0\rangle_A + |1\rangle_A) = 0. \tag{1.a.23}$$

Thus the ancilla stores the same answer as circuit (ii), but in a different orthonormal basis:

$$\text{first-qubit alternative } 0 \longleftrightarrow |+\rangle_A , \qquad \text{first-qubit alternative } 1 \longleftrightarrow |-\rangle_A . \tag{1.a.24}$$

This is the relevant **phase kickback**. On the first-qubit-1 branch, the -1 eigenvalue of Z becomes the relative minus sign that changes the ancilla from $|+\rangle$ to $|-\rangle$. This is the same phase-kickback mechanism used in the phase-estimation algorithm (PEA). The phase-kickback discussion in Appendix A.1.ii places this instance in the general controlled-unitary mechanism and shows how the resulting phase can be read.

Now apply the two branch calculations linearly to the complete pure input from (1.a.9). The entire circuit evolution is

$$\begin{aligned}(\alpha|0\rangle|\phi_0\rangle + \beta|1\rangle|\phi_1\rangle)|0\rangle_A &\xrightarrow{H_A} (\alpha|0\rangle|\phi_0\rangle + \beta|1\rangle|\phi_1\rangle)|+\rangle_A \\ &\xrightarrow{\text{CZ}} \alpha|0\rangle|\phi_0\rangle|+\rangle_A + \beta|1\rangle|\phi_1\rangle|-\rangle_A =: |\Psi_{\text{iii}}\rangle .\end{aligned} \tag{1.a.25}$$

The 0 branch therefore carries record $|+\rangle_A$, while the 1 branch carries the orthogonal record $|-\rangle_A$.

Discarding the phase-basis record. The trash symbol again tells us to retain only the original n wires. We now apply the partial-trace definition directly to the actual phase-basis record in (1.a.25), rather than replacing this circuit by an abstract record map. Expanding only as much as this calculation requires gives

$$\begin{aligned}\text{Tr}_A(|\Psi_{\text{iii}}\rangle\langle\Psi_{\text{iii}}|) &= |\alpha|^2 |0\rangle|\phi_0\rangle\langle 0|\langle\phi_0| + \alpha\overline{\beta}\ \langle -|+\rangle\ |0\rangle|\phi_0\rangle\langle 1|\langle\phi_1| \\ &\quad + \overline{\alpha}\beta\ \langle +|-\rangle\ |1\rangle|\phi_1\rangle\langle 0|\langle\phi_0| + |\beta|^2 |1\rangle|\phi_1\rangle\langle 1|\langle\phi_1| \\ &= |\alpha|^2 |0\rangle|\phi_0\rangle\langle 0|\langle\phi_0| + |\beta|^2 |1\rangle|\phi_1\rangle\langle 1|\langle\phi_1| .\end{aligned} \tag{1.a.26}$$

Here the two coherence terms vanish because their record endpoints are orthogonal: $\langle -|+\rangle = \langle +|-\rangle = 0$. The surviving pure-state output is therefore identical to that of circuit (ii), even though the record basis is different.

For completeness, we carry out the extension rather than merely asserting it. Let U_{iii} denote the Hadamard followed by controlled-Z on the original system and the ancilla. Substituting (1.a.18), distributing the unitary

multiplication, and then distributing the partial trace gives

$$\begin{aligned}\rho_{\text{out}}^{(\text{iii})} &:= \operatorname{Tr}_A\left[U_{\text{iii}}(\rho\otimes|0\rangle\langle 0|_A)U_{\text{iii}}^\dagger\right] = \operatorname{Tr}_A\left[U_{\text{iii}}\left(\sum_k q_k\,|\psi_k\rangle\langle\psi_k|\otimes|0\rangle\langle 0|_A\right)U_{\text{iii}}^\dagger\right]\\ &= \sum_k q_k\,\operatorname{Tr}_A\left[U_{\text{iii}}(|\psi_k\rangle\langle\psi_k|\otimes|0\rangle\langle 0|_A)U_{\text{iii}}^\dagger\right] = \sum_k q_k\,(P_0\,|\psi_k\rangle\langle\psi_k|\,P_0 + P_1\,|\psi_k\rangle\langle\psi_k|\,P_1)\\ &= P_0\rho P_0 + P_1\rho P_1 = \rho_{\text{out}}^{(\text{i})}. \end{aligned} \tag{1.a.27}$$

Thus circuit (iii), after its actual phase-basis record is discarded, also has the same retained output as circuit (i).

Relation to the standard deferred-measurement circuit. One final Hadamard converts these records to the computational basis:

$$H_A\,|+\rangle_A = |0\rangle_A\,, \qquad H_A\,|-\rangle_A = |1\rangle_A\,. \tag{1.a.28}$$

This gives the standard identity

$$\text{CNOT}_{1\to A} = H_A\,\text{CZ}_{1,A}\,H_A, \tag{1.a.29}$$

where the rightmost operation acts first. Circuit (iii) omits the final Hadamard because the record is immediately discarded. Applying it would merely rewrite the orthogonal records $|+\rangle_A\,, |-\rangle_A$ as $|0\rangle_A\,, |1\rangle_A$; the two explicit trace calculations above show that either record basis removes the same cross terms.

1.a.iv Equality of the three retained outputs

Each circuit takes the input density matrix ρ on the original n qubits to the same output density matrix:

$$\boxed{\rho_{\text{out}}^{(\text{i})} = \rho_{\text{out}}^{(\text{ii})} = \rho_{\text{out}}^{(\text{iii})} = P_0\rho P_0 + P_1\rho P_1 = \mathcal{D}(\rho).} \tag{1.a.30}$$

Circuit (i) performs the measurement without retaining its classical result. Circuit (ii) stores the same result temporarily in the orthogonal records $|0\rangle\,, |1\rangle$. Circuit (iii) stores it in the orthogonal records $|+\rangle\,, |-\rangle$. Once the record is discarded, all three remove the same cross-alternative coherence pair and retain the same two within-alternative terms. ■

Scope of the equivalence. Since n and ρ are arbitrary, the identity applies to a measured qubit even when it is entangled with the remaining wires. It can therefore replace an intermediate measurement whose outcome is unused. When later operations depend on that outcome, the unitary simulation must retain the ancillary record and use it as a coherent control, as developed in part (b).

1.b Deferring all measurements over a universal set S

PROBLEM

Let S be a finite universal set of unitary gates, closed under adjoints, and let C be a circuit containing gates from S, intermediate standard-basis measurements, and classical circuits made of standard one- and two-bit gates. Earlier classical inputs and measurement outcomes may determine which later unitaries or measurements are performed. Using only gates from S, defer all measurements to the end, make the new output distribution ε-close to that of C for every input, and bound the additional qubits and unitary operations. CNOT and controlled-Z may not be assumed to belong to S.

The challenge is to postpone measurements while preserving the information that controls the rest of the computation. Earlier outcomes may determine later gates, either directly or through classical processing. Such a circuit is often called *adaptive*.[1] Here *classical feed-forward* means that the available classical input bits and earlier measurement outcomes are processed into bits that select later unitary or measurement instructions.

I will first replace these outcomes by quantum records and implement the classical processing reversibly, so that subsequent operations can still use the information they need. This gives an ideal circuit with no intermediate measurements. I will then express its additional operations approximately using gates from the prescribed set S, and bound the resulting change in the output distribution.

[1] For this use of *adaptive quantum circuit*, see M. Foss-Feig et al., *Experimental demonstration of the advantage of adaptive quantum circuits*, arXiv:2302.03029.

The two steps are

$$C \xrightarrow{\text{exact unitary replacement}} C_{\text{ideal}} \xrightarrow{\text{approximate compilation into } S} C'. \tag{1.b.1}$$

The first step removes the intermediate measurements and classical processing exactly, while temporarily permitting convenient fixed-size unitary gates. The second step replaces only those convenient gates by circuits over S. Within the ideal, noiseless circuit model, deferring the measurements preserves the output distribution exactly; all approximation error comes from compiling the additional unitary operations into the prescribed gate set S.

To state the required accuracy precisely, write $\Pr[C(\rho, x) = y]$ for the probability of output string y on quantum input ρ and classical input string x. We require one circuit C' whose requested output distribution has statistical distance less than ε from that of C for every ρ and x:

$$\mathrm{SD}(C(\rho, x), C'(\rho, x)) := \frac{1}{2} \sum_{y} |\Pr[C(\rho, x) = y] - \Pr[C'(\rho, x) = y]| < \varepsilon. \tag{1.b.2}$$

Fix an arbitrary x for the proof and suppress it in the probability notation.

1.b.i Exact unitary replacement

I now construct C_{ideal}, starting with unitary replacements for individual measurements and classical control operations, then assembling them into the full circuit.

Coherent measurement record. Suppose C measures a qubit q. Relative to that qubit, a pure state has the form $|\psi\rangle = |0\rangle_q |\phi_0\rangle + |1\rangle_q |\phi_1\rangle$, where the two states on the remaining wires may be unnormalized. Using the recorder from Part (a), append a fresh qubit r in $|0\rangle$ and apply CNOT from q to r:

$$|\psi\rangle |0\rangle_r \longmapsto |0\rangle_q |\phi_0\rangle |0\rangle_r + |1\rangle_q |\phi_1\rangle |1\rangle_r . \tag{1.b.3}$$

In each computational-basis branch, r holds the value that the measurement would have returned and can supply the same later control, as shown in Figure 2. The components remain coherent; C_{ideal} measures neither q nor r here.



Figure 2. A measurement result used later as a control and its exact unitary replacement.

Part (a) shows that discarding the record reproduces measurement with the outcome forgotten. Here the outcome is needed for later control, so the record must remain available through its last use: replacing "measured" by "unmeasured" does not mean ignoring the would-be outcome. If a stored qubit c decides whether U is performed, define the *controlled unitary* $\mathrm{ctrl}(U)$ by

$$\mathrm{ctrl}(U) := |0\rangle\langle 0|_c \otimes I + |1\rangle\langle 1|_c \otimes U, \qquad |c\rangle |\phi\rangle \longmapsto |c\rangle\ U^c |\phi\rangle . \tag{1.b.4}$$

Here U may act on several qubits; its number of target qubits is bounded because S is fixed and finite. CNOT and $\mathrm{ctrl}(U)$ define the ideal construction; neither is assumed to lie in S. The control can be a measurement record, a computed Boolean output, or a supplied classical-input qubit.

Reversible classical control. The classical processing in C may be irreversible. For example, the AND map in Figure 3 sends (m_1, m_2) to $m_1 \wedge m_2$. It is many-to-one and therefore cannot be unitary. Its reversible repair preserves both inputs and XORs the answer into a fresh target:

$$|m_1, m_2\rangle |z\rangle \longmapsto |m_1, m_2\rangle |z \oplus (m_1 \wedge m_2)\rangle , \tag{1.b.5}$$

which is the Toffoli operation.

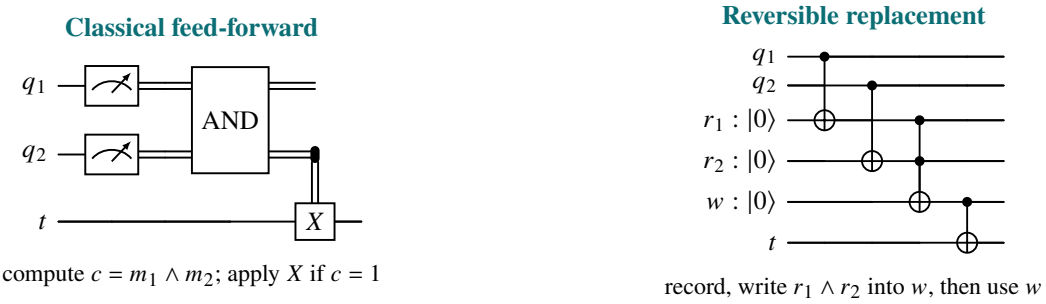


Figure 3. The AND case study. Retaining the inputs and writing the answer into a fresh qubit makes the classical update reversible.

The same repair works for every allowed Boolean gate $f : \{0,1\}^k \to \{0,1\}$, $k \in \{1,2\}$:

$$R_f |x\rangle |z\rangle = |x\rangle |z \oplus f(x)\rangle . \tag{1.b.6}$$

Applying R_f twice gives the identity, so it permutes the computational basis and is therefore unitary. On a fresh target it writes $f(x)$ while preserving its inputs. Since $k \leq 2$, it acts on at most three qubits. The discussion of reversible Boolean gates in Appendix A.2.i gives an explicit construction using X, CNOT, and Toffoli.

Conditional measurements. If a measurement instruction is itself enabled by an earlier bit e, use the controlled recorder

$$|e\rangle |z\rangle_q |0\rangle_r \longmapsto |e\rangle |z\rangle_q |e \wedge z\rangle_r . \tag{1.b.7}$$

For $e = 1$ this records the would-be outcome, while for $e = 0$ it leaves $r = 0$. The retained pair (e, r) distinguishes "inactive" from "active with outcome 0."

Circuit-wide construction. I now assemble these local replacements into C_{ideal}.

Definition 1.2 (Registers and histories)**.** Let m and b count the intermediate measurements and Boolean gates. Define D to be the quantum data register, initially containing ρ, R a record register with one fresh qubit per measurement, and W a work register with one fresh qubit per Boolean-gate output. Each classical input bit is represented by a computational-basis qubit; the resulting $|x\rangle$ is retained separately and unchanged.

After t instructions, define a *history* h of C to be a possible list of its measurement outcomes and Boolean outputs, using zero for an inactive measurement. In C_{ideal}, this same classical string is encoded as $|h\rangle_{\text{RW}}$ on the used record/work qubits. The evolving quantum data is tracked separately in D, and its state may depend on h. Together with the supplied classical input x, h determines the classical control decisions at this stage. In expressions such as $f(h)$ and $c(h)$, dependence on the fixed x is understood. The unchanged factor $|x\rangle$ and unused fresh zero qubits are omitted from subsequent state expressions and the schematic figure.

We retain all stored values until the end, so no uncomputation is needed. An active measurement appends its outcome to h; for an inactive one, the retained activation bit distinguishes its zero placeholder from an actual outcome zero. A Boolean gate appends the deterministic value $f(h)$.

In Figure 4, $s_{\ell,*}$ denotes the one or two stored inputs selected for f_ℓ; R_{f_ℓ} acts on those inputs and the fresh target w_ℓ. The left episode uses the record r_1 to control U_{t_1} directly; the right uses the computed work bit w_ℓ to control U_t. Each unitary acts only on its indicated target wires in D, leaving the other data wires unchanged. A supplied classical-input qubit can also provide a control. The ellipsis preserves the original chronological order.

Process the instructions of C chronologically:

1. Copy an unconditional $U \in S$ unchanged on the same data wires.
2. Replace a measurement by the recorder in (1.b.3), or by the controlled recorder in (1.b.7) when it is conditional.
3. Replace a Boolean gate f by R_f on its one or two stored inputs and a fresh work qubit.
4. Replace a classically controlled U by $\text{ctrl}(U)$, controlled by the qubit holding the same Boolean value.

For a pure input, linearity performs these replacements on all histories at once. Fix a step t whose instruction applies a controlled unitary U. Let h range over the histories just before this instruction, with corresponding data

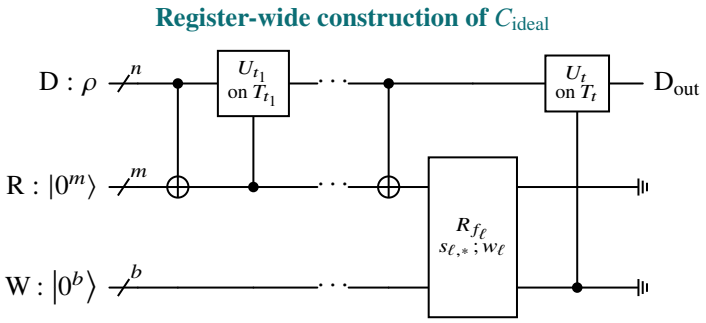


Figure 4. Schematic examples of the register-wide construction: a control may come directly from R or from a computed bit in W. The ellipsis contains the remaining instructions in their original order. Requested record or work bits are measured as outputs at the end; only the remaining auxiliary bits are discarded.

states $|\phi_h\rangle$, suppressing their dependence on t. If the control on history h is $c(h)$, then

$$\sum_h |h\rangle_{\text{RW}} |\phi_h\rangle_{\text{D}} \longmapsto \sum_h |h\rangle_{\text{RW}} U^{c(h)} |\phi_h\rangle_{\text{D}} . \tag{1.b.8}$$

This is a coherent superposition tagged by orthogonal basis states, not a classical mixture; each branch nevertheless receives exactly the instruction selected by C.

1.b.ii Exactness of the full circuit

I now use Part (a) to prove that the constructed circuit preserves the output distribution. Its local measurement calculation supplies the starting point; the remaining work is to justify adaptive control and final-only readout. For that purpose, I compare measuring each record immediately after it is written with measuring the records at the end.

Part (a)'s recorded-state calculation, (1.a.14), gives the local fact needed here: measuring the CNOT record and obtaining z selects the unnormalized data state $P_z \sigma P_z$, where σ is the current data state and $P_z = |z\rangle\langle z|_q \otimes I$. This is the corresponding diagonal record block; the statement extends to mixed σ by linearity. Its trace and normalization give exactly the measurement probability and conditional state from (1.a.3) and (1.a.5). The fact applies even when q is entangled with the other data wires.

With immediate record measurements, agreement with C follows by induction on the processed instructions. The invariant is equality of history probabilities and, for each nonzero-probability history, equality of the conditional data state and stored classical values. It holds initially because both circuits start with (ρ, x) and the empty history. For a measurement, Part (a)'s local fact preserves the invariant. For a conditional measurement, the same fact applies when enabled; when disabled, both descriptions leave the data unchanged and use the zero placeholder, distinguished by the retained enable bit. Reversible Boolean gates write the same classical values, so controlled unitaries select the same operations; unconditional unitaries are copied unchanged. These cases complete the induction and establish equality of the joint history and final-readout statistics.

It remains to move the record measurements to the end. Once written, a record r is used only as a control or Boolean input, or is left alone. Every later unitary V therefore preserves its two basis subspaces and commutes with $P_z^r = |z\rangle\langle z|_r \otimes I$. For any joint state σ,

$$V(P_z^r \sigma P_z^r)V^\dagger = P_z^r (V \sigma V^\dagger) P_z^r, \qquad z \in \{0, 1\}. \tag{1.b.9}$$

The two sides describe measuring r before or after V. Equality of the unnormalized joint states preserves the outcome probability, conditional state when defined, and all subsequent statistics. Measurements of different records also commute. Moving each record measurement past the later operations thus preserves the joint outcome distribution at every step. The measurement being moved is that of the *record* r; subsequent gates may change the original data qubit q.

The resulting procedure is C_{ideal} with final readout of the requested data wires and stored classical outputs. The general partial-trace rule in Appendix A.1.iii, equation (A.1.10), also justifies measuring and then forgetting an unrequested record in place of discarding it. The probability of a requested output y is the sum over all complete outcomes consistent with y. Since every complete outcome probability is preserved, so is this sum. Hence, for

every quantum input ρ, classical input x, and requested output string y,

$$\Pr[C_{\text{ideal}}(\rho, x) = y] = \Pr[C(\rho, x) = y]. \tag{1.b.10}$$

This proves the first arrow of (1.b.1), with zero error. An alternative calculation using history weights and conditional data states is retained in Appendix A.2.ii.

1.b.iii Compilation error and resource bounds

The ideal circuit has no intermediate measurements and reproduces the original output distribution exactly. It remains to express its operations using only the prescribed gate set S. I first count the added operations and workspace. I will then replace each added gate by a short sequence over S, bound how much these approximations change the output distribution, and determine the final gate count for the chosen accuracy.

Let $N := \text{size}(C)$ count all quantum gates, measurement instructions, and Boolean gates. The empty circuit ($N = 0$) needs no replacement, so the resource bounds below assume $N \geq 1$. Let a count the classically controlled quantum gates, with m and b as in Definition 1.2. Let L count the newly introduced auxiliary-gate occurrences that will require compilation: one recorder (CNOT or Toffoli) per measurement, one R_f per Boolean gate, and one $\text{ctrl}(U)$ per controlled quantum gate. Each original instruction adds at most one such operation and at most one record/work qubit, so

$$\text{additional record/work qubits} = m + b = O(N), \qquad L = m + b + a = O(N). \tag{1.b.11}$$

If the classical input is not already encoded in computational-basis qubits, its encoding requires one additional qubit per input bit, separately from the record and work qubits counted here. Here L counts auxiliary operations before compilation, not their cost in gates from S. Unconditional gates already in S are retained unchanged. Compilation below determines the final S-gate count and any additional padding qubits.

The original circuit C uses unitary gates from S, by assumption. Thus, whenever an unconditional instruction applies U, the ideal circuit applies the same U, and no replacement is needed. A conditional instruction is different: its replacement is the larger gate $\text{ctrl}(U)$, acting on both the stored control and the original target wires. Although $U \in S$, the question does not guarantee that $\text{ctrl}(U) \in S$. The recording gates and reversible Boolean gates introduced above likewise need not belong to S.

I therefore replace these additional gates by approximating sequences of gates from S. This replacement is called *compilation*. The possible additional gate types form the fixed finite library

$$\mathcal{G} = \{\text{CNOT}, \text{Toffoli}\} \cup \{R_f : f \text{ belongs to the fixed Boolean basis}\} \cup \{\text{ctrl}(U) : U \in S\}. \tag{1.b.12}$$

The count in (1.b.11) is $L = m+b+a \leq N$: m recording gates, b reversible Boolean gates, and a controlled-unitary gates. This count includes every such occurrence, even when the gate already belongs to S. Each member of $\mathcal{G}$ acts on a constant number of qubits, since S and the Boolean basis are fixed: CNOT on two, Toffoli and every R_f on at most three, and $\text{ctrl}(U)$ on one more qubit than U.

Let δ be the error allowed in *one* replacement; ε is the allowed error in the *final output distribution*. I will show that L replacements with error at most δ change that distribution by at most $L\delta$, and then choose δ to meet the required ε.

For each added gate G, compilation uses a fixed-size block large enough for universal synthesis over S of every gate in $\mathcal{G}$. If extra wires are needed, a shared $O_S(1)$-qubit register supplies them. The desired operation on the block is $G \otimes I$: apply G on its intended wires and leave the spare wires alone. The block size depends only on S.

I use the Solovay–Kitaev approximation theorem from Lecture 2 as a course result. Since S is finite, universal, and closed under adjoints, for every $G \in \mathcal{G}$ and $0 < \delta < 1/2$ it supplies a circuit $\widetilde{G}$ over S with the following accuracy and gate count:[2]

$$\left\| G \otimes I - e^{i\theta}\widetilde{G} \right\| \leq \delta, \qquad |\widetilde{G}| = O_S\left(\log^4 \frac{1}{\delta}\right). \tag{1.b.13}$$

The norm bound holds uniformly over all input states on the block. The factor $e^{i\theta}$ is an unobservable global phase; for a conditional instruction, the target being approximated is the complete gate $\text{ctrl}(U)$, including its control.

[2] Course Lecture 2: Solovay–Kitaev approximation, operator norm, and gate-set replacement.

Replacing the L added occurrences produces C', whose unitary gates all belong to S. To compare it with C_{ideal}, I use the error-accumulation result proved in Lecture 3: the error of a sequence of unitary operations is at most the sum of the individual replacement errors.[3] Here the unchanged original gates contribute zero, and each of the L added occurrences contributes at most δ. Writing U_{ideal} and U' for the full unitary circuits, G_j for each added gate including its identity extension, and $e^{i\Theta}$ for the combined global phase,

$$\left\| U_{\text{ideal}} - e^{i\Theta} U' \right\| \le \sum_{j=1}^{L} \left\| G_j - e^{i\theta_j} \widetilde{G}_j \right\| \le L\delta. \tag{1.b.14}$$

The question concerns measurement probabilities. The estimate from Lecture 2 bounds their statistical distance by the distance between the normalized state vectors; keeping only the requested outputs cannot increase it.[4]

For any normalized complete input $|\psi\rangle$, including $|x\rangle$ and all fresh zero qubits, let p and q be the requested output distributions of C_{ideal} and C'. Applying that estimate gives

$$\mathrm{SD}(p, q) \le \left\| (U_{\text{ideal}} - e^{i\Theta} U') \, |\psi\rangle \right\|_2 \le L\delta. \tag{1.b.15}$$

Linearity of output probabilities and convexity of statistical distance extend this bound to every mixed input. Appendix A.2.iii collects the detailed error calculations and the phase and spare-wire conventions.

Thus $L\delta$ bounds the total output error, and I can now choose the accuracy of each replacement to meet the required ε. If $L = 0$, take $C' = C_{\text{ideal}}$. Otherwise choose $\delta = \varepsilon_0/(2L)$, where $\varepsilon_0 = \min\{\varepsilon, 1/2\}$. This satisfies $0 < \delta < 1/2$. Combining the exact first arrow with (1.b.15) gives

$$\mathrm{SD}(C(\rho, x), C'(\rho, x)) \le L\delta = \frac{\varepsilon_0}{2} < \varepsilon \qquad \text{for every } \rho, x. \tag{1.b.16}$$

Finally, substitute this value of δ into the cost of one replacement. For $L \ge 1$, each of the L auxiliary occurrences is replaced by $O_S(\log^4(1/\delta))$ gates. At most N original unitary gates are copied unchanged. Since $L \le N$, the total number of unitary gates is

$$N + O_S\left(L \log^4 \frac{2L}{\varepsilon_0} \right) = O_S\left(N \log^4 \frac{2N}{\varepsilon_0} \right). \tag{1.b.17}$$

The $L = 0$ case uses at most N gates and satisfies the same upper bound. The additional ancillas are $m + b + O_S(1) = O_S(N)$: the record and work qubits plus any shared padding, with $m + b \le N$.

Thus, for every $\varepsilon > 0$, the constructed circuit C' uses only gates from S before its final measurements, with

$$\boxed{\begin{aligned} &\text{additional workspace qubits} = O_S(N), \\ &\text{total unitary gates} = O_S\left(N \log^4 \frac{2N}{\varepsilon_0} \right). \end{aligned}} \tag{1.b.18}$$

Here $\varepsilon_0 = \min\{\varepsilon, 1/2\}$, and $\mathrm{SD}(C(\rho, x), C'(\rho, x)) < \varepsilon$ for every input ρ, x. The workspace bound assumes that the classical inputs are supplied in computational-basis encoding. If their encoding must also be counted, it adds one qubit per classical input bit, giving $n_{\text{cl}} + O_S(N)$ additional qubits.

[3] Course Lecture 3: additive accumulation of approximation errors and the choice of per-gate accuracy.

[4] Course Lecture 2: the relation between state-vector distance and the statistical distance of measurement outcomes.

QUESTION 2

Quantum Error Correcting Codes & Leakage

Quantum error correction stores logical information in correlations among the physical qubits of an encoding, rather than in any sufficiently small subset of them. Thus a small set T of physical qubits should reveal no logical information. This is made precise by its reduced state $\rho_T = \mathrm{Tr}_{\overline{T}} |\phi\rangle\langle\phi|$, which determines every measurement on T: no leakage means that ρ_T is independent of the encoded state $|\phi\rangle$.

P2a tracks how transversal Hadamard exchanges the two CSS descriptions, and P2b shows that diagonality in both the computational and Hadamard bases forces maximal mixing. Part (a) combines these facts with the two classical distance bounds to prove $\rho_T = I/2^{|T|}$ for $|T| < d$; part (b) uses the Pauli expansion and stabilizer distance to prove the general, weaker conclusion that ρ_T is codeword-independent but need not be maximally mixed.

P2a The code $B = H^{\otimes n}C$

PROBLEM

Let $C \subset \mathcal{H}_2^{\otimes n}$ be a **CSS code** or, more generally, a **stabilizer code** with parameters $[[n, k, d]]$, and let $B = H^{\otimes n}C = \{H^{\otimes n} |\phi\rangle : |\phi\rangle \in C\}$. What can you say about B? (CSS, stabilizer, parameters.)

I first examine how applying a Hadamard gate to every physical qubit changes the description of a CSS or stabilizer code, and which properties it preserves.

The operator $H^{\otimes n}$ applies one Hadamard to each physical qubit and couples no two of them. It is a reversible change of basis that swaps X- and Z-type behaviour at each coordinate without adding qubits or moving a nonidentity Pauli to another coordinate. We should therefore expect the same encoded dimension and protection strength, with the two CSS sides exchanged. The proof makes this expectation exact.

We prove that B is a stabilizer code with the same parameters as C, namely $[[n, k, d]]$. For brevity, define $U := H^{\otimes n}$; then $U^\dagger = U$ and $U^\dagger U = I$. The stabilizer group of C is defined by $S = \mathrm{Stab}(C) := \{g \in \mathcal{P}_n : g |\psi\rangle = |\psi\rangle \text{ for every } |\psi\rangle \in C\}$. We claim that the stabilizer transforms by conjugation and, if $C = \mathrm{CSS}(C_1, C_2)$ with $C_2 \subseteq C_1$, the CSS pair transforms as follows:

$$\mathrm{Stab}(B) = USU^\dagger, \qquad B = \mathrm{CSS}(C_2^\perp, C_1^\perp). \tag{P2a.1}$$

Here $C_1^\perp \subseteq C_2^\perp$.

P2a.i Code-space notation

For any stabilizer group G, define its common +1 eigenspace by

$$C(G) := \{|\phi\rangle : g |\phi\rangle = |\phi\rangle \text{ for every } g \in G\}. \tag{P2a.2}$$

Since C is a stabilizer code with stabilizer S, $C = C(S)$.

P2a.ii Constructing a valid transformed stabilizer group

I define the candidate transformed checks by conjugating each original check with U:

$$S_B := USU^\dagger = \{UgU^\dagger : g \in S\}. \tag{P2a.3}$$

There are two tasks: first verify that S_B is a valid stabilizer group, then identify its common +1 eigenspace with B and its elements with the full Pauli stabilizer of B. This subsection establishes the group requirements.

For an arbitrary Pauli string

$$P = \omega P_1 \otimes \cdots \otimes P_n, \qquad P_j \in \{I, X, Y, Z\}, \quad \omega \in \{\pm 1, \pm i\}, \tag{P2a.4}$$

the bitwise form of U gives

$$UPU^\dagger = \omega(HP_1H) \otimes \cdots \otimes (HP_nH), \qquad HIH = I, \quad HXH = Z, \quad HZH = X, \quad HYH = -Y. \tag{P2a.5}$$

Thus conjugation by this particular U maps Pauli strings bijectively to Pauli strings; its inverse is conjugation by $U^\dagger$. It also preserves products:

$$(UgU^\dagger)(UhU^\dagger) = UghU^\dagger. \tag{P2a.6}$$

Consequently, the image of the subgroup S is again a subgroup. Since S is abelian, $gh = hg$; the same product identity therefore gives equal conjugated products in either order, so S_B is abelian as well. Finally, $-I \notin S_B$: otherwise $UgU^\dagger = -I$ for some $g \in S$, and conjugating back would give $g = -I$, contrary to $-I \notin S$. Hence S_B is a valid stabilizer group.

P2a.iii Identifying its code space and full stabilizer

I now identify the common +1 eigenspace of these checks with B. The two inclusions in $B = C(S_B)$ have different jobs. The inclusion $B \subseteq C(S_B)$ says that every intended transformed codeword passes every transformed check. The reverse inclusion says that the checks admit no additional unintended vectors.

Every intended image vector passes the transformed checks. Let $h \in S_B$ and $|\beta\rangle \in B$ be arbitrary. By the definitions of these two sets, write $h = UgU^\dagger$ and $|\beta\rangle = U|\psi\rangle$ for some $g \in S$ and $|\psi\rangle \in C$. Conjugation transports the original check to the transformed state: since $g|\psi\rangle = |\psi\rangle$,

$$h|\beta\rangle = (UgU^\dagger)U|\psi\rangle = U(g|\psi\rangle) = U|\psi\rangle = |\beta\rangle\,. \tag{P2a.7}$$

Both h and $|\beta\rangle$ were arbitrary, so every vector in B is fixed by every element of S_B. Hence $B \subseteq C(S_B)$.

The transformed checks admit no extra vectors. Now take an arbitrary $|\beta\rangle \in C(S_B)$. To show that $|\beta\rangle \in B$, choose an arbitrary $g \in S$. Its transformed operator $h = UgU^\dagger$ belongs to S_B, and hence fixes $|\beta\rangle$:

$$(UgU^\dagger)|\beta\rangle = |\beta\rangle\,. \tag{P2a.8}$$

Multiplying by $U^\dagger$ on the left gives

$$g\left(U^\dagger|\beta\rangle\right) = U^\dagger|\beta\rangle\,. \tag{P2a.9}$$

Since $g \in S$ was arbitrary, $U^\dagger|\beta\rangle$ is fixed by every element of S and therefore lies in $C(S) = C$. Applying U again gives $|\beta\rangle \in UC = B$. Thus $C(S_B) \subseteq B$. Combining the two directions proves the exact code-space identity

$$\boxed{B = C(S_B) = C(USU^\dagger).} \tag{P2a.10}$$

Why this is also the full stabilizer of B. The code-space identity identifies the vectors passing the checks. To identify the full stabilizer directly, I also account for every Pauli that fixes all of B. The forward calculation already gives $S_B \subseteq \mathrm{Stab}(B)$. Conversely, let $P \in \mathrm{Stab}(B)$. Conjugation preserves Paulis, so $U^\dagger PU$ is a Pauli. For every $|\psi\rangle \in C$, the vector $U|\psi\rangle$ belongs to B and is fixed by P; hence

$$(U^\dagger PU)|\psi\rangle = U^\dagger P(U|\psi\rangle) = U^\dagger U|\psi\rangle = |\psi\rangle\,. \tag{P2a.11}$$

Thus $U^\dagger PU$ belongs to the full pointwise stabilizer $\mathrm{Stab}(C) = S$, which implies $P \in USU^\dagger = S_B$. Therefore

$$\boxed{\mathrm{Stab}(B) = U\,\mathrm{Stab}(C)U^\dagger = S_B.} \tag{P2a.12}$$

P2a.iv Preservation of code parameters

Physical length n. The parameter n counts physical qubits. Since $U = H^{\otimes n}$ acts on the same n tensor factors, B is again an n-qubit code.

Logical dimension k. The parameter k is determined by $\dim C = 2^k$. The restriction $U : C \to B$ is onto by the definition $B = UC$. It is one-to-one because $U^\dagger U = I$: if $U|\psi\rangle = U|\phi\rangle$, multiplying by $U^\dagger$ gives $|\psi\rangle = |\phi\rangle$. Hence this restriction is a bijection and

$$\dim B = \dim C = 2^k. \tag{P2a.13}$$

Thus B also encodes k logical qubits.

Distance d. The distance is the minimum physical weight of a Pauli that acts as a nontrivial logical operation. In stabilizer language, such operators normalize the stabilizer but are not stabilizers themselves. To make the usual global-phase convention explicit, define

$$S^{\mathrm{ph}} := \{\omega s : \omega \in \{\pm 1, \pm i\},\ s \in S\}. \tag{P2a.14}$$

The operators in S^{ph} act on the code as an irrelevant scalar phase, whereas the nontrivial logical Paulis are $N(S) \setminus S^{\mathrm{ph}}$, where the Pauli normalizer is defined by

$$N(S) := \{P \in \mathcal{P}_n : PSP^\dagger = S\}. \tag{P2a.15}$$

Equivalently, when global phases are suppressed, this is the familiar shorthand $N(S) \setminus S$. Therefore

$$d = \min_{P \in N(S) \setminus S^{\mathrm{ph}}} \mathrm{wt}(P). \tag{P2a.16}$$

We must transport two facts: which Paulis are nontrivial logical operations, and how many physical coordinates they affect. Let $Q = UPU^\dagger$. Since $S_B = USU^\dagger$,

$$QS_BQ^\dagger = (UPU^\dagger)(USU^\dagger)(UP^\dagger U^\dagger) = U(PSP^\dagger)U^\dagger. \tag{P2a.17}$$

It follows that

$$P \in N(S) \quad \Longleftrightarrow \quad UPU^\dagger \in N(S_B), \tag{P2a.18}$$

where the reverse direction follows by conjugating back with $U^\dagger$. The same conjugation maps S^{ph} bijectively onto S_B^{ph}. Hence it maps the nontrivial logical Paulis $N(S) \setminus S^{\mathrm{ph}}$ bijectively onto $N(S_B) \setminus S_B^{\mathrm{ph}}$.

It remains to compare their weights. This is where the coordinatewise form of U, not merely its unitarity, is essential. At each physical coordinate, Hadamard maps I to I and each of X, Y, Z to a nonidentity Pauli. It may change the Pauli type or its sign, but it never creates or removes a nonidentity tensor factor. Therefore

$$\mathrm{wt}(UPU^\dagger) = \mathrm{wt}(P), \tag{P2a.19}$$

and the two minima agree:

$$d_B = \min_{Q \in N(S_B) \setminus S_B^{\mathrm{ph}}} \mathrm{wt}(Q) = \min_{P \in N(S) \setminus S^{\mathrm{ph}}} \mathrm{wt}(UPU^\dagger) = \min_{P \in N(S) \setminus S^{\mathrm{ph}}} \mathrm{wt}(P) = d. \tag{P2a.20}$$

This completes the general stabilizer case: B is a stabilizer code with the same parameters $[[n, k, d]]$.

P2a.v Transformed CSS pair

When the original code is CSS, one part of the question remains: does B retain CSS structure, and which classical codes describe it? I now identify the transformed pair. Write

$$C = \mathrm{CSS}(C_1, C_2), \qquad C_2 \subseteq C_1. \tag{P2a.21}$$

Its logical computational-basis states are the uniform coset states

$$|x + C_2\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{c \in C_2} |x + c\rangle, \qquad x \in C_1. \tag{P2a.22}$$

This description makes the two stabilizer families concrete. For $c_0 \in C_2$, the operator X^{c_0} adds c_0 to every basis label and only permutes the words within the same coset:

$$X^{c_0} |x + C_2\rangle = |x + c_0 + C_2\rangle = |x + C_2\rangle. \tag{P2a.23}$$

For $z \in C_1^\perp$, every label $x + c$ belongs to C_1, so $z \cdot (x + c) = 0$ and Z^z contributes phase +1 to every term. Hence the CSS stabilizer is generated by

$$X^c \quad (c \in C_2), \qquad Z^z \quad (z \in C_1^\perp), \tag{P2a.24}$$

where $X^c = \bigotimes_j X^{c_j}$ and $Z^z = \bigotimes_j Z^{z_j}$.

Hadamard exchanges these two kinds of checks coordinatewise:

$$UX^cU^\dagger = Z^c \quad (c \in C_2), \qquad UZ^zU^\dagger = X^z \quad (z \in C_1^\perp). \tag{P2a.25}$$

To identify the transformed CSS pair, compare these families with a general $\mathrm{CSS}(D_1, D_2)$ code:

	$\mathrm{CSS}(D_1, D_2)$	transformed code B
X-check indices	D_2	$C_1^\perp$
Z-check indices	$D_1^\perp$	C_2

Matching the rows gives

$$D_2 = C_1^\perp, \qquad D_1^\perp = C_2, \qquad D_1 = C_2^\perp. \tag{P2a.26}$$

The required nesting also holds. Since $C_2 \subseteq C_1$, every vector orthogonal to all of C_1 is certainly orthogonal to all of C_2; hence

$$C_1^{\perp} \subseteq C_2^{\perp}. \tag{P2a.27}$$

Thus

$$\boxed{H^{\otimes n}\,\mathrm{CSS}(C_1, C_2) = \mathrm{CSS}(C_2^{\perp}, C_1^{\perp}).} \tag{P2a.28}$$

The CSS pair identifies the transformed code space. At the level of individual states, Hadamard transfers the logical coset label from shifts of the computational strings to relative phases. Appendix B.1 gives the explicit Fourier formula and explains why the image need not be a single canonical coset-basis state.

In conclusion, transversal Hadamard gives the unitary equivalence

$$(C, S) \longmapsto (B, USU^{\dagger}) \tag{P2a.29}$$

between stabilizer codes with identical parameters $[[n, k, d]]$. In the CSS case it reverses and dualizes the classical pair. If $B = C$, this equivalence is a logical gate on the original code; otherwise it is a reversible map between two different codes. In either case, the transformed CSS description is exactly the complementary-basis bridge required later in Question 2.

P2b Diagonal in both bases implies maximally mixed

PROBLEM

Let ρ be a density operator on $\mathcal{H}_2^{\otimes n}$. Show that if ρ and $\rho' = H^{\otimes n}\rho H^{\otimes n}$ are both diagonal then $\rho = I_{2^n}/2^n$.

P2b.i Terminology and intuition

This preliminary isolates the mechanism used later in the leakage argument. A matrix diagonal in one basis may scale those basis directions differently. If it is also diagonal in a second basis containing a superposition of all those directions, that superposition must retain its direction; this forces all the scalings to agree.

The computational basis is not essential to this mechanism: one could begin with any orthonormal basis and designate it as the reference basis. I use the computational basis here because it is the basis singled out by the statement and by the later CSS application. What is essential is the relation between the two bases. An arbitrary second basis does not suffice; it must mix the reference directions strongly enough to force their scaling factors to agree. Appendix B.2 gives the precise overlap condition and a counterexample for an insufficiently mixing pair.

In the computational basis, "ρ is diagonal" means

$$\langle x|\,\rho\,|y\rangle = 0 \quad (x \neq y), \qquad \rho = \sum_{x\in\{0,1\}^n} p_x\,|x\rangle\langle x|\,. \tag{P2b.1}$$

Here

$$p_x = \langle x|\,\rho\,|x\rangle\,, \qquad \rho\,|x\rangle = p_x\,|x\rangle\,, \qquad \sum_x p_x = 1. \tag{P2b.2}$$

Thus p_x is at once the diagonal entry, the eigenvalue associated with $|x\rangle$, and the probability of outcome x in a computational-basis measurement.

Diagonality alone does not make the weights uniform: for example, $\mathrm{diag}(0.9, 0.1)$ is diagonal but biased. By contrast, the maximally mixed state has no preferred basis or outcome:

$$\rho_{\mathrm{mix}} = \frac{I_{2^n}}{2^n} = \frac{1}{2^n}\sum_x |x\rangle\langle x|\,, \qquad V\rho_{\mathrm{mix}}V^{\dagger} = \rho_{\mathrm{mix}} \quad \text{for every unitary } V. \tag{P2b.3}$$

Define the Hadamard-basis vectors by $|\widetilde{x}\rangle = H^{\otimes n}\,|x\rangle$. Since $H^{\otimes n}$ is its own adjoint, their matrix entries satisfy

$$\langle\widetilde{x}|\,\rho\,|\widetilde{y}\rangle = \langle x|\,H^{\otimes n}\rho H^{\otimes n}\,|y\rangle = \langle x|\,\rho'\,|y\rangle\,. \tag{P2b.4}$$

Thus the second hypothesis says exactly that ρ is also diagonal in the Hadamard basis: its off-diagonal entries in that basis vanish.

P2b.ii The one-qubit mechanism

For one qubit the mechanism is already complete. Write $\rho = \left(\begin{smallmatrix} p & 0 \\ 0 & 1-p \end{smallmatrix}\right)$. The condition that $H\rho H$ is diagonal is equivalent to requiring $|+\rangle = H|0\rangle$ and $|-\rangle = H|1\rangle$ to be eigenvectors of ρ. In particular,

$$\rho|+\rangle = \frac{p|0\rangle + (1-p)|1\rangle}{\sqrt{2}}. \tag{P2b.5}$$

For this vector to remain proportional to $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, its two coefficients must agree. Hence $p = 1-p = 1/2$. The same mechanism appears in the compact matrix calculation

$$H\rho H = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} p & 0 \\ 0 & 1-p \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 2p-1 \\ 2p-1 & 1 \end{pmatrix}. \tag{P2b.6}$$

In every chosen basis, diagonal entries are measurement populations and off-diagonal entries are coherences relative to that basis. Here $H\rho H$ is the matrix of the same operator in the Hadamard basis. The last display shows that the computational populations $p, 1-p$ become equal Hadamard populations $1/2, 1/2$, while their bias becomes the Hadamard-basis coherence $p - 1/2$. Requiring $H\rho H$ to be diagonal removes precisely that remaining bias.

P2b.iii General n-qubit eigenvector proof

The same eigenvector argument works for all n. Put $U = H^{\otimes n}$. Since $H^\dagger = H$ and $H^2 = I$, we have $U^\dagger = U$ and $U^\dagger U = I$. Thus the conjugation $\rho' = U\rho U^\dagger$ agrees with $U\rho U$ in the statement.

A matrix diagonal in the computational basis maps each basis vector to a scalar multiple of itself. To apply this fact explicitly, write

$$\rho' = \sum_{x\in\{0,1\}^n} p'_x |x\rangle\langle x| . \tag{P2b.7}$$

The vector $|0^n\rangle = |0\rangle^{\otimes n}$ is the computational basis state labelled by the all-zero string. Orthonormality gives $\langle x|0^n\rangle = \delta_{x,0^n}$, so

$$\rho'|0^n\rangle = \sum_x p'_x |x\rangle \langle x|0^n\rangle = p'_{0^n}|0^n\rangle = \lambda|0^n\rangle, \qquad \lambda := p'_{0^n}. \tag{P2b.8}$$

This is precisely the eigenvector equation for $|0^n\rangle$. Multiplying by $U^\dagger = U$ and using $U^\dagger U = I$ gives

$$\rho U|0^n\rangle = \lambda U|0^n\rangle . \tag{P2b.9}$$

But

$$U|0^n\rangle = |+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}|x\rangle , \tag{P2b.10}$$

Substituting this expansion into the eigenvector equation yields

$$\frac{1}{\sqrt{2^n}}\sum_x p_x|x\rangle = \rho|+\rangle^{\otimes n} = \lambda|+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}}\sum_x \lambda|x\rangle . \tag{P2b.11}$$

The computational-basis vectors are linearly independent, so two expansions in that basis are equal only when the coefficient of each $|x\rangle$ agrees. Therefore

$$p_x = \lambda \qquad \text{for every } x \in \{0,1\}^n. \tag{P2b.12}$$

There are exactly 2^n such binary strings. Moreover, because ρ is a density operator, its diagonal probabilities satisfy $\sum_x p_x = 1$. Since every one of those 2^n probabilities equals λ,

$$1 = \sum_x p_x = \sum_x \lambda = 2^n\lambda, \qquad \text{so} \qquad \lambda = \frac{1}{2^n}. \tag{P2b.13}$$

Substituting this common value back into the diagonal expansion of ρ gives

$$\rho = \sum_x p_x|x\rangle\langle x| = \frac{1}{2^n}\sum_x |x\rangle\langle x| = \frac{I_{2^n}}{2^n}. \tag{P2b.14}$$

Conclusion. The eigenvector argument has already proved P2b. Computational-basis diagonality allows one weight p_x for each basis state, while Hadamard-basis diagonality makes the equal superposition $|+\rangle^{\otimes n}$ an

eigenvector and thereby forces all those weights to be equal. Normalization then fixes their common value to 2^{-n}, so ρ is maximally mixed.

P2b.iv Equivalent Pauli-basis proof

I now express the same mechanism through Pauli expectation values, preparing the language used in Question 2(b). Computational diagonality permits diagonal Pauli biases; the second diagonality condition will force every nonidentity bias coefficient to vanish.

For $s = (s_1, \ldots, s_n) \in \{0,1\}^n$, define

$$Z^s := Z^{s_1} \otimes \cdots \otimes Z^{s_n}, \qquad X^s := X^{s_1} \otimes \cdots \otimes X^{s_n}, \tag{P2b.15}$$

where $Z^0 = X^0 = I$. Since

$$Z^s |x\rangle = (-1)^{s\cdot x} |x\rangle, \qquad \langle x| Z^s |y\rangle = (-1)^{s\cdot y}\delta_{x,y}, \tag{P2b.16}$$

every Z^s is diagonal. These 2^n operators are mutually orthogonal:

$$\mathrm{Tr}(Z^s Z^t) = 2^n \delta_{s,t}. \tag{P2b.17}$$

Indeed, if $s = t$ their product is $I^{\otimes n}$; if $s \neq t$, at least one tensor factor in the product is Z, whose trace is zero. The space of $2^n \times 2^n$ diagonal matrices has dimension 2^n, one free entry for each computational string. Hence the 2^n nonzero orthogonal matrices Z^s form an orthogonal basis of that space.

The first diagonality assumption therefore gives the unique expansion

$$\rho = \frac{1}{2^n} \sum_{s\in\{0,1\}^n} a_s Z^s, \qquad a_s = \mathrm{Tr}(\rho Z^s). \tag{P2b.18}$$

Conjugating term by term and using $HZH = X$ yields

$$\rho' = H^{\otimes n} \rho H^{\otimes n} = \frac{1}{2^n} \sum_s a_s X^s. \tag{P2b.19}$$

For $s \neq 0$, the operator X^s flips at least one bit:

$$X^s |x\rangle = |x \oplus s\rangle, \qquad \langle x| X^s |x\rangle = 0. \tag{P2b.20}$$

Because ρ' is diagonal, write $\rho' = \sum_x p'_x |x\rangle\langle x|$. Orthogonality of the Pauli basis then identifies the same coefficient as

$$a_s = \mathrm{Tr}(\rho' X^s) = \sum_x p'_x \langle x| X^s |x\rangle = 0 \qquad (s \neq 0). \tag{P2b.21}$$

Only $s = 0$ remains. Since $Z^0 = I^{\otimes n}$ and $a_0 = \mathrm{Tr}(\rho) = 1$, it follows that

$$\boxed{\rho = \frac{I_{2^n}}{2^n}.} \tag{P2b.22}$$

Connection to Question 2. The two proofs describe the same mechanism: the Hadamard condition forces all computational-basis weights to agree, equivalently removing every nonidentity diagonal Pauli coefficient. Question 2(a) establishes the two diagonality conditions for a small reduced subsystem and invokes P2b to conclude maximal mixing. Question 2(b) uses the broader Pauli-basis method: it determines the reduced state's Pauli expectations from the stabilizer. Those expectations are fixed across encoded states, although some can remain nonzero. The Pauli calculation here prepares that distinction.

2.a CSS codes: any $d-1$ qubits are maximally mixed

PROBLEM

Let C be the CSS code from $C_2 \subset C_1$ with $\mathrm{dist}(C_1), \mathrm{dist}(C_2^\perp) \geq d$, and let

$$|\phi\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{\ell=1}^{2^k} \alpha_\ell \sum_{c\in C_2} |x_\ell + c\rangle, \tag{2.a.1}$$

where $x_1, \ldots, x_{2^k}$ represent the cosets of C_1/C_2. Show that tracing out any $n-t$ coordinates, $t = d-1$, leaves $I_{2^t}/2^t$. (You may assume the t qubits are the first ones — why?)

2.a.i Setup and proof strategy

The nested binary codes $C_2 \subseteq C_1 \subseteq \mathbb{F}_2^n$ define the CSS code, and the problem gives the complementary bounds

$$\operatorname{dist}(C_1) \geq d, \qquad \operatorname{dist}(C_2^\perp) \geq d. \tag{2.a.2}$$

The first will control the computational description; the second will control the description obtained after transversal Hadamard.

For each coset representative x_ℓ, write

$$|\ell_L\rangle := \frac{1}{\sqrt{|C_2|}} \sum_{c \in C_2} |x_\ell + c\rangle \,. \tag{2.a.3}$$

Here ℓ labels a coset of C_2 in C_1, x_ℓ is its chosen representative, and c runs through every word of C_2. Hence $|\phi\rangle = \sum_{\ell=1}^{2^k} \alpha_\ell |\ell_L\rangle$, with $\sum_\ell |\alpha_\ell|^2 = 1$, is an arbitrary normalized state in the CSS code.

I will index the same state directly by its cosets:

$$|\phi\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{L \in C_1/C_2} \alpha_L \sum_{x \in L} |x\rangle \,. \tag{2.a.4}$$

This is only a re-indexing: $L = x_\ell + C_2$, $\alpha_L = \alpha_\ell$, and every $x \in L$ is uniquely $x = x_\ell + c$ for some $c \in C_2$.

Now choose any $t = d-1$ physical positions S to retain and let $R = \{1, \ldots, n\} \setminus S$ be the positions traced out. The state visible on S is

$$\rho_S := \operatorname{Tr}_R |\phi\rangle\langle\phi| \,. \tag{2.a.5}$$

This is only a definition; no diagonality, input-independence, or uniformity has yet been proved. The target is

$$\boxed{\rho_S = \frac{I_{2^t}}{2^t}}. \tag{2.a.6}$$

The error-correction intuition uses the same partition in the opposite direction: inspecting S keeps S and traces out R, whereas erasing S loses S and leaves R for recovery. The distance bounds make this erasure correctable, suggesting that S alone carries no logical information. This motivates the proof; the distance/recovery explanation is preserved in Remark B.1. An input-independent subsystem could still have a fixed bias. Our direct calculation proves the stronger claim that its state is maximally mixed.

The direct proof uses each ingredient once: $\operatorname{dist}(C_1) \geq d$ makes ρ_S computational-basis diagonal; after P2a, $\operatorname{dist}(C_2^\perp) \geq d$ makes its Hadamard conjugate diagonal; P2b then yields $\rho_S = I_{2^t}/2^t$. The two-basis argument in P2b explains why these two diagonalities force uniformity.

2.a.ii Why the selected qubits may be placed first

The following calculation is simplest when the retained subsystem occupies the first t tensor factors. This is only a relabelling of physical positions, as I now justify.

Relabelling the coordinates. Write

$$S = \{s_1 < \cdots < s_t\} \tag{2.a.7}$$

and choose a coordinate permutation π satisfying

$$\pi(s_j) = j \qquad (1 \leq j \leq t). \tag{2.a.8}$$

Thus the original qubit s_j becomes qubit j after the permutation. For a binary word $x = (x_1, \ldots, x_n)$, let πx denote the correspondingly reordered word, characterized by

$$(\pi x)_{\pi(r)} = x_r. \tag{2.a.9}$$

Let U_π perform the same reordering on the physical qubits:

$$U_\pi |x\rangle = |\pi x\rangle \,. \tag{2.a.10}$$

Hence πx and $U_\pi |x\rangle$ are the classical and quantum descriptions, respectively, of the same coordinate relabelling.

Preservation of the CSS hypotheses. Define

$$C_i^\pi := \{\pi x : x \in C_i\}. \tag{2.a.11}$$

Coordinate permutation is a linear bijection preserving Hamming weight and the binary inner product. Therefore

$$C_2^\pi \subseteq C_1^\pi, \qquad \mathrm{wt}(\pi x) = \mathrm{wt}(x), \qquad \mathrm{dist}(C_1^\pi) = \mathrm{dist}(C_1). \tag{2.a.12}$$

It also preserves the dual code. Indeed, for $z \in \mathbb{F}_2^n$,

$$\begin{aligned} \pi z \in (C_2^\pi)^\perp &\iff (\pi z)\cdot(\pi c) = 0 \quad \text{for every } c \in C_2 \\ &\iff z \cdot c = 0 \quad \text{for every } c \in C_2 \iff z \in C_2^\perp. \end{aligned} \tag{2.a.13}$$

Consequently,

$$(C_2^\pi)^\perp = \{\pi z : z \in C_2^\perp\}, \qquad \mathrm{dist}\big((C_2^\pi)^\perp\big) = \mathrm{dist}(C_2^\perp). \tag{2.a.14}$$

Thus both distance hypotheses remain unchanged.

For each coset $L \in C_1/C_2$, write $\pi L := \{\pi x : x \in L\}$. These image sets are precisely the cosets of C_2^π in C_1^π. Therefore

$$U_\pi |\phi\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{L \in C_1/C_2} \alpha_L \sum_{y \in \pi L} |y\rangle \tag{2.a.15}$$

is an encoded state of $\mathrm{CSS}(C_1^\pi, C_2^\pi)$, with the same logical amplitudes. The permuted code and state are therefore exactly another instance of the stated problem, satisfying the same hypotheses.

The first t tensor factors of $U_\pi |\phi\rangle$ are the original qubits $s_1, \ldots, s_t$, in that order. Identifying the retained subsystem in this order, their reduced state is

$$\mathrm{Tr}_{\{t+1,\ldots,n\}}\left(U_\pi |\phi\rangle\langle\phi| U_\pi^\dagger\right) = \rho_S. \tag{2.a.16}$$

If the claim is proved for the first t positions, its left-hand side equals $I_{2^t}/2^t$, and therefore

$$\rho_S = \frac{I_{2^t}}{2^t}. \tag{2.a.17}$$

It consequently suffices to continue with $S = \{1, \ldots, t\}$.

2.a.iii Computational-basis diagonality

For each word in (2.a.4), write $x = (x_S, x_R)$, so $|x\rangle = |x_S\rangle \otimes |x_R\rangle$: the first factor is retained, and the second is discarded. Using the basis-sum definition of partial trace (Definition 1.1) with the computational basis of R, one density-matrix term gives

$$|x\rangle\langle x'| = \big(|x_S\rangle\langle x'_S|\big) \otimes \big(|x_R\rangle\langle x'_R|\big), \qquad \mathrm{Tr}_R(|x\rangle\langle x'|) = \underbrace{\langle x'_R|x_R\rangle}_{\substack{\text{scalar from the} \\ \text{traced-out rest}}} \underbrace{|x_S\rangle\langle x'_S|}_{\substack{\text{operator on the} \\ \text{retained subsystem}}}. \tag{2.a.18}$$

The scalar is 0 when the discarded labels differ and 1 when they agree. For example, keeping the first qubit and tracing out the second gives

$$\mathrm{Tr}_2(|00\rangle\langle 11|) = |0\rangle\langle 1| \, \langle 1|0\rangle = 0, \qquad \mathrm{Tr}_2(|00\rangle\langle 10|) = |0\rangle\langle 1| \, \langle 0|0\rangle = |0\rangle\langle 1|. \tag{2.a.19}$$

Partial trace alone does not eliminate every off-diagonal term: equal discarded labels can leave coherence on S. The distance assumption will exclude precisely that case for this CSS state.

Applying (2.a.18) term by term to $|\phi\rangle\langle\phi|$ gives

$$\rho_S = \mathrm{Tr}_R |\phi\rangle\langle\phi| = \frac{1}{|C_2|} \sum_{L,L' \in C_1/C_2} \alpha_L \overline{\alpha_{L'}} \sum_{\substack{x \in L \\ x' \in L'}} \langle x'_R|x_R\rangle \, |x_S\rangle\langle x'_S|. \tag{2.a.20}$$

The role of the distance bound is easiest to see before introducing the difference vector. Since $\mathrm{dist}(C_1) \geq d$, any two distinct words of C_1 differ in at least d coordinates. A surviving off-diagonal term, however, would require physical words $x, x' \in C_1$ with

$$x_R = x'_R \quad \text{(the term survives the trace)}, \qquad x_S \neq x'_S \quad \text{(the surviving term is off-diagonal)}. \tag{2.a.21}$$

These words would be distinct, but every coordinate on which they differ would lie in S. Because $|S| = d - 1$, they could differ in at most $d - 1$ coordinates, contradicting the distance bound.

Equivalently, set $z = x + x'$ to make the same contradiction algebraic. Since C_1 is linear, $z \in C_1$; the inequality $x_S \neq x'_S$ gives $z \neq 0$; and the equality $x_R = x'_R$ gives $z_R = 0$. Thus every nonzero coordinate of z lies in S. The two sides of the contradiction are therefore

$$\mathrm{wt}(z) \geq \mathrm{dist}(C_1) \geq d, \qquad \mathrm{wt}(z) \leq |S| = d - 1. \tag{2.a.22}$$

No such pair x, x' can exist. Hence every term that survives the partial trace is diagonal on S, and

$$\rho_S \quad \text{is computational-basis diagonal.} \tag{2.a.23}$$

Diagonal here does not yet mean maximally mixed: its diagonal probabilities may still be biased.

2.a.iv Hadamard-basis diagonality

P2a states that transversal Hadamard sends the entire CSS code according to

$$H^{\otimes n}\,\mathrm{CSS}(C_1, C_2) = \mathrm{CSS}(C_2^\perp, C_1^\perp), \qquad C_1^\perp \subseteq C_2^\perp, \tag{2.a.24}$$

while preserving its parameters $[[n, k, d_{\mathrm{CSS}}]]$. Define the transformed global state and its reduced state on the same retained qubits S by

$$|\phi_H\rangle := H^{\otimes n}\,|\phi\rangle\,, \qquad \widetilde{\rho}_S := \mathrm{Tr}_R\,|\phi_H\rangle\langle\phi_H|\,. \tag{2.a.25}$$

Thus P2a gives $|\phi_H\rangle \in \mathrm{CSS}(C_2^\perp, C_1^\perp)$.

The proof of (2.a.23) used only that all basis strings lie in the outer code C_1 and that two distinct such strings differ in at least d positions. For $|\phi_H\rangle$, the outer code is $C_2^\perp$, and the given bound $\mathrm{dist}(C_2^\perp) \geq d$ supplies the same distance property. Hence a surviving off-diagonal term in $\widetilde{\rho}_S$ would require two distinct words of $C_2^\perp$ that differ only among the $d - 1$ positions of S, which is impossible. Therefore

$$\widetilde{\rho}_S \quad \text{is computational-basis diagonal.} \tag{2.a.26}$$

It remains to connect $\widetilde{\rho}_S$ to the original ρ_S.

Fact 2.1 (Local unitaries and partial trace)**.** For any state ρ_{SR} and unitaries U_S, V_R on the indicated subsystems,

$$\mathrm{Tr}_R\Big[(U_S \otimes V_R)\rho_{SR}(U_S^\dagger \otimes V_R^\dagger)\Big] = U_S\,\mathrm{Tr}_R(\rho_{SR})U_S^\dagger. \tag{2.a.27}$$

Proof. Choose any orthonormal basis $\{|j\rangle_R\}$ of the discarded subsystem. By the definition of partial trace,

$$\begin{aligned}
&\mathrm{Tr}_R\Big[(U_S \otimes V_R)\rho_{SR}(U_S^\dagger \otimes V_R^\dagger)\Big] \\
&= \sum_j (I_S \otimes \langle j|_R)(U_S \otimes V_R)\rho_{SR}(U_S^\dagger \otimes V_R^\dagger)(I_S \otimes |j\rangle_R) \\
&= U_S\left[\sum_j (I_S \otimes \langle j|_R\, V_R)\rho_{SR}(I_S \otimes V_R^\dagger\,|j\rangle_R)\right] U_S^\dagger.
\end{aligned} \tag{2.a.28}$$

The last equality pulls out the operators on S. Put $\big|\widetilde{j}\big\rangle_R := V_R^\dagger\,|j\rangle_R$. Unitarity makes $\{\big|\widetilde{j}\big\rangle_R\}$ another orthonormal basis, with $\langle j|_R\, V_R = \big\langle\widetilde{j}\big|_R$. The bracket is therefore

$$\sum_j (I_S \otimes \big\langle\widetilde{j}\big|_R)\rho_{SR}(I_S \otimes \big|\widetilde{j}\big\rangle_R) = \mathrm{Tr}_R(\rho_{SR}), \tag{2.a.29}$$

which proves (2.a.27). A unitary on the retained subsystem remains as a conjugation; a unitary acting solely on the subsystem that is traced out amounts only to a change of basis in the summation. ∎

Write $H_S := H^{\otimes t}$ and $H_R := H^{\otimes(n-t)}$ for the Hadamards on the retained and discarded qubits, respectively. Since $H^{\otimes n} = H_S \otimes H_R$, Fact 2.1 gives

$$\widetilde{\rho}_S = \mathrm{Tr}_R[(H_S \otimes H_R)\,|\phi\rangle\langle\phi|\,(H_S \otimes H_R)] = H_S\,\mathrm{Tr}_R(|\phi\rangle\langle\phi|)H_S = H^{\otimes t}\rho_S H^{\otimes t}. \tag{2.a.30}$$

Thus (2.a.26) says exactly that the Hadamard conjugate $H^{\otimes t}\rho_S H^{\otimes t}$ is diagonal, as P2b requires.

2.a.v Applying P2b

P2b states that every t-qubit density operator σ satisfies

$$\left.\begin{array}{l}\sigma \text{ is computational-basis diagonal,}\\ H^{\otimes t}\sigma H^{\otimes t} \text{ is computational-basis diagonal}\end{array}\right\} \quad\Longrightarrow\quad \sigma = \frac{I_{2^t}}{2^t}. \tag{2.a.31}$$

For $\sigma = \rho_S$, the two required hypotheses have now been established:

$$\begin{array}{rl}\rho_S \text{ is diagonal} & \text{by (2.a.23),}\\ H^{\otimes t}\rho_S H^{\otimes t} = \widetilde{\rho}_S \text{ is diagonal} & \text{by (2.a.26) and the calculation above.}\end{array} \tag{2.a.32}$$

Therefore P2b gives exactly the desired reduced state:

$$\boxed{\rho_S = \frac{I_{2^t}}{2^t}}. \tag{2.a.33}$$

Since S was arbitrary, every set of $d-1$ physical qubits has this same reduced state. Its computational-basis outcomes are uniform, and no measurement on it can reveal the logical amplitudes α_L. The proof map in Appendix B.3.ii records where each hypothesis was used.

2.b Stabilizer codes: ρ_T is codeword-independent

PROBLEM

Let C be a stabilizer code of distance d, $t \le d-1$, $T \subset \{1,\dots,n\}$ with $|T| = t$, $W = \overline{T}$, and $\rho = \mathrm{Tr}_W(|\phi\rangle\langle\phi|)$ for $|\phi\rangle \in C$. Show that the $2^t \times 2^t$ matrix ρ may depend on T but is otherwise the same for all $|\phi\rangle \in C$. Hint: first show that any Hermitian ρ can be written $\rho = 2^{-t}\sum_P a_P P$ over the t-qubit Paulis $\{I,X,Y,Z\}^{\otimes t}$ with $a_P = \mathrm{Tr}(\rho P)$, noting the orthogonality $2^{-t}\,\mathrm{Tr}(PQ) = \delta_{PQ}$; then use the stabilizer formalism to show that for our particular ρ, $\mathrm{Tr}(\rho P)$ can be either 0 or 1, the same for any state in C. Finally, to show the dependence on T, give an example of a code where this happens.

2.b.i Proof strategy

For each encoded state $|\phi\rangle \in C$, write

$$\rho_T^{\phi} := \mathrm{Tr}_W\, |\phi\rangle\langle\phi|\,. \tag{2.b.1}$$

The claim is

$$\boxed{\rho_T^{\phi} = \rho_T^{\psi} \quad \text{for every } |\phi\rangle, |\psi\rangle \in C,} \tag{2.b.2}$$

although the common reduced state may change when the physical set T changes.

This is weaker than saying that ρ_T^{ϕ} is maximally mixed. The subsystem T may retain correlations fixed by the code, but none may depend on the logical state. The proof will describe ρ_T^{ϕ} by its Pauli coordinates, lift each local Pauli basis element to the full code block, and use the distance bound to show that the corresponding coordinate is fixed across the code space.

2.b.ii Pauli reconstruction of the reduced state

I use the Pauli operator basis encountered in the course's treatment of error discretization. In the Pauli proof of P2b, a diagonal state was expanded in Z-strings; here the reduced state need not be diagonal, so I use the full family of phase-free Hermitian Pauli strings

$$\mathcal{B}_t := \{I,X,Y,Z\}^{\otimes t}. \tag{2.b.3}$$

Thus $P \in \mathcal{B}_t$ means $P = P_1 \otimes \cdots \otimes P_t$ with each $P_j \in \{I,X,Y,Z\}$.

Fact 2.2 (Pauli reconstruction)**.** Every $2^t \times 2^t$ matrix A has the unique expansion

$$A = 2^{-t} \sum_{P \in \mathcal{B}_t} \mathrm{Tr}(AP)P. \tag{2.b.4}$$

Consequently, two such matrices are equal exactly when their traces against every $P \in \mathcal{B}_t$ agree.

Proof. For single-qubit Paulis, $\mathrm{Tr}(P_jQ_j) = 2$ when $P_j = Q_j$, whereas for $P_j \neq Q_j$ the product P_jQ_j is, up to

phase, a traceless nonidentity Pauli. Since the trace of a tensor product is the product of the traces, for $P, Q \in \mathcal{B}_t$,

$$\mathrm{Tr}(PQ) = \prod_{j=1}^{t} \mathrm{Tr}(P_j Q_j) = 2^t \delta_{P,Q}. \tag{2.b.5}$$

These are 4^t nonzero orthogonal matrices in a matrix space of dimension $(2^t)^2 = 4^t$, so they form a basis.

Let $A = \sum_{P\in\mathcal{B}_t} c_P P$, and fix one $Q \in \mathcal{B}_t$. Taking the trace against Q extracts its coordinate. Indeed, by (2.b.5), every term with $P \neq Q$ vanishes, and only the $P = Q$ term remains:

$$\mathrm{Tr}(AQ) = \sum_{P\in\mathcal{B}_t} c_P \,\mathrm{Tr}(PQ) = c_Q \,\mathrm{Tr}(Q^2) = 2^t c_Q, \qquad \therefore \quad c_Q = 2^{-t}\,\mathrm{Tr}(AQ). \tag{2.b.6}$$

The factor 2^{-t} therefore comes from dividing by $\mathrm{Tr}(Q^2) = 2^t$. Since Q was arbitrary, this determines every coordinate uniquely and proves the expansion. ∎

Applying Fact 2.2 to the reduced state gives

$$\rho_T^{\phi} = 2^{-t} \sum_{P\in\mathcal{B}_t} a_P^{\phi} P, \qquad a_P^{\phi} := \mathrm{Tr}(\rho_T^{\phi} P). \tag{2.b.7}$$

Because P is Hermitian and $P^2 = I$, its measurement outcomes lie in $\{+1, -1\}$. If $p_{\pm}^{\phi}$ denote their probabilities, then $a_P^{\phi} = (+1)p_+^{\phi} + (-1)p_-^{\phi} = p_+^{\phi} - p_-^{\phi}$ is their expectation value, while the coordinate multiplying P in (2.b.7) is $2^{-t} a_P^{\phi}$. The fact therefore reduces the claim to proving that every a_P^{ϕ} is independent of $|\phi\rangle$. Appendix B.4.i develops the Pauli coordinates step by step and derives the projector probabilities and trace expectations from the Born rule, beginning with a concrete measurement example.

2.b.iii Lifting a local Pauli to the full code block

Write $T = \{j_1 < \cdots < j_t\}$ and fix $P = P_1 \otimes \cdots \otimes P_t \in \mathcal{B}_t$. Let $\widehat{P}$ be the n-qubit Pauli whose factor on physical qubit j_r is P_r, and whose factor at every position in W is the identity. Applying Definition 1.1 successively to the qubits of W, and temporarily grouping the factors as $T \otimes W$, gives the required local-to-global bridge. In product bases $\{|i\rangle\}$ of T and $\{|w\rangle\}$ of W,

$$a_P^{\phi} = \mathrm{Tr}_T[(\mathrm{Tr}_W |\phi\rangle\langle\phi|)P] = \sum_{i,w} \langle i, w| \,|\phi\rangle\langle\phi|\, \widehat{P}\, |i, w\rangle = \mathrm{Tr}_{TW}(|\phi\rangle\langle\phi|\, \widehat{P}) = \langle\phi|\, \widehat{P}\, |\phi\rangle\,. \tag{2.b.8}$$

The final equality uses $\mathrm{Tr}(|u\rangle\langle v|\, A) = \langle v|\, A\, |u\rangle$. Thus measuring P after discarding W has the same expectation as measuring P on T and doing nothing on W in the full state. This is the bridge to the distance bound: by construction,

$$\mathrm{supp}(\widehat{P}) \subseteq T, \qquad \mathrm{wt}(\widehat{P}) \leq |T| = t < d. \tag{2.b.9}$$

The calculation in Appendix B.4.ii expands the partial-trace identity and illustrates the embedding when the selected positions are not consecutive.

2.b.iv Code distance fixes each Pauli coefficient

Let $S = \mathrm{Stab}(C)$, and use the phase closure S^{ph} from (P2a.14). The distance characterization (P2a.16) gives

$$d = \min_{Q\in N(S)\setminus S^{\mathrm{ph}}} \mathrm{wt}(Q); \tag{2.b.10}$$

the operators in this set are precisely the nontrivial logical Paulis, up to phase. Since $\mathrm{wt}(\widehat{P}) < d$, $\widehat{P}$ cannot be one of them. Two cases remain.

If $\widehat{P} \in S^{\mathrm{ph}}$, write $\widehat{P} = \lambda_P s$ for some $s \in S$. Hermiticity forces $\lambda_P \in \{+1, -1\}$, and every codeword is fixed by s. Therefore

$$a_P^{\phi} = \langle\phi|\, \widehat{P}\, |\phi\rangle = \lambda_P, \tag{2.b.11}$$

a value determined by P, T, and the code, not by $|\phi\rangle$.

If $\widehat{P} \notin S^{\mathrm{ph}}$, it cannot commute with every stabilizer: otherwise it would lie in $N(S) \setminus S^{\mathrm{ph}}$, contradicting (2.b.10) and $\mathrm{wt}(\widehat{P}) < d$. Hence some $g \in S$ anticommutes with $\widehat{P}$. Using $g\,|\phi\rangle = |\phi\rangle$, $\langle\phi|\, g = \langle\phi|$, and $g^2 = I$,

$$a_P^{\phi} = \langle\phi|\, \widehat{P}\, |\phi\rangle = \langle\phi|\, g\widehat{P}g\, |\phi\rangle = \langle\phi|\, (-\widehat{P})\, |\phi\rangle = -a_P^{\phi}. \tag{2.b.12}$$

Thus $a_P^\phi = 0$ for every encoded state.

Geometrically, $|\phi\rangle$ belongs to the +1 eigenspace of the Hermitian stabilizer check g, whereas anticommutation gives

$$g(\widehat{P}\,|\phi\rangle) = -\widehat{P}g\,|\phi\rangle = -\widehat{P}\,|\phi\rangle\,. \tag{2.b.13}$$

Thus $\widehat{P}\,|\phi\rangle$ belongs to its orthogonal -1 eigenspace, so the overlap $\langle\phi|\,\widehat{P}\,|\phi\rangle$ vanishes.

We have proved that every a_P^ϕ is fixed by P, T, and the code: it is a fixed sign when $\widehat{P}$ is a stabilizer up to phase, and it is zero otherwise. The hint suppresses the sign subtlety and states the nonzero case as 1; with the fixed phase-free representatives in $\mathcal{B}_t$, it may also be -1. If $\widehat{P} = -s$, it commutes with the same operators as s and simply has eigenvalue -1 throughout the code space. Fact 2.2 now gives

$$\boxed{\rho_T^\phi = \rho_T^\psi \qquad \text{for all } |\phi\rangle, |\psi\rangle \in C.} \tag{2.b.14}$$

This is the no-leakage counterpart of erasure correction. The distinction between erasing the small subsystem and observing it is explained in Remark B.1, with its retained set S corresponding to T here and its complement R to W.

2.b.v How the reduced state depends on T

Distance d rules out nontrivial *logical* Paulis below weight d. It does not rule out stabilizers of low weight. Such a stabilizer fixes a local correlation for every codeword. The correlation reveals a property of the code but no information about which logical state was encoded.

The same calculation identifies the remaining T-dependence. Define

$$S[T] := \{g \in S : \mathrm{supp}(g) \subseteq T\}, \tag{2.b.15}$$

For $g \in S[T]$, let g_T be the signed Pauli on T obtained by deleting the identity factors in W; equivalently, after grouping the factors as $T \otimes W$, $g = g_T \otimes I_W$. If a Pauli coordinate is nonzero, then $\widehat{P} = \lambda_P g$ for some $g \in S[T]$; restriction to T gives $P = \lambda_P g_T$, and hence

$$a_P^\phi P = \lambda_P P = \lambda_P^2 g_T = g_T. \tag{2.b.16}$$

Conversely, for each $g \in S[T]$, its signed restriction has a unique form $g_T = \lambda_P P$ with $P \in \mathcal{B}_t$ and $\lambda_P \in \{+1, -1\}$. Then $\widehat{P} = \lambda_P g$, so this P contributes exactly the term $a_P^\phi P = g_T$. All other coordinates vanish, and (2.b.7) becomes

$$\boxed{\rho_T^\phi = 2^{-t} \sum_{g \in S[T]} g_T.} \tag{2.b.17}$$

Thus T can matter through nonidentity stabilizers whose entire support lies inside it. Different physical subsets can contain different stabilizers, so their common codeword-independent reduced states can differ.

For example, the standard $[[9, 1, 3]]$ Shor code has stabilizer generators

$$\begin{aligned} &Z_1Z_2,\ Z_2Z_3,\ Z_4Z_5,\ Z_5Z_6,\ Z_7Z_8,\ Z_8Z_9, \\ &X_1X_2X_3X_4X_5X_6, \qquad X_4X_5X_6X_7X_8X_9. \end{aligned} \tag{2.b.18}$$

Its distance is $d = 3$, so I compare two subsets of size $t = 2 < 3$.

First take $T = \{1, 2\}$. The stabilizer Z_1Z_2 is supported completely inside this pair. Within the first three-qubit block, the Z-type generators produce exactly

$$I, \quad Z_1Z_2, \quad Z_2Z_3, \quad Z_1Z_3. \tag{2.b.19}$$

Only the first two fit inside T. Every nontrivial product of the two X-type generators occupies six positions; multiplying by Z-type stabilizers can turn X into Y up to phase, but cannot remove that support. Hence no stabilizer with a nonidentity X component fits inside the pair, and $S[T] = \{I, Z_1Z_2\}$. Equation (2.b.17) gives

$$\rho_{\{1,2\}} = \frac{1}{4}(I \otimes I + Z \otimes Z) = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|). \tag{2.b.20}$$

These qubits are not maximally mixed as a pair: their computational-basis outcomes are 00 and 11, each with probability $1/2$, so they have fixed even Z-parity. This correlation is identical for every encoded logical state.

Now take $T' = \{1, 4\}$. The Z-type stabilizer subgroup has even Z-parity within each of the three blocks $\{1, 2, 3\}$, $\{4, 5, 6\}$, and $\{7, 8, 9\}$. Only one position is available in each of the first two blocks. Any nonidentity Z string

supported inside T' would therefore have odd parity in at least one block, which is impossible. A stabilizer with a nonidentity X component again has support larger than two. Thus $S[T'] = \{I\}$: every nonidentity Pauli expectation on this pair vanishes, and

$$\rho_{\{1,4\}} = \frac{1}{4} I_4. \tag{2.b.21}$$

Equations (2.b.20) and (2.b.21) show that $\rho_{\{1,2\}} \neq \rho_{\{1,4\}}$, although each reduced state is individually independent of the encoded logical state. The first pair reveals a fixed even-parity constraint of the code; the second pair is maximally mixed. This is how the reduced state can depend on T without revealing which logical state was encoded. The enumeration in Appendix B.4.iii records the block stabilizers and the three nonidentity X-type products explicitly.

2.b.vi Why part (a) is stronger

Quantum distance excludes low-weight nontrivial logical Paulis, but it does not exclude low-weight stabilizers. The two classical distance assumptions in part (a) impose this additional restriction. Indeed, every stabilizer of that CSS code has the form

$$X(u)Z(v), \qquad X(u) := \bigotimes_{j=1}^{n} X^{u_j}, \quad u \in C_2, \qquad Z(v) := \bigotimes_{j=1}^{n} Z^{v_j}, \quad v \in C_1^{\perp}. \tag{2.b.22}$$

Since $C_2 \subseteq C_1$ and $C_1^{\perp} \subseteq C_2^{\perp}$, the assumptions $\mathrm{dist}(C_1), \mathrm{dist}(C_2^{\perp}) \geq d$ imply

$$u \neq 0 \Longrightarrow \mathrm{wt}(u) \geq d, \qquad v \neq 0 \Longrightarrow \mathrm{wt}(v) \geq d. \tag{2.b.23}$$

Overlapping X and Z factors become Y up to phase rather than cancelling, so

$$\mathrm{supp}(X(u)Z(v)) = \mathrm{supp}(u) \cup \mathrm{supp}(v). \tag{2.b.24}$$

Consequently, every nonidentity stabilizer in part (a) has weight at least d. For $|T| < d$, this forces $S[T] = \{I\}$, and (2.b.17) reduces to

$$\rho_T = 2^{-t} I. \tag{2.b.25}$$

This is the exact distinction: general stabilizer distance still permits fixed local stabilizers, so part (b) proves codeword-independence; part (a)'s additional classical bounds remove those terms and force maximal mixing.

QUESTION 3

Quantum State Complexity

In this question I examine the resources needed to encode and prepare quantum states. Parts (A)–(C) establish depth lower bounds for quantum codes and for the GHZ and W states, using the limited spread of information through bounded-arity unitary circuits. Part (D) shows how measurements and classical feed-forward allow GHZ preparation with constant quantum depth. Finally, part (E) turns from depth to total size: a counting argument shows that some states require exponentially many operations even when measurements and classical computation are allowed.

3.A Encoding circuits have depth $\Omega(\log d)$

PROBLEM

Let C be a $[[n, k, d]]$ code with $k \geq 1$, and let QC_A map $b \in \{0,1\}^k$ together with $n-k$ qubits in $|0\rangle$ to $|S_b\rangle \in C$, distinct b's going to orthogonal logical states. Show $\text{depth}(QC_A) = \Omega(\log d)$. Note: do not use undefined notions such as "the state of a subset of the qubits" without defining them.

3.A.i Probing the first logical input bit with Z

Let $U = QC_A$ denote the unitary implemented by the encoder. For a logical input string $b = (b_1, \ldots, b_k) \in \{0,1\}^k$, define

$$|\text{in}_b\rangle := |b\rangle\,|0\rangle^{\otimes(n-k)}, \qquad U\,|\text{in}_b\rangle = |S_b\rangle \in C. \tag{3.A.1}$$

The problem assumes $k \geq 1$, so a first logical input qubit exists, and it states that distinct logical inputs are mapped to distinct orthogonal code states.

To relate circuit depth to code distance, we need one object to which both constraints apply. It should begin on very few input qubits, so depth limits how far it can spread, but after encoding it should still carry logical information, so distance limits how locally it can be accessed. Pauli Z on the first logical input qubit is a natural choice. Define

$$Z_1 := Z \otimes I^{\otimes(n-1)}. \tag{3.A.2}$$

Since $Z\,|0\rangle = |0\rangle$ and $Z\,|1\rangle = -\,|1\rangle$,

$$Z_1\,|\text{in}_b\rangle = (-1)^{b_1}\,|\text{in}_b\rangle\,. \tag{3.A.3}$$

Thus Z_1 initially acts on only one qubit, while its eigenvalue distinguishes $b_1 = 0$ from $b_1 = 1$.

3.A.ii Transporting the first-bit observable

Define

$$\widetilde{Z}_1 := UZ_1U^\dagger. \tag{3.A.4}$$

Here "transporting" means that $\widetilde{Z}_1$ is the output-side operator representing the same logical question that Z_1 asked on the input side. Equivalently,

$$\widetilde{Z}_1 U = (UZ_1U^\dagger)U = UZ_1, \tag{3.A.5}$$

so applying Z_1 before the encoder has the same final effect as applying $\widetilde{Z}_1$ after the encoder. Indeed, for every b,

$$\widetilde{Z}_1\,|S_b\rangle = UZ_1U^\dagger U\,|\text{in}_b\rangle = UZ_1\,|\text{in}_b\rangle = (-1)^{b_1}U\,|\text{in}_b\rangle = (-1)^{b_1}\,|S_b\rangle\,. \tag{3.A.6}$$

Hence encoded states with $b_1 = 0$ have eigenvalue $+1$ under $\widetilde{Z}_1$, while those with $b_1 = 1$ have eigenvalue -1.

3.A.iii Circuit depth bounds the physical support

For $T \subseteq \{1, \ldots, n\}$, an operator A is **supported on** T if

$$A = A_T \otimes I_{\bar{T}}; \tag{3.A.7}$$

that is, it acts as the identity outside T. The support $\text{supp}(A)$ is the smallest such set.

The depth claim is understood in the standard bounded-arity circuit model. Let $r \geq 2$ be the fixed maximum number of qubits touched by one gate, independent of n, k, d; equivalently, $r = O(1)$. Gates within a layer act on pairwise disjoint sets of qubits. Under the one-qubit and two-qubit convention stated explicitly in part (B), $r = 2$. Without a fixed arity bound, the entire encoder could be treated as one n-qubit gate, so no growing depth lower bound would be possible.

I first explain the growth mechanism for two-qubit gates. Initially Z_1 acts nontrivially on only the first qubit. A gate on two other qubits sees only identity factors of this observable, so conjugating by it changes nothing. To spread the support, a gate must touch a qubit that is already in the support. A two-qubit gate can then bring in at most one new partner.

The layer condition is crucial: each qubit participates in at most one gate in that layer. A newly reached partner therefore cannot immediately pass the support on through another gate; that must wait for the next layer. Starting with one supported qubit, the first layer can therefore leave at most two. In the second layer, each of those two can meet one new partner, giving at most four; the next layer can similarly reach at most eight:

$$1 \longrightarrow 2 \longrightarrow 4 \longrightarrow 8 \longrightarrow \cdots . \tag{3.A.8}$$

These are upper bounds, not guaranteed increases. A gate between two already supported qubits adds no new wire, and a gate may fail to enlarge the support at all.

For the formal proof, let D be the number of circuit layers. For $j = 1, \ldots, D$, define U_j to be the unitary on all n qubits implemented by layer j: it applies that layer's disjoint gates and acts as the identity on every untouched qubit. Thus U_j represents the entire layer, which may contain several gates. Layer 1 acts first, so

$$U = U_D U_{D-1} \cdots U_1. \tag{3.A.9}$$

Define L_j to be the observable obtained from Z_1 after transport through the first j layers. Equivalently, the definition is recursive:

$$L_0 := Z_1, \qquad L_j := U_j L_{j-1} U_j^\dagger \quad (1 \leq j \leq D). \tag{3.A.10}$$

After all D layers this gives

$$L_D = U Z_1 U^\dagger = \widetilde{Z}_1. \tag{3.A.11}$$

Set $s_j := |\operatorname{supp}(L_j)|$. If an operator A is supported on S and a gate G acts on R, then every factor in $GAG^\dagger$ acts as the identity outside $S \cup R$; hence $\operatorname{supp}(GAG^\dagger) \subseteq S \cup R$. In particular, a gate in layer j whose qubits are disjoint from $\operatorname{supp}(L_{j-1})$ commutes with L_{j-1}, so conjugating by that gate leaves the operator unchanged. Because the gates in a layer are pairwise disjoint, at most s_{j-1} gates can meet the current support. Each such gate acts on at most r qubits and can therefore introduce at most $r - 1$ new qubits. Hence

$$s_j \leq s_{j-1} + (r-1)s_{j-1} = r s_{j-1}. \tag{3.A.12}$$

Since $s_0 = |\operatorname{supp}(Z_1)| = 1$, induction gives $s_j \leq r^j$. In particular,

$$|\operatorname{supp}(\widetilde{Z}_1)| = |\operatorname{supp}(L_D)| \leq r^D. \tag{3.A.13}$$

For one-qubit and two-qubit gates this specializes to $|\operatorname{supp}(\widetilde{Z}_1)| \leq 2^D$.

3.A.iv Distance forces the support to be large

Let

$$T := \operatorname{supp}(\widetilde{Z}_1). \tag{3.A.14}$$

By the definition of support,

$$\widetilde{Z}_1 = (\widetilde{Z}_1)_T \otimes I_{\bar{T}}. \tag{3.A.15}$$

Choose two inputs that differ only in their first logical bit, for example

$$b^{(0)} := 0^k, \qquad b^{(1)} := (1, 0, \ldots, 0). \tag{3.A.16}$$

Their reduced output density matrices on T are

$$\rho_T^{(j)} := \operatorname{Tr}_{\bar{T}}\left(\left|S_{b^{(j)}}\right\rangle\left\langle S_{b^{(j)}}\right|\right), \qquad j \in \{0, 1\}. \tag{3.A.17}$$

This is the formal meaning of the state of the output subsystem T.

The defining property of partial trace and equation (3.A.6) give

$$\mathrm{Tr}_T\left(\rho_T^{(0)}(\widetilde{Z}_1)_T\right) = \left\langle S_{b^{(0)}}\right| \widetilde{Z}_1 \left|S_{b^{(0)}}\right\rangle = 1, \qquad \mathrm{Tr}_T\left(\rho_T^{(1)}(\widetilde{Z}_1)_T\right) = \left\langle S_{b^{(1)}}\right| \widetilde{Z}_1 \left|S_{b^{(1)}}\right\rangle = -1. \tag{3.A.18}$$

The same reduced density matrix cannot give two different expectation values for the same operator. Therefore

$$\rho_T^{(0)} \neq \rho_T^{(1)}. \tag{3.A.19}$$

The physical subsystem T can distinguish two code states carrying different values of the first logical bit.

Question 2(c) states directly that

$$|T| < d \quad \Longrightarrow \quad \text{all code states have the same reduced density matrix on } T. \tag{3.A.20}$$

The two reduced states above are different, so $|T| \geq d$. Thus the same support set T is forced by code distance to contain at least d qubits and is limited by circuit depth to at most r^D qubits. Equivalently,

$$d \leq |\operatorname{supp}(\widetilde{Z}_1)| \leq r^D. \tag{3.A.21}$$

Therefore

$$D \geq \log_r d = \frac{\log d}{\log r}, \qquad \boxed{D = \Omega(\log d)}, \tag{3.A.22}$$

because r is fixed. In the one-qubit and two-qubit model, this reads

$$D \geq \log_2 d. \tag{3.A.23}$$

3.B State preparation needs depth $t \geq \log_2 d$

PROBLEM

Let QC_B be a depth-t circuit of 1-qubit and 2-qubit gates with $QC_B\,|0\rangle^{\otimes n} = |\phi\rangle \in C$, a $[[n, k, d]]$ code with $k \geq 1$. Prove $t \geq \log_2 d$. You may use Q2(c) without proof: the trace leaving strictly fewer than d qubits is the same for all states in the code space.

3.B.i Transported input conditions

Let $U = QC_B$. Unlike part (A), where the circuit received a variable logical input, here the circuit starts from one fixed input and prepares one particular code state:

$$U\,|0^n\rangle = |\phi\rangle \in C. \tag{3.B.1}$$

The assumption $k \geq 1$ in the problem gives $\dim C = 2^k \geq 2$. We may therefore choose a normalized state in the same code space that is orthogonal to the prepared state:

$$|\psi\rangle \in C, \qquad \langle\psi|\phi\rangle = 0. \tag{3.B.2}$$

For each input qubit $i \in \{1, \ldots, n\}$, define

$$Z_i := I^{\otimes(i-1)} \otimes Z \otimes I^{\otimes(n-i)}. \tag{3.B.3}$$

Since $Z\,|0\rangle = |0\rangle$, every one of these operators fixes the all-zero input:

$$Z_i\,|0^n\rangle = |0^n\rangle \qquad \text{for every } i. \tag{3.B.4}$$

We transport all n input conditions to the output by defining them explicitly as

$$A_1 := UZ_1U^\dagger, \qquad A_2 := UZ_2U^\dagger, \qquad \ldots \qquad A_n := UZ_nU^\dagger. \tag{3.B.5}$$

Here A_i is the output-side version of the input condition $Z_i = +1$. Indeed, for every i,

$$A_i\,|\phi\rangle = UZ_iU^\dagger U\,|0^n\rangle = UZ_i\,|0^n\rangle = U\,|0^n\rangle = |\phi\rangle\,. \tag{3.B.6}$$

Thus

$$A_i\,|\phi\rangle = |\phi\rangle \qquad \text{for every } i. \tag{3.B.7}$$

When we say that A_i stabilizes $|\phi\rangle$, we mean only this displayed equation; no stabilizer-code assumption is being made.

3.B.ii The contradiction if the circuit is too shallow

Assume, for contradiction, that

$$2^t < d. \tag{3.B.8}$$

For each i, let $T_i := \mathrm{supp}(A_i)$. The support-growth lemma proved in part (A) applies directly to the initially one-qubit operator Z_i, and gives

$$|T_i| = |\,\mathrm{supp}(UZ_iU^\dagger)| \le 2^t < d. \tag{3.B.9}$$

Thus each transported condition A_i can be tested using fewer than d physical output qubits.

Both $|\phi\rangle$ and $|\psi\rangle$ are normalized states in C. Therefore Q2(c), which the problem explicitly permits us to use, gives equality of their reduced states on T_i:

$$\rho^{\phi}_{T_i} := \mathrm{Tr}_{\overline{T_i}} |\phi\rangle\langle\phi| = \mathrm{Tr}_{\overline{T_i}} |\psi\rangle\langle\psi| =: \rho^{\psi}_{T_i}. \tag{3.B.10}$$

Because A_i is supported on T_i, its expectation depends only on this reduced state. Hence

$$\langle\psi|\, A_i\, |\psi\rangle = \langle\phi|\, A_i\, |\phi\rangle = 1 \qquad \text{for every } i, \tag{3.B.11}$$

where the final equality follows from $A_i\,|\phi\rangle = |\phi\rangle$.

Each A_i is a Hermitian unitary because it is unitarily conjugate to Z_i; in particular, $A_i^2 = I$. Since $|\psi\rangle$ is normalized, the preceding expectation gives the compact identity

$$\big\|(I - A_i)\,|\psi\rangle\big\|^2 = \langle\psi|\,(I - A_i)^2\,|\psi\rangle = 2 - 2\,\langle\psi|\,A_i\,|\psi\rangle = 0. \tag{3.B.12}$$

Consequently,

$$A_i\,|\psi\rangle = |\psi\rangle \qquad \text{for every } i. \tag{3.B.13}$$

It remains to undo the transport. Substituting $A_i = UZ_iU^\dagger$ into the last equation and multiplying by $U^\dagger$ gives

$$Z_i\big(U^\dagger\,|\psi\rangle\big) = U^\dagger\,|\psi\rangle \qquad \text{for every } i. \tag{3.B.14}$$

Thus $U^\dagger\,|\psi\rangle$ is a simultaneous $+1$ eigenstate of $Z_1, \ldots, Z_n$. For a computational-basis vector,

$$Z_i\,|x_1 \cdots x_n\rangle = (-1)^{x_i}\,|x_1 \cdots x_n\rangle\,. \tag{3.B.15}$$

The n requirements are separate, so eigenvalue $+1$ for every Z_i forces $x_i = 0$ for every i. A superposition cannot evade this conclusion: every basis component containing a 1 violates at least one of the conditions and must have zero coefficient. Hence the common $+1$ eigenspace is precisely $\mathrm{span}\{|0^n\rangle\}$, and therefore

$$U^\dagger\,|\psi\rangle = e^{i\theta}\,|0^n\rangle \tag{3.B.16}$$

for some global phase $e^{i\theta}$. Applying U now gives

$$|\psi\rangle = e^{i\theta}U\,|0^n\rangle = e^{i\theta}\,|\phi\rangle\,, \tag{3.B.17}$$

contradicting $\langle\psi|\phi\rangle = 0$.

The assumption $2^t < d$ is therefore impossible. It follows that

$$2^t \ge d, \qquad \mathrm{depth}(QC_B) = t \ge \log_2 d. \tag{3.B.18}$$

Discussion: other pure product inputs. The choice of $|0^n\rangle$ is conventional rather than essential. For an arbitrary pure product input $|\alpha\rangle = |\alpha_1\rangle \otimes \cdots \otimes |\alpha_n\rangle$, let P_i act as $2\,|\alpha_i\rangle\langle\alpha_i| - I$ on qubit i and as the identity on all other qubits. The state $|\alpha\rangle$ is the unique common $+1$ eigenstate of $P_1, \ldots, P_n$. Replacing Z_i by P_i leaves the support-growth and Q2(c) arguments unchanged, so the same lower bound holds for every pure product input.

3.C GHZ and W need depth $\Omega(\log n)$

PROBLEM

With $|\mathrm{GHZ}_n\rangle = \frac{1}{\sqrt{2}}(|0\ldots0\rangle + |1\ldots1\rangle)$ and $|W_n\rangle = \frac{1}{\sqrt{n}}\sum_{k=1}^{n}|e_k\rangle$, let C take n (or more) qubits in $|0\rangle$ to $|\phi\rangle$ (or $|\phi\rangle \otimes |\mathrm{aux}\rangle$) for $|\phi\rangle$ either state, using arbitrary 1-, 2- and 3-qubit gates. Give a formal proof that $\mathrm{depth}(C) = \Omega(\log n)$. A light-cone or other hand-waving argument is insufficient. The best bound known to the lecturer is $\frac{1}{2}\log_3 n$, not $\log_3 n$.

3.C.i Circuit setup and causal reachability

Let the circuit use $m \geq n$ input qubits and have depth t. All its inputs are initially unentangled and in the state

$$|0^m\rangle . \tag{3.C.1}$$

Label the n designated target output qubits by $1, \ldots, n$, and denote the unitary implemented by the circuit by U. The preparation requirement in the problem is

$$U|0^m\rangle = |\phi\rangle_{1\cdots n} \otimes |\text{aux}\rangle , \qquad |\phi\rangle \in \{|\text{GHZ}_n\rangle , |W_n\rangle\}. \tag{3.C.2}$$

This is not a superposition of the GHZ and W states: the circuit is required to prepare one of those two kinds of target. The additional pure state $|\text{aux}\rangle$ merely allows arbitrary workspace among the remaining $m - n$ outputs. Because the final state is a tensor product across the target and auxiliary systems, tracing out the auxiliary outputs leaves the designated qubits exactly in the state $|\phi\rangle$.

The proof will compare two simple notions of causal reach. The **forward light cone** $F(x)$ of an input wire x is the set of output wires that could possibly be influenced by x through a chain of gates. To find it, mark x at the input, move through the circuit one layer at a time, and after each layer mark every wire in a gate that touches an already marked wire.

The **backward light cone** $B(i)$ of an output wire i is the set of input wires from which i can be reached through a chain of gates. It is obtained by the same marking procedure in the opposite direction: start by marking output i and move from the last circuit layer back toward the input.

These are not two different forms of influence. They are the same reachability relation read from opposite ends of the circuit:

$$j \in F(x) \quad \Longleftrightarrow \quad x \in B(j). \tag{3.C.3}$$

Indeed, either statement says that there is a chain of gates connecting input x to output j while respecting the order of the layers.

Every gate in the problem touches at most three wires, and gates within one layer act on disjoint wire sets. Consequently, in one layer each currently marked wire can select at most one gate containing at most three wires, so the marked set can grow by at most a factor of 3. Starting from one marked wire gives the progression

$$1 \longrightarrow 3 \longrightarrow 9 \longrightarrow \cdots . \tag{3.C.4}$$

A direct induction on the number of traversed layers therefore proves

$$|F(x)| \leq 3^t, \qquad |B(i)| \leq 3^t. \tag{3.C.5}$$

One- and two-qubit gates can only make these sets smaller, so allowing them in addition to three-qubit gates does not change the bound.

These size bounds describe how far causal influence can spread, but they do not yet explain why limited causal reach prevents the preparation of GHZ or W. The missing connection concerns pairs of output qubits. We will first show that every target pair is correlated, then prove that two outputs with disjoint backward cones must instead have a product reduced state because the input is a product state. Finally, the cone-size bounds above will force such a disjoint pair whenever the circuit is too shallow.

3.C.ii The correlation required by the target states

The property needed from GHZ and W is not pairwise entanglement. In particular, when $n \geq 3$, the two-qubit GHZ marginal is separable. What matters is the weaker statement

$$\boxed{\text{Every pair of distinct designated target qubits has a non-product reduced state.}} \tag{3.C.6}$$

Thus every pair retains a correlation, whether or not that correlation is itself quantum entanglement.

For the GHZ state and distinct target qubits i, j, tracing out the remaining target qubits gives, when $n \geq 3$,

$$\rho_{ij}^{\text{GHZ}} = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11| . \tag{3.C.7}$$

The off-diagonal terms disappear because the traced-out states $\left|0^{n-2}\right\rangle$ and $\left|1^{n-2}\right\rangle$ are orthogonal. Each individual qubit is maximally mixed,

$$\rho_i^{\text{GHZ}} = \rho_j^{\text{GHZ}} = \frac{I}{2}, \qquad \rho_i^{\text{GHZ}} \otimes \rho_j^{\text{GHZ}} = \frac{I_4}{4}, \tag{3.C.8}$$

which is not equal to ρ_{ij}^{GHZ}: the actual pair produces only 00 or 11 in the computational basis, whereas the product marginals would produce all four outcomes. (For $n = 2$ the pair is the full Bell state and is still non-product; for $n = 1$ the claimed bound $t \geq 0$ is trivial.)

For $|W_n\rangle$ with $n \geq 2$, each designated qubit contains the unique excitation with probability

$$\Pr(i = 1) = \Pr(j = 1) = \frac{1}{n}. \tag{3.C.9}$$

If the two-qubit reduced state were the product $\rho_i^W \otimes \rho_j^W$, computational-basis measurements on the two qubits would be independent and would satisfy

$$\Pr(i = 1, j = 1) = \Pr(i = 1)\Pr(j = 1) = \frac{1}{n^2} > 0. \tag{3.C.10}$$

But every basis string occurring in $|W_n\rangle$ contains exactly one 1, so two distinct qubits can never both be 1:

$$\Pr(i = 1, j = 1) = 0. \tag{3.C.11}$$

Hence $\rho_{ij}^W \neq \rho_i^W \otimes \rho_j^W$. This proves equation (3.C.6) for either permitted target.

3.C.iii Disjoint backward cones force a product pair

We now prove the formal implication that will conflict with the target-state calculation. Suppose two designated outputs i and j have disjoint backward cones:

$$B(i) \cap B(j) = \varnothing. \tag{3.C.12}$$

Let $P, Q \in \{I, X, Y, Z\}$ be arbitrary one-qubit Pauli operators, and let P_i and Q_j denote their extensions to all m output wires by identities. Define their input-side conjugates by

$$\widetilde{P} := U^\dagger P_i U, \qquad \widetilde{Q} := U^\dagger Q_j U. \tag{3.C.13}$$

This notation does not describe a physical evolution running backward in time. It is the algebraic identity used to calculate an output expectation with the known input state. For example, if $|\mathrm{out}\rangle = U|0^m\rangle$, then

$$\langle \mathrm{out}| P_i |\mathrm{out}\rangle = \langle 0^m| U^\dagger P_i U |0^m\rangle = \langle 0^m| \widetilde{P} |0^m\rangle . \tag{3.C.14}$$

Because P_i initially acts only on output i, pulling it across the layers can reach only the wires marked by the backward-cone procedure. A gate that does not touch the current marked set commutes past the current operator and cancels with its adjoint; a gate that does touch it can enlarge the support only to the other wires of that same gate. Therefore $\widetilde{P}$ is supported on $B(i)$. The identical argument shows that $\widetilde{Q}$ is supported on $B(j)$. Here an operator is supported on a set when it acts as the identity on every wire outside that set.

The joint output expectation involves the product of these two input-side operators. Inserting $UU^\dagger = I$ makes this explicit:

$$U^\dagger P_i Q_j U = U^\dagger P_i (UU^\dagger) Q_j U = (U^\dagger P_i U)(U^\dagger Q_j U) = \widetilde{P}\widetilde{Q}. \tag{3.C.15}$$

Let $R = B(i)$ and $S = B(j)$. Since $R \cap S = \varnothing$, group the input Hilbert space as

$$\mathcal{H}_R \otimes \mathcal{H}_S \otimes \mathcal{H}_{\mathrm{rest}}. \tag{3.C.16}$$

The initialized input factors across this grouping:

$$|0^m\rangle = |0_R\rangle \otimes |0_S\rangle \otimes |0_{\mathrm{rest}}\rangle . \tag{3.C.17}$$

Because $\widetilde{P}$ is supported on R and $\widetilde{Q}$ on S, operators P_R and Q_S satisfy

$$\widetilde{P} = P_R \otimes I_S \otimes I_{\mathrm{rest}}, \qquad \widetilde{Q} = I_R \otimes Q_S \otimes I_{\mathrm{rest}}, \tag{3.C.18}$$

so the tensor-product multiplication rule gives

$$\widetilde{P}\widetilde{Q} = P_R \otimes Q_S \otimes I_{\mathrm{rest}}. \tag{3.C.19}$$

Taking the expectation in the product input gives

$$\langle 0^m| \widetilde{P}\widetilde{Q} |0^m\rangle = \langle 0_R| P_R |0_R\rangle \langle 0_S| Q_S |0_S\rangle \langle 0_{\mathrm{rest}}| I |0_{\mathrm{rest}}\rangle = \langle 0^m| \widetilde{P} |0^m\rangle \langle 0^m| \widetilde{Q} |0^m\rangle . \tag{3.C.20}$$

This is precisely where the initial product-state hypothesis is used. Disjoint operator supports alone would not imply factorization in an entangled input state, which could already contain correlations between R and S.

Let $|\text{out}\rangle = U|0^m\rangle$ and define the reduced state of outputs i, j formally by

$$\rho_{ij} := \text{Tr}_{\overline{\{i,j\}}}\big(|\text{out}\rangle\langle\text{out}|\big). \tag{3.C.21}$$

Let ρ_i and ρ_j be its one-qubit marginals. Translating equations (3.C.15) and (3.C.20) back to output expectations gives, for all one-qubit Paulis P, Q,

$$\text{Tr}\big(\rho_{ij}(P \otimes Q)\big) = \text{Tr}(\rho_i P)\,\text{Tr}(\rho_j Q). \tag{3.C.22}$$

It remains to justify why these expectation identities determine the entire two-qubit state. As in the Pauli reconstruction used in Q2(b), the sixteen matrices $P \otimes Q$, with $P, Q \in \{I, X, Y, Z\}$, form an orthogonal basis of the 4×4 matrices. Consequently,

$$\rho_{ij} = \frac{1}{4} \sum_{P,Q \in \{I,X,Y,Z\}} \text{Tr}\big(\rho_{ij}(P \otimes Q)\big)(P \otimes Q), \tag{3.C.23}$$

The sixteen traces $\text{Tr}(\rho_{ij}(P \otimes Q))$ determine all Pauli-basis coefficients of ρ_{ij}; the actual coefficient of $P \otimes Q$ is one quarter of that trace. For the product state, the identities $(A \otimes B)(C \otimes D) = AC \otimes BD$ and $\text{Tr}(M \otimes N) = \text{Tr}(M)\,\text{Tr}(N)$ give

$$\text{Tr}\big[(\rho_i \otimes \rho_j)(P \otimes Q)\big] = \text{Tr}\big[(\rho_i P) \otimes (\rho_j Q)\big] = \text{Tr}(\rho_i P)\,\text{Tr}(\rho_j Q). \tag{3.C.24}$$

By equation (3.C.22), ρ_{ij} and $\rho_i \otimes \rho_j$ have all sixteen Pauli-basis expectation values, and therefore all Pauli coefficients, in common. Since the matrices $P \otimes Q$ form a basis, the two states are equal:

$$\boxed{B(i) \cap B(j) = \varnothing \quad \Longrightarrow \quad \rho_{ij} = \rho_i \otimes \rho_j.} \tag{3.C.25}$$

3.C.iv Counting which outputs can share an input ancestor

Only now do we use both cone-size bounds. Fix one designated target output i. It has at most 3^t possible input ancestors because $|B(i)| \le 3^t$. For each input $x \in B(i)$, its forward cone contains at most 3^t outputs. Define

$$S_i := \bigcup_{x \in B(i)} F(x). \tag{3.C.26}$$

The set S_i contains every output that could share at least one input ancestor with i. Indeed, if output j shares an ancestor x with i, then $x \in B(i)$ and $x \in B(j)$. Cone reciprocity (3.C.3) turns the latter statement into $j \in F(x)$, so $j \in S_i$.

Using the union bound and equation (3.C.5),

$$|S_i| \le \sum_{x \in B(i)} |F(x)| \le |B(i)|3^t \le 3^t 3^t = 3^{2t}. \tag{3.C.27}$$

If $n > 3^{2t}$, then among the n designated target outputs there is some $j \notin S_i$. By the definition of S_i, no input ancestor of i can also influence j. Hence

$$B(i) \cap B(j) = \varnothing. \tag{3.C.28}$$

The disjoint-cone lemma (3.C.25) would then give $\rho_{ij} = \rho_i \otimes \rho_j$. But i and j are distinct designated target qubits, and equation (3.C.6) established that every such pair has a non-product reduced state for either allowed target. This contradiction proves

$$n \le 3^{2t}. \tag{3.C.29}$$

3.C.v Depth lower bound

Taking base-3 logarithms gives the quantitative lower bound requested in the problem:

$$\boxed{t \ge \frac{1}{2}\log_3 n,} \tag{3.C.30}$$

Hence $\text{depth}(C) = \Omega(\log n)$. A shallow circuit would force some target pair to have disjoint causal pasts and therefore a product reduced state, but every pair in GHZ or W is correlated.

3.D Quantum depth with classical feed-forward

PROBLEM

Use parity measurements, classical computation, and feed-forward to prepare $|\mathrm{GHZ}_n\rangle$ for any $n \geq 3$, with quantum depth 2, classical depth $O(\log n)$, quantum size $O(n)$ and total size $O(n)$.

The public edition develops the construction through the circuit figures below.

Part (C) proved a logarithmic-depth lower bound for purely unitary bounded-arity circuits. Here intermediate measurements produce classical bits that control later one-qubit corrections. I will show that this feed-forward model prepares

$$|\mathrm{GHZ}_n\rangle = \frac{|0^n\rangle + |1^n\rangle}{\sqrt{2}} \tag{3.D.1}$$

with the resource bounds stated in the problem.

3.D.i Adjacent parities leave one global choice coherent

Start the n data qubits in the uniform superposition

$$|+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle\,. \tag{3.D.2}$$

For every neighboring pair, we will measure only the relative bit

$$p_i = x_i \oplus x_{i+1}, \qquad i = 1, \ldots, n-1. \tag{3.D.3}$$

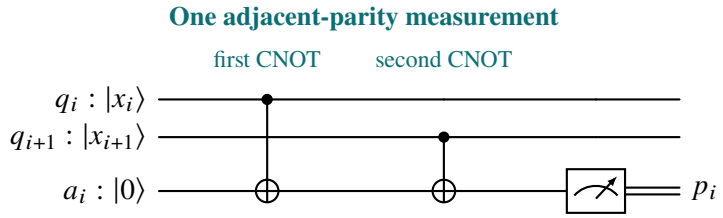


Figure 5. Nondestructive measurement of the adjacent parity $x_i \oplus x_{i+1}$. The individual data bits are not measured.

The gadget in Figure 5 computes this parity without measuring either data qubit. On a computational-basis input, its CNOTs act as

$$|x_i, x_{i+1}, 0\rangle \longmapsto |x_i, x_{i+1}, x_i\rangle \longmapsto |x_i, x_{i+1}, x_i \oplus x_{i+1}\rangle\,. \tag{3.D.4}$$

Measuring the ancilla returns only whether the data bits agree or disagree: $p_i = 0$ does not distinguish 00 from 11, and $p_i = 1$ does not distinguish 01 from 10. The measurement therefore learns a relation between two bits, not either bit separately.

Let D denote all data qubits and A the $n-1$ ancillas, each initially zero. Applying all parity gadgets to the uniform input gives the following joint state before any ancilla is measured:

$$\frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle_{\mathrm{D}}\, |x_1 \oplus x_2, \ldots, x_{n-1} \oplus x_n\rangle_{\mathrm{A}}\,. \tag{3.D.5}$$

Obtaining the record $p = (p_1, \ldots, p_{n-1})$ projects this sum onto exactly the terms satisfying $x_i \oplus x_{i+1} = p_i$ for every i. The compatible strings below are precisely the data labels of those surviving terms.

One parity outcome is only one constraint and still leaves 2^{n-1} compatible strings. Each successive adjacent parity introduces the next bit and halves the number again. Only after all $n-1$ parities have been measured are two strings left:

$$2^n \longrightarrow 2^{n-1} \longrightarrow 2^{n-2} \longrightarrow \cdots \longrightarrow 2. \tag{3.D.6}$$

For a fixed parity record, we now justify why exactly two basis strings are compatible with all $n-1$ equations. Since $p_i = x_i \oplus x_{i+1}$, XORing both sides with x_i gives

$$x_{i+1} = x_i \oplus p_i. \tag{3.D.7}$$

Thus either value of x_1 determines x_2, then x_3, and eventually the entire string uniquely. The choices $x_1 = 0$ and $x_1 = 1$ therefore give the two compatible strings, denoted x and $\bar{x}$. They are complements because complementing both bits leaves every adjacent parity unchanged:

$$(1 \oplus x_i) \oplus (1 \oplus x_{i+1}) = x_i \oplus x_{i+1} = p_i. \tag{3.D.8}$$

Thus the complete parity record still does not reveal the one remaining global choice between x and $\bar{x}$. Because the initial amplitudes are equal and positive, the CNOTs only permute basis states, and the measurements project onto the two compatible strings. Conditioning on $p = (p_1, \ldots, p_{n-1})$ therefore leaves

$$\boxed{|\psi_p\rangle = \frac{|x\rangle + |\bar{x}\rangle}{\sqrt{2}}.} \tag{3.D.9}$$

No outcome-dependent relative phase is introduced.

For example, when $n = 4$ and $(p_1, p_2, p_3) = (1, 0, 1)$, the recursion gives

$$x_1 = 0 \quad \Longrightarrow \quad x = 0110, \qquad x_1 = 1 \quad \Longrightarrow \quad \bar{x} = 1001. \tag{3.D.10}$$

The conditional state is therefore $(|0110\rangle + |1001\rangle)/\sqrt{2}$.

At this point the data are in an equal superposition of two complementary strings, but those strings need not yet be 0^n and 1^n. The remaining task is to determine from the measured parities a pattern of local X corrections that maps one branch to 0^n and the other to 1^n without disturbing their coherent superposition.

3.D.ii Classical prefix XORs determine the corrections

To straighten each branch into a constant string, qubit j should be flipped exactly when x_j disagrees with the first bit x_1. The XOR of the measured adjacent parities along the path from qubit 1 to qubit j gives precisely this correction bit. Define

$$c_1 = 0, \qquad c_j = p_1 \oplus \cdots \oplus p_{j-1} \quad (2 \le j \le n). \tag{3.D.11}$$

To see why, substitute $p_i = x_i \oplus x_{i+1}$:

$$c_j = (x_1 \oplus x_2) \oplus (x_2 \oplus x_3) \oplus \cdots \oplus (x_{j-1} \oplus x_j) = x_1 \oplus x_j. \tag{3.D.12}$$

Every intermediate bit $x_2, \ldots, x_{j-1}$ appears twice and cancels because $y \oplus y = 0$. Only the two endpoint bits remain. Consequently, $c_j = 0$ precisely when $x_j = x_1$, and $c_j = 1$ precisely when they differ.

The earlier $n = 4$ example makes this meaning concrete. From $p = (1, 0, 1)$ we obtain

$$c = (c_1, c_2, c_3, c_4) = (0, p_1, p_1 \oplus p_2, p_1 \oplus p_2 \oplus p_3) = (0, 1, 1, 0). \tag{3.D.13}$$

This correction string is 0110, so it sends the two compatible strings to opposite constant strings:

$$0110 \oplus 0110 = 0000, \qquad 1001 \oplus 0110 = 1111. \tag{3.D.14}$$

In general, use the classical bit c_j to apply X^{c_j} to data qubit j; here $X^0 = I$ and $X^1 = X$. On a computational-basis bit this applies the map $x_j \mapsto x_j \oplus c_j$. For the branch x,

$$x_j \longmapsto x_j \oplus c_j = x_j \oplus x_1 \oplus x_j = x_1. \tag{3.D.15}$$

Thus the branch x becomes the constant string x_1^n. Crucially, the complementary branch has the same parity record by equation (3.D.8), so the controller applies the same correction bits to it. Since $\bar{x}_j = 1 \oplus x_j$,

$$\bar{x}_j \oplus c_j = (1 \oplus x_j) \oplus (x_1 \oplus x_j) = 1 \oplus x_1. \tag{3.D.16}$$

The complementary branch therefore becomes the opposite constant string. If $x_1 = 0$, the two branches become 0^n and 1^n; if $x_1 = 1$, their order is reversed. In either case,

$$\frac{|x\rangle + |\bar{x}\rangle}{\sqrt{2}} \longmapsto \frac{|0^n\rangle + |1^n\rangle}{\sqrt{2}} = |\mathrm{GHZ}_n\rangle\,, \tag{3.D.17}$$

This proves correctness for every measurement outcome. Conceptually, the parity measurements retain one coherent global bit, and the prefix XORs tell each physical qubit whether it must be flipped to agree with that bit.

3.D.iii Quantum and classical resource bounds

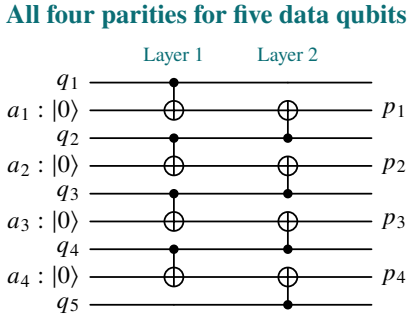


Figure 6. All $n-1$ adjacent parities are extracted using only two parallel layers of two-qubit gates. The displayed $n = 5$ instance makes the scheduling explicit: no wire participates in two gates within either layer.

Figure 6 gives the general two-layer schedule. First apply

$$q_i \longrightarrow a_i, \qquad i = 1, \ldots, n-1. \tag{3.D.18}$$

and then apply

$$q_{i+1} \longrightarrow a_i, \qquad i = 1, \ldots, n-1. \tag{3.D.19}$$

Within each layer the CNOTs are pairwise disjoint: each uses a different data qubit and a different ancilla. Afterwards every ancilla contains

$$a_i = x_i \oplus x_{i+1} = p_i. \tag{3.D.20}$$

Thus parity extraction has exactly two layers of two-qubit gates:

$$\boxed{\text{quantum depth} = 2.} \tag{3.D.21}$$

Here quantum depth counts these two-qubit layers; parallel $|+\rangle$ preparation, measurement, and classically controlled one-qubit Pauli corrections do not add quantum-depth layers under the exercise's convention.

The classical circuit must compute all prefixes in equation (3.D.11). Computing them from left to right via $c_{j+1} = c_j \oplus p_j$ would take $n-1$ sequential steps, so it would have linear classical depth. A parallel-prefix scan computes the same values with a balanced tree.

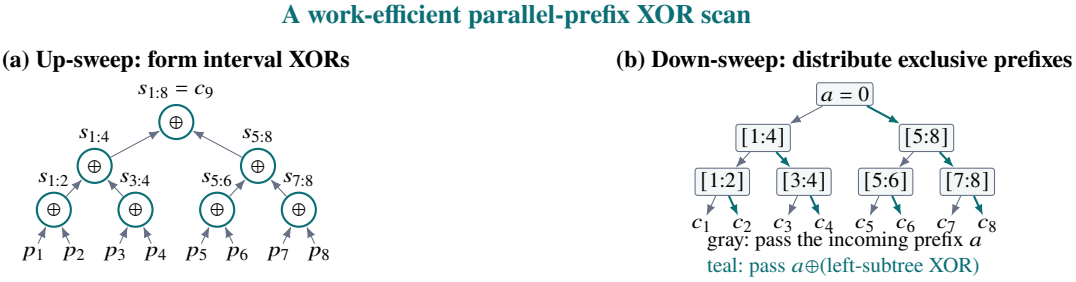


Figure 7. A work-efficient scan for eight parity bits. The up-sweep forms interval XORs in three parallel levels. The down-sweep starts with prefix 0, passes it unchanged to each left child, and XORs it with the stored left subtotal before sending it to the right child. Its leaves give $c_1, \ldots, c_8$, while the up-sweep total gives $c_9 = p_1 \oplus \cdots \oplus p_8$. For general n, the two sweeps use $O(\log n)$ depth and $O(n)$ XOR gates.

Figure 7 illustrates the scan for eight parity bits. It has two parts.

In the *up-sweep*, neighboring parities are first combined in parallel, then neighboring two-bit totals are combined, and so on. The first level forms $p_1 \oplus p_2$, $p_3 \oplus p_4$, and the other two-bit interval XORs; the next level forms the four-bit interval XORs; the root finally holds $p_1 \oplus \cdots \oplus p_8$. The length of an interval doubles at every level, so only $O(\log n)$ levels are required.

The *down-sweep* turns those stored interval XORs into the exclusive prefixes that we need. The symbol a in the figure is a temporary classical prefix value, not one of the measured parity ancillas a_i. At any tree node, suppose the interval is divided into a left half L and a right half R, and a is the XOR of all parity bits lying before this whole interval. Then

$$\text{prefix sent to } L = a, \qquad \text{prefix sent to } R = a \oplus s_L, \tag{3.D.22}$$

where s_L is the left-half XOR already stored by the up-sweep. The right half receives the extra s_L because every bit in the left half precedes it.

For example, with four parity leaves the up-sweep has stored $s_{1:2} = p_1 \oplus p_2$ and $s_{3:4} = p_3 \oplus p_4$. The root starts with prefix 0. It sends 0 to the left pair and $s_{1:2} = p_1 \oplus p_2$ to the right pair. Splitting those two pairs once more gives the four leaf values

$$c_1 = 0, \qquad c_2 = p_1, \qquad c_3 = p_1 \oplus p_2, \qquad c_4 = p_1 \oplus p_2 \oplus p_3. \tag{3.D.23}$$

The full up-sweep total is c_5. The same reasoning at every scale shows that the value arriving at leaf p_j is $p_1 \oplus \cdots \oplus p_{j-1} = c_j$. For eight parities the leaves in Figure 7 therefore give $c_1, \ldots, c_8$, while the root total gives c_9; with $n-1$ parities we obtain all $c_1, \ldots, c_n$.

If $n-1$ is not a power of two, append zero leaves up to the next power of two. This does not change any prefix and uses fewer than twice as many leaves. Each sweep has $O(\log n)$ levels, and a binary tree with $O(n)$ leaves has $O(n)$ nodes. The scan therefore has $O(\log n)$ depth and $O(n)$ XOR gates:

$$\boxed{\text{classical depth} = O(\log n), \qquad \text{classical size} = O(n).} \tag{3.D.24}$$

The quantum part uses $n-1$ parity ancillas, $2(n-1)$ CNOTs, and $O(n)$ local preparations, measurements, and corrections. Together with the $O(n)$-gate prefix circuit, this gives

$$\boxed{\text{quantum size} = O(n), \qquad \text{total size} = O(n).} \tag{3.D.25}$$

Why feed-forward evades the unitary-depth bound. The parity measurements reveal the relative bits but preserve the coherent choice between x and $\bar{x}$. The $O(\log n)$-depth scan computes $c_j = x_1 \oplus x_j$, and local X corrections turn those branches into 0^n and 1^n. Classical feed-forward thus supplies the nonlocal coordination that a shallow unitary circuit cannot.

3.E States requiring exponential total size

> **PROBLEM**
>
> Let S be a finite universal set and $0 < \varepsilon < 1/10$. Show that with gates from S, 1-qubit measurements, and classical Boolean circuits over all ≤ 2-input gates, there are n-qubit states $|\phi\rangle$ requiring a feed-forward circuit of exponential total size to produce a $|\psi\rangle$ with $\||\phi\rangle - |\psi\rangle\| \leq \varepsilon$. Hint: there are at most $2^{O(s \log sm)}$ Boolean circuits of size s on m inputs.

I now use a counting argument with two sides. First I construct an enormous family of mutually separated target states. I then count the possible outputs of all circuits below a given total-size limit. Because one output cannot be ε-close to two sufficiently separated targets, fewer circuit outputs than targets will leave at least one target that none of those circuits can ε-approximate.

3.E.i Many separated target states

Set $N := 2^n$, and let

$$z = (z_1, \ldots, z_N) \in \{0,1\}^N \tag{3.E.1}$$

be an N-bit string. With $|e_1\rangle, \ldots, |e_N\rangle$ denoting the computational basis of n qubits, define the **sign state**

$$\boxed{|\phi_z\rangle := \frac{1}{\sqrt{N}} \sum_{j=1}^{N} (-1)^{z_j} |e_j\rangle .} \tag{3.E.2}$$

There is one independent sign bit for each of the $N = 2^n$ equal-magnitude amplitudes. Thus, before removing duplicates or nearby choices, there are $2^N = 2^{2^n}$ sign patterns. Geometrically, these states are normalized corners of $\{\pm 1\}^N$. For instance, $N = 4$ and $z = (0,1,0,1)$ give $|\phi_z\rangle = \frac{1}{2}(|e_1\rangle - |e_2\rangle + |e_3\rangle - |e_4\rangle)$. The string z determines the sign of every amplitude. Its bitwise complement replaces each z_j by $\bar{z}_j = 1 - z_j$, and therefore $(-1)^{\bar{z}_j} = (-1)^{1-z_j} = -(-1)^{z_j}$ for every j. Thus every amplitude changes sign at once, giving $|\phi_{\bar{z}}\rangle = -|\phi_z\rangle$. These two kets differ only by the global phase -1, so they describe the same physical state. We should therefore not count both as different targets. To select one representative, use z_1: this is the sign-label bit controlling the

amplitude of $|e_1\rangle$, not the value of the first physical qubit. Since $\bar{z}_1 = 1 - z_1$, exactly one member of each pair $\{z, \bar{z}\}$ has first sign-label bit zero. Restricting to $z_1 = 0$ therefore leaves 2^{N-1} nonduplicated candidates.

For two sign strings at Hamming distance $h := d_H(z, z')$, the coefficient of $|e_j\rangle$ in the difference vector is

$$\frac{(-1)^{z_j} - (-1)^{z'_j}}{\sqrt{N}}.$$

If $z_j = z'_j$, the two signs are equal and this coefficient is zero. If $z_j \neq z'_j$, the signs are opposite, so the coefficient is either $2/\sqrt{N}$ or $-2/\sqrt{N}$ and therefore has magnitude $2/\sqrt{N}$. By definition, $h = d_H(z, z')$ is exactly the number of coordinates at which the strings differ. Hence exactly h coordinates make this nonzero contribution, while the remaining $N - h$ coordinates contribute zero. Therefore

$$\| |\phi_z\rangle - |\phi_{z'}\rangle \|^2 = h \left(\frac{2}{\sqrt{N}} \right)^2 = \frac{4h}{N}. \tag{3.E.3}$$

Hence

$$\boxed{\| |\phi_z\rangle - |\phi_{z'}\rangle \| = 2\sqrt{\frac{h}{N}}.} \tag{3.E.4}$$

In particular, if $d_H(z, z') \geq N/16$, then

$$\| |\phi_z\rangle - |\phi_{z'}\rangle \| \geq 2\sqrt{\frac{1}{16}} = \frac{1}{2}. \tag{3.E.5}$$

Since $\varepsilon < 1/10$, one state $|\psi\rangle$ cannot be ε-close to two targets separated by $1/2$, because the triangle inequality would give

$$\| |\phi_z\rangle - |\phi_{z'}\rangle \| \leq \| |\phi_z\rangle - |\psi\rangle \| + \| |\psi\rangle - |\phi_{z'}\rangle \| \leq 2\varepsilon, \tag{3.E.6}$$

contradicting equation (3.E.5).

This is the counting lever: one particular branch-output state—the state associated with one circuit and one measurement transcript—can ε-approximate at most one member of a pairwise $1/2$-separated family. Later we will count all such output states across all small circuits and all their transcripts. We therefore need to show that our 2^{N-1} candidate sign states contain a very large pairwise-separated family.

By equation (3.E.5), it is enough to find many sign labels whose pairwise Hamming distances are at least $N/16$. Let $C_N := \{z \in \{0,1\}^N : z_1 = 0\}$ be the pool of 2^{N-1} candidates.

We construct the desired family $\mathcal{Z}_N$ directly. Initially every label in C_N is unprocessed. In each round, choose one unprocessed label z, admit it to $\mathcal{Z}_N$, and discard every other still-unprocessed label w satisfying $d_H(z, w) < N/16$. Repeat until no unprocessed labels remain. Thus every original candidate is eventually processed exactly once: it is either admitted to $\mathcal{Z}_N$ or discarded for being too close to an admitted label. Only an admitted label starts a round; a discarded label can never be selected later.

The admitted labels are pairwise separated by construction. If z is admitted before z', then every still-unprocessed label too close to z is discarded in that round. Since z' remains available for later admission, it must satisfy $d_H(z, z') \geq N/16$.

It remains to show that many labels are admitted. To bound how many candidates one admission can process, define, for $z \in C_N$,

$$\mathrm{Close}(z) := \left\{ w \in C_N : d_H(z, w) < \frac{N}{16} \right\}. \tag{3.E.7}$$

The candidates processed in the round started by z—namely z itself and the still-unprocessed candidates discarded in that round—form a subset of $\mathrm{Close}(z)$. Exactly $\binom{N}{r}$ binary strings differ from z in exactly r coordinates, because such a string is determined by choosing which r coordinates of z to flip. Counting in the entire cube and possibly including the boundary only enlarges the collection. The standard estimate $\binom{N}{r} \leq (eN/r)^r$ therefore gives, for all sufficiently large N,

$$|\mathrm{Close}(z)| \leq \sum_{r=0}^{\lfloor N/16 \rfloor} \binom{N}{r} \leq (N+1)(16e)^{N/16} < 2^{0.05N} 2^{0.3402N} < 2^{0.4N}. \tag{3.E.8}$$

The intermediate bounds can be seen directly. For $1 \leq r \leq N/16$, put $f(r) = (eN/r)^r$. This function increases

throughout the range because

$$\frac{d}{dr}\log f(r) = \log(N/r) > 0. \tag{3.E.9}$$

Consequently every positive-r term is at most $f(N/16) = (16e)^{N/16}$. The $r = 0$ term is 1, and there are at most $N + 1$ terms in total. To compare the exponential factor with the candidate pool, write

$$(16e)^{N/16} = 2^{\alpha N}, \qquad \alpha = \frac{\log_2(16e)}{16} \approx 0.34017 < 0.3402. \tag{3.E.10}$$

Also $\log_2(N+1)/N \to 0$, so $N + 1 < 2^{0.05N}$ for sufficiently large N; this explains the final two inequalities. The decisive point is that $0.4 < 1$: the number processed by one admission has a smaller exponential rate than the original 2^{N-1} candidates.

Hence one admission processes fewer than $2^{0.4N}$ candidates.

Let $M := |\mathcal{Z}_N|$. Since exactly one label is admitted in each round, M is also the number of rounds. Let a_j be the number of candidates processed in round j, including the label admitted in that round. The processed groups are disjoint, and together they contain all 2^{N-1} original candidates. Thus

$$2^{N-1} = \sum_{j=1}^{M} a_j < M\, 2^{0.4N}, \qquad \text{so} \qquad M = |\mathcal{Z}_N| > 2^{0.6N-1} = 2^{\Omega(N)}. \tag{3.E.11}$$

The corresponding sign states therefore form a family $\mathcal{F}_n$ with

$$\boxed{|\mathcal{F}_n| = 2^{\Omega(N)} = 2^{\Omega(2^n)},} \tag{3.E.12}$$

and its members are pairwise separated by at least $1/2 > 2\varepsilon$. This completes the "many targets" side of the argument. We now count how many different outputs can be produced by all adaptive circuits below a general size limit. If that number is smaller than $|\mathcal{F}_n|$, equation (3.E.6) ensures that at least one member of $\mathcal{F}_n$ is not ε-approximated by any such output.

3.E.ii Counting small adaptive circuits

Let s be an arbitrary positive integer, and consider every circuit whose total size is at most s. Here S remains the fixed finite gate set from the problem; the lowercase s is only the circuit-size limit used in the counting argument.

We use the usual state-preparation convention: quantum inputs and work qubits start in fixed computational-basis states, and measurements are in the fixed computational basis; other bases require gates from S. Thus no continuously parameterized state or measurement is free.

We first count the possible quantum parts of these circuits. This is useful because every circuit has a finite description, so the number of possible descriptions is an upper bound on the number of circuits. If several descriptions happen to specify the same circuit, we have only overcounted.

By the *quantum part* we mean the ordered list of quantum gates and measurements, the wires on which they act, and any references to earlier classical control values. The Boolean circuitry that computes those values will be counted separately. Since S is fixed and finite, its maximum gate arity $r := \max_{G\in S} \text{arity}(G)$ is a constant. At most s operations can touch at most $rs = O(s)$ distinct ancillary wires; an untouched ancilla can be omitted. Together with the n designated output wires, there are therefore at most $n + O(s)$ relevant quantum wires.

The operation type requires only $O(1)$ bits, while naming its constantly many wires and, when relevant, an earlier classical value requires $O(\log(n + s))$ bits. Thus $O(s)$ quantum instructions have a description of length $L = O(s \log(n + s))$. An L-bit description has at most 2^L possible values, so the number of possible quantum parts is at most

$$2^{O(s\log(n+s))}. \tag{3.E.13}$$

We next count the classical controllers. Suppose the circuit makes m measurements, so $m \le s$. A Boolean gate has constantly many possible types, and each of its at most two inputs can be chosen from the m measurement bits or the outputs of earlier gates. Hence each gate has at most $C(m + s)^2$ descriptions for a constant C. For at most s Boolean gates, this gives the bound suggested in the hint:

$$\#\{\text{Boolean controllers}\} \le \left(C(m+s)^2\right)^s = 2^{O(s\log(m+s))} \le 2^{O(s\log(sm))} \le 2^{O(s\log s)}. \tag{3.E.14}$$

This controller already specifies the response to every measurement history; its gates count toward total size, and no separate decision tree is needed.

A complete hybrid circuit is specified by choosing both a quantum part and a classical controller. Therefore, using $\log s \le \log(n+s)$, the number of possible pairs is at most

$$\#\{\text{hybrid circuit descriptions}\} \le 2^{O(s\log(n+s))}\, 2^{O(s\log s)} = \boxed{2^{O(s\log(n+s))}}. \tag{3.E.15}$$

One adaptive circuit may have different conditional output states for different measurement transcripts, so counting circuits alone is not yet enough. At most s binary measurements give at most 2^s transcripts. For a safe upper bound, we pretend that every transcript of every circuit occurs and produces a different useful output state. This deliberately overcounts: some transcripts may have probability zero, and different transcripts or different circuits may produce the same state. Consequently,

$$\#\{\text{possible branch-output states}\} \le 2^{O(s\log(n+s))}\, 2^{s} = 2^{O(s\log(n+s))+s} = \boxed{2^{O(s\log(n+s))}}, \tag{3.E.16}$$

because $s = O(s\log(n+s))$. Requiring every branch to prepare the same pure state would only reduce the number of possible outputs.

This completes the "few outputs" side for a general size limit s. It remains to choose an exponential limit for which this output count is still smaller than $|\mathcal{F}_n|$.

3.E.iii The counting comparison

The problem asks us to show that some state requires exponential total size. It is therefore enough to find some exponential size threshold in n such that not every target can be ε-approximated by circuits below that threshold. We choose

$$\boxed{s_n := \lfloor 2^{n/2} \rfloor}. \tag{3.E.17}$$

s_n is the total-size cutoff: we now consider every allowed preparation circuit of total size at most s_n. If some target is not ε-close to any of their branch outputs, then preparing that target within error ε requires total size greater than $s_n = 2^{\Omega(n)}$, exactly the lower bound sought. We now verify that the two counts leave such a target. Writing $s = s_n = \Theta(2^{n/2})$, we have

$$\log(n+s) = O(n), \qquad s\log(n+s) = O(n2^{n/2}). \tag{3.E.18}$$

Equation (3.E.16) therefore bounds all candidate outputs of size-$\le s_n$ adaptive circuits by

$$\boxed{2^{O(n2^{n/2})}}. \tag{3.E.19}$$

We are therefore comparing $2^{\Omega(2^n)}$ separated targets with only $2^{O(n2^{n/2})}$ possible outputs. Since both quantities are powers of 2, the key comparison is between their exponents, 2^n and $n2^{n/2}$. Factoring out their common $2^{n/2}$ leaves $2^{n/2}$ against merely n:

$$2^n = 2^{n/2}2^{n/2}, \qquad \frac{n2^{n/2}}{2^n} = \frac{n}{2^{n/2}} \longrightarrow 0. \tag{3.E.20}$$

Formally, equation (3.E.12) supplies constants $a > 0$ and n_0 such that $|\mathcal{F}_n| \ge 2^{a2^n}$ for $n \ge n_0$, while equation (3.E.19) supplies a constant $C > 0$ such that the number of candidate outputs is at most $2^{Cn2^{n/2}}$. Equation (3.E.20) gives $Cn2^{n/2} < a2^n$ for all sufficiently large n, and hence

$$\boxed{2^{Cn2^{n/2}} < 2^{a2^n} \le |\mathcal{F}_n|.} \tag{3.E.21}$$

By equation (3.E.6), one output can ε-approximate at most one member of $\mathcal{F}_n$. Since there are fewer size-$\le s_n$ branch outputs than targets, some $|\phi\rangle \in \mathcal{F}_n$ is farther than ε from every such output. Preparing an ε-approximation to this state therefore requires size greater than $s_n = \Theta(2^{n/2}) = 2^{\Omega(n)}$. The argument proves existence rather than identifying the hard state explicitly.

QUESTION 4

Discrete Log via Phase Estimation

PROBLEM

Let p be an n-bit prime, g a generator of $\mathbb{Z}_p^*$, and $y \in \mathbb{Z}_p^*$; find x with $y = g^x \pmod p$. Using **phase estimation**, give an efficient quantum algorithm, reducing the number of qubits as far as possible while keeping polynomial complexity. State the qubit count, the number of quantum gates, and the classical complexity, as functions of n. Hint: use phase estimation for U_g and U_y (notation of Lecture 4).

Since p is encoded using n bits, $p < 2^n$; the required complexity is polynomial in n, the bit length of p.

The discrete-logarithm problem is important in cryptography: an efficient algorithm for computing discrete logarithms would compromise several classical public-key cryptographic schemes [2, §5.4.2, pp. 238–240]. This motivates the question of what quantum computation makes possible. Here I use phase estimation to recover the unknown exponent x, with quantum and classical computational costs polynomial in n, following the quantum discrete-logarithm approach of Shor [11, §6].

4.I Spectral construction and the ideal algorithm

The aim is to turn the hidden exponent x into a phase that phase estimation can read. View the nonzero residues modulo p in the cyclic order

$$1, g, g^2, \ldots, g^{p-2}. \tag{4.1}$$

Multiplication by g moves one position around this cycle. Since the given element is $y = g^x \bmod p$, multiplication by y moves x positions.

The idea is to use this relationship between the two multiplication operations to learn the unknown shift x. I first define their unitary operators and derive their common eigenstates, identifying the phases that phase estimation can reveal. I then show how to obtain useful measurement results starting from an easily prepared state, and how classical arithmetic turns those results into equations determining x.

The next part accounts for finite measurement precision and verifies the recovered answer. Finally, I justify the efficient implementation of the required operations, count the quantum and classical resources, and explain how reusing a single phase-estimation qubit reduces quantum memory. The detailed reversible-arithmetic construction is provided in Appendix C.2.

4.I.i Modular multiplication as a cyclic shift

We now formalize this picture. Let p be the given n-bit prime and let $\mathbb{Z}_p^* = \{1, 2, \ldots, p-1\}$ denote the multiplicative group of nonzero residues modulo p. Its order is $N = p - 1$. The given $g \in \mathbb{Z}_p^*$ is a generator, so every $a \in \mathbb{Z}_p^*$ has a unique representation $a = g^j \bmod p$ with $j \in \{0, \ldots, N-1\}$. The other input is $y \in \mathbb{Z}_p^*$, and the unknown to be recovered is the unique $x \in \{0, \ldots, N-1\}$ such that $y = g^x \bmod p$. Throughout, $\left|g^j\right\rangle$ means the computational-basis state labelled by the residue $g^j \bmod p$, and exponents of g are understood modulo N.

Phase estimation needs reversible multiplication by g and by y, as well as controlled powers of those operations. All are instances of the same construction, so define a general multiplier U_c and later specialize to $c = g$ and $c = y$. On the group basis its intended action is $U_c \left|a\right\rangle = \left|ca \bmod p\right\rangle$. This action is reversible: because $c \in \mathbb{Z}_p^*$, it has an inverse $c^{-1} \bmod p$, and multiplication by c^{-1} undoes multiplication by c.

To define U_c on the physical register as a whole, put $M = 2^n$. The n-qubit register has basis labels $0, \ldots, M-1$, whereas only $1, \ldots, p-1$ represent elements of $\mathbb{Z}_p^*$: the label 0 is outside the multiplicative group, and the labels $p, \ldots, M-1$ are unused. We simply fix all these outside labels:

$$U_c \left|a\right\rangle = \begin{cases} \left|ca \bmod p\right\rangle, & 1 \le a \le p-1, \\ \left|a\right\rangle, & a = 0 \text{ or } p \le a < M. \end{cases} \tag{4.2}$$

Thus U_c is a permutation of the full computational basis and hence a unitary. The second line has no algorithmic

role; it only completes the definition outside the group subspace.

Now specialize. Multiplication by g gives

$$U_g \left|g^j\right\rangle = \left|g\, g^j \bmod p\right\rangle = \left|g^{j+1} \bmod p\right\rangle, \tag{4.3}$$

so the exponent label changes from j to $j+1$. For y, the crucial step is the ordinary exponent law, with the substitution $y = g^x \bmod p$ written out:

$$y(g^j \bmod p) \equiv (g^x \bmod p)(g^j \bmod p) \equiv g^x g^j = g^{x+j} \pmod p. \tag{4.4}$$

Therefore

$$U_y \left|g^j\right\rangle = \left|y g^j \bmod p\right\rangle = \left|g^{x+j} \bmod p\right\rangle. \tag{4.5}$$

In the ordered cycle, this is exactly the permutation $j \mapsto j+x \pmod N$: multiplication by y advances x positions, with wraparound because $g^N = 1 \bmod p$. Write

$$U_g^x := \underbrace{U_g \circ U_g \circ \cdots \circ U_g}_{x \text{ successive applications}}, \qquad U_g^0 := I. \tag{4.6}$$

Thus the superscript x denotes an operator power: applying the one-position shift U_g exactly x times. For every group-basis state,

$$U_g^x \left|g^j\right\rangle = \left|g^{j+x} \bmod p\right\rangle = U_y \left|g^j\right\rangle, \qquad \text{hence} \qquad U_y = U_g^x \tag{4.7}$$

on the group subspace.

Phase estimation requires an eigenstate of the unitary being measured. Here the relevant unitaries are U_g and U_y; but since $U_y = U_g^x$, any relation $U_g |\psi\rangle = \lambda |\psi\rangle$ immediately gives $U_y |\psi\rangle = \lambda^x |\psi\rangle$. Thus one eigenstate of U_g can be retained for both phase-estimation runs, producing correlated phases of their eigenvalues. To derive such states, write a general state on the group cycle as

$$|\psi\rangle = \sum_{j=0}^{N-1} a_j \left|g^j\right\rangle \tag{4.8}$$

and impose $U_g |\psi\rangle = \lambda |\psi\rangle$. Both λ and the coefficients a_j are unknown: I seek a nonzero vector whose coefficients, after the cyclic shift, equal their original values multiplied by the same number λ. Written in the same ordered basis, this condition is

$$\underbrace{a_{N-1}\left|g^0\right\rangle + a_0\left|g^1\right\rangle + a_1\left|g^2\right\rangle + \cdots + a_{N-2}\left|g^{N-1}\right\rangle}_{U_g|\psi\rangle} = \underbrace{\lambda a_0\left|g^0\right\rangle + \lambda a_1\left|g^1\right\rangle + \cdots + \lambda a_{N-1}\left|g^{N-1}\right\rangle}_{\lambda|\psi\rangle}. \tag{4.9}$$

The displayed computational-basis states are distinct, so equality of the vectors is equivalent to equality of their coefficients:

$$a_{N-1} = \lambda a_0, \qquad a_{j-1} = \lambda a_j \quad (1 \le j < N). \tag{4.10}$$

Since a unitary eigenvalue is nonzero, multiplying the second equality by λ^{-1} gives

$$a_j = \lambda^{-1} a_{j-1} = \lambda^{-2} a_{j-2} = \cdots = \lambda^{-j} a_0. \tag{4.11}$$

Thus specifying λ and a_0 determines every coefficient. This pattern must also satisfy the remaining equation, $a_{N-1} = \lambda a_0$, at the point where the cycle wraps around to $\left|g^0\right\rangle$. Substituting $a_{N-1} = \lambda^{-(N-1)} a_0$ into that equation, and using $a_0 \neq 0$ (otherwise the recurrence would make every a_j zero), gives

$$\lambda^{-(N-1)} a_0 = a_{N-1} = \lambda a_0 \quad \Longrightarrow \quad \lambda^{-(N-1)} = \lambda \quad \Longrightarrow \quad 1 = \lambda^N. \tag{4.12}$$

Every eigenvalue must therefore be an N-th root of unity. I first list these possible eigenvalues, then construct a vector for each one and verify that it works. Put $\omega = e^{2\pi i/N}$. Since U_g is unitary, $|\lambda| = 1$, so write $\lambda = e^{i\varphi}$ with the angle $\varphi \in \mathbb{R}$ defined modulo 2π. The root condition becomes

$$\begin{aligned} \lambda^N = 1 &\Longleftrightarrow (e^{i\varphi})^N = 1 \Longleftrightarrow e^{iN\varphi} = 1 \Longleftrightarrow N\varphi \in 2\pi\mathbb{Z} \\ &\Longleftrightarrow N\varphi = 2\pi s \quad (s \in \mathbb{Z}) \Longleftrightarrow \varphi = \frac{2\pi s}{N} \Longleftrightarrow \lambda = e^{2\pi i s/N} = \omega^s. \end{aligned} \tag{4.13}$$

Replacing s by $s+N$ leaves the root unchanged, so label the N distinct roots by $\lambda_s := \omega^s$, for $s \in \{0, \ldots, N-1\}$. Every one satisfies $\lambda_s^N = e^{2\pi i s} = 1$. Here s is an integer label and λ_s is the complex number it selects; these are our possible eigenvalues.

I now fix one label s and construct the vector associated with λ_s. Substituting this choice into the coefficient recurrence gives

$$a_j = \lambda_s^{-j} a_0 = (\omega^s)^{-j} a_0 = \omega^{-sj} a_0. \tag{4.14}$$

All N coefficients have magnitude $|a_0|$. Since the states $\left|g^j\right\rangle$ are orthonormal, normalization means summing the squared magnitudes of these coefficients:

$$1 = \langle\psi|\psi\rangle = \sum_{j=0}^{N-1} \left|a_j\right|^2 = \sum_{j=0}^{N-1} |a_0|^2 = N\,|a_0|^2\,. \tag{4.15}$$

Thus $|a_0| = 1/\sqrt{N}$; the overall phase is irrelevant, so choose $a_0 = 1/\sqrt{N}$. This fixes every coefficient of the vector associated with our chosen s. I denote the resulting normalized state by $|\chi_s\rangle$:

$$|\chi_s\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \omega^{-sj} \left|g^j\right\rangle = \frac{1}{\sqrt{N}} \left(|1\rangle + \omega^{-s}\,|g\rangle + \omega^{-2s} \left|g^2\right\rangle + \cdots + \omega^{-(N-1)s} \left|g^{N-1}\right\rangle\right). \tag{4.16}$$

Here s stays fixed while j runs over all the group basis states. The notation $|\chi_s\rangle$ names the whole vector: choosing a different s changes the relative phases of its coefficients, while the same basis states occur with equal coefficient magnitude. Increasing j by one multiplies the coefficient by ω^{-s}, so its phase changes by the same amount at each step. This is the sense in which I informally call $|\chi_s\rangle$ a "phase wave" around the exponent cycle.

It remains to verify that this constructed state has the chosen eigenvalue $\lambda_s = \omega^s$. Apply the one-position shift and reindex the sum by $k = j + 1 \pmod N$:

$$U_g\,|\chi_s\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \omega^{-sj} \left|g^{j+1}\right\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{-s(k-1)} \left|g^k\right\rangle = \omega^s \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{-sk} \left|g^k\right\rangle = \omega^s\,|\chi_s\rangle\,. \tag{4.17}$$

The wraparound term obeys the same reindexing because $\omega^{sN} = 1$. Since s was arbitrary, this proves that every listed root $\lambda_s = \omega^s$ actually occurs as an eigenvalue of U_g, with the normalized eigenstate $|\chi_s\rangle$ constructed above. Thus the N possible eigenvalues each have a corresponding state. Since $U_y = U_g^x$—that is, multiplication by y equals x successive applications of U_g—each of these states is also an eigenstate of U_y:

$$U_y\,|\chi_s\rangle = U_g^x\,|\chi_s\rangle = (\omega^s)^x\,|\chi_s\rangle = \omega^{sx}\,|\chi_s\rangle\,. \tag{4.18}$$

For an eigenvalue written as $e^{2\pi i\theta}$, the number $\theta \pmod 1$ is called its *eigenphase*: it is the phase of the eigenvalue, not of the eigenvector. Here $\omega^s = e^{2\pi i s/N}$ and $\omega^{sx} = e^{2\pi i (sx \bmod N)/N}$, so the eigenphases of U_g and U_y on $|\chi_s\rangle$ are s/N and $(sx \bmod N)/N$, respectively.

Until now, I fixed s and expressed one eigenstate as a sum over the computational-basis states indexed by j. I now sum over the different eigenstates themselves, indexed by s. The following identity shows that their equal superposition is the easily prepared state $|1\rangle = \left|g^0\right\rangle$. Thus the algorithm need not prepare any chosen $|\chi_s\rangle$: the state $|1\rangle$ already contains a component along every one of them. Explicitly,

$$\frac{1}{\sqrt{N}} \sum_{s=0}^{N-1} |\chi_s\rangle = \frac{1}{N} \sum_{j=0}^{N-1} \left(\sum_{s=0}^{N-1} \omega^{-sj}\right) \left|g^j\right\rangle = \left|g^0\right\rangle = |1\rangle\,, \tag{4.19}$$

because the finite geometric sum is, explicitly,

$$\sum_{s=0}^{N-1} \omega^{-sj} = \begin{cases} 1 + \cdots + 1 = N, & j = 0, \\ 1 + \omega^{-j} + \omega^{-2j} + \cdots + \omega^{-(N-1)j} = \dfrac{1 - \omega^{-Nj}}{1 - \omega^{-j}} = 0, & 1 \le j < N. \end{cases} \tag{4.20}$$

4.I.ii What the two phase estimations reveal

Before following the registers, fix the roles of the two numbers we want the quantum part to produce. The unknown of the original problem is still x, defined by $y = g^x \bmod p$. The quantum algorithm will not output x directly; instead, it will produce a modular linear equation containing x. The symbol $s \in \{0, \ldots, N-1\}$ labels one of the common eigenstates $|\chi_s\rangle$. It is not an input chosen by the algorithm: the prepared state $|1\rangle$ contains every label s, and the first phase estimation will reveal one uniformly random label. After that measurement, the same target state $|\chi_s\rangle$ is retained.

On this selected state, U_g has eigenphase s/N. The action of U_y on the same state has eigenphase $(sx \bmod N)/N$; we give its numerator the name $t := sx \bmod N$. Thus t is not a second unrelated index: it is the residue obtained by multiplying the selected label s by the unknown x. The desired quantum output is therefore the correlated pair

$$U_g : \frac{s}{N} \rightsquigarrow s, \qquad U_y : \frac{t}{N} \rightsquigarrow t, \qquad t \equiv sx \pmod{N}. \tag{4.21}$$

This relation is useful because it contains the sought value: if $\gcd(s, N) = 1$, then $x \equiv s^{-1}t \pmod{N}$ immediately, while the recovery section below explains how to combine samples when s is not invertible. The purpose of the two phase estimations is precisely to obtain s and t for the *same* eigenstate. We now justify that mechanism register by register.

For the moment, I treat the phase readout as an exact label, so that the relationship between the two estimations can be seen before accounting for finite precision. If $U|\phi_r\rangle = e^{2\pi i\theta_r}|\phi_r\rangle$, linearity gives the schematic transformation

$$|0\rangle_P \sum_r \alpha_r |\phi_r\rangle_T \xrightarrow{\mathrm{PEA}(U)} \sum_r \alpha_r |\theta_r\rangle_P |\phi_r\rangle_T . \tag{4.22}$$

Here P is the phase-estimation ancilla register and T the target register. In this ideal picture, distinct eigenphases have perfectly distinguishable output labels. When the θ_r are distinct, as they are for U_g on the $|\chi_s\rangle$, measuring P selects branch r with probability $|\alpha_r|^2$ and leaves T in $|\phi_r\rangle$. A finite register instead attaches an output superposition to each eigenstate. Those outputs can overlap, so measuring the phase register need not select a single eigenstate. Part II treats this finite readout explicitly.

Apply this rule to the already prepared state $|1\rangle = N^{-1/2}\sum_{s=0}^{N-1}|\chi_s\rangle$ and to U_g. Since $U_g|\chi_s\rangle = \omega^s|\chi_s\rangle = e^{2\pi i(s/N)}|\chi_s\rangle$, the first estimation acts as

$$|0\rangle_{P_g}|1\rangle_T = \frac{1}{\sqrt{N}}\sum_{s=0}^{N-1}|0\rangle_{P_g}|\chi_s\rangle_T \xrightarrow{\mathrm{PEA}(U_g)} \frac{1}{\sqrt{N}}\sum_{s=0}^{N-1}|s/N\rangle_{P_g}|\chi_s\rangle_T . \tag{4.23}$$

The N branches have equal squared amplitude $1/N$. Thus, in this ideal picture, measuring P_g returns one uniformly random value s/N and collapses the target to the corresponding state:

$$\frac{1}{\sqrt{N}}\sum_{r=0}^{N-1}|r/N\rangle_{P_g}|\chi_r\rangle_T \xrightarrow{\text{measure } P_g} |s/N\rangle_{P_g}|\chi_s\rangle_T , \qquad \Pr[s] = \frac{1}{N}. \tag{4.24}$$

The measurement has simply selected one of the N eigenstate branches, and the first phase value tells us which label s was selected. Crucially, we keep the target register in this particular $|\chi_s\rangle$.

Now reset only the phase workspace and run phase estimation for U_y on that same target. The eigenvalue relation proved above can be written

$$U_y|\chi_s\rangle = \omega^{sx}|\chi_s\rangle = e^{2\pi i(sx \bmod N)/N}|\chi_s\rangle . \tag{4.25}$$

If $t := sx \bmod N$, the second estimation therefore gives

$$|0\rangle_{P_y}|\chi_s\rangle_T \xrightarrow{\mathrm{PEA}(U_y)} |t/N\rangle_{P_y}|\chi_s\rangle_T , \qquad t = sx \bmod N. \tag{4.26}$$

The phase is the fraction t/N; its numerator t is the residue that will be recovered from the measured bits. Hence one complete run produces the pair (s, t) satisfying $t = sx \pmod{N}$. The first estimation has named the selected wave; the second has read how the unknown x-step shift acts on that same wave. Repreparing $|1\rangle$ between them would instead select an independent s' and sever this relation. This is Shor's broad pattern—modular multiplication becomes phase information for classical number theory—but two commuting multipliers now produce the related residues s and sx.

4.II Finite precision, verification, and recovery

4.II.i Finite precision and candidate residues

From approximate phase measurements to exact integer numerators.

The ideal argument produced two related phases whose integer numerators give an equation for x. I now show how finite phase measurements can recover those numerators. Because their denominator N is known, a sufficiently accurate estimate can be rescaled and rounded to an exact integer. I first explain what changes when the readout

is finite, determine the accuracy needed for rounding, and prove that phase estimation supplies it with constant probability. The next subsection then establishes the relation between the two readings when they use the same target.

What changes in finite phase estimation. An m-qubit phase register returns an integer $z \in \{0, \ldots, M-1\}$, interpreted as the phase estimate z/M, where $M = 2^m$. I keep the choice $m = n$, so $M = 2^n > N = p - 1$.

If the measured bit string is $b_1 \cdots b_m$, it represents the binary integer

$$z = \sum_{j=1}^{m} b_j 2^{m-j}. \tag{4.27}$$

Dividing by $M = 2^m$ changes the place values from integer powers of two to binary fractional places:

$$\frac{z}{M} = \sum_{j=1}^{m} b_j 2^{-j} = (0.b_1 \cdots b_m)_2. \tag{4.28}$$

For example, the readout 011 gives $z = 3$ and the phase estimate $3/8 = 0.011_2$. The circuit measures the bits; the fraction z/M is their interpretation as a phase, measured in turns.

The earlier ideal picture treated different phases as perfectly distinguishable output labels. In this PEA circuit, all amplitude reaches one label only when the phase lies exactly on its binary grid. Then the inverse Fourier transform makes the contributions reinforce at that label and cancel at all others. Between grid points, the cancellation is incomplete: several output labels retain nonzero amplitudes, even with perfectly implemented gates.

For example, a three-bit register represents multiples of $1/8$, so it cannot represent $1/6$ exactly. The true eigenphase is still $1/6$; finite precision changes the available readout labels, not the eigenvalue of the target. Before measurement, the circuit produces a coherent superposition of those labels. The random classical outcome appears only when the phase register is measured.

There are two consequences. For one fixed eigenstate, measurement can return different estimates, while the target eigenstate stays unchanged. For a superposition of eigenstates, different eigenstates can produce the same measured label. The measurement then leaves a superposition of those surviving eigenstates rather than selecting just one. Rounding the measured number to one candidate does not remove these remaining quantum terms. I first analyze one fixed eigenstate; the correlation argument below will handle the actual superposition.

Accuracy needed for exact rounding. Write the phase as $\theta = r/N$, where $r \in \{0, \ldots, N-1\}$ denotes the integer numerator we want to recover. This calculation will apply to either of the two phases.

There are two grids with different roles. The possible true phases lie on the N-point grid, while the measured estimates lie on the M-point grid:

$$\begin{aligned} &0, \frac{1}{N}, \frac{2}{N}, \ldots, \frac{N-1}{N} && \text{possible true phases,} \\ &0, \frac{1}{M}, \frac{2}{M}, \ldots, \frac{M-1}{M} && \text{possible phase estimates.} \end{aligned} \tag{4.29}$$

The comparison is $z/M \approx r/N$: the integer z itself is not an approximation to r. Consecutive possible true phases are $1/N$ apart, so an estimate less than $1/(2N)$ from r/N identifies its numerator uniquely. Because $M > N$, half a measurement-grid step satisfies

$$\frac{1}{2M} < \frac{1}{2N}. \tag{4.30}$$

Thus a nearest measurement-grid point is sufficiently accurate. Rescaling by N translates this separation of phases into separation of integers.

Since z/M estimates r/N, multiplying it by N estimates r. Define the classical rounding operation by

$$\mathrm{Round}_N(z) := \left\lfloor \frac{Nz}{M} + \frac{1}{2} \right\rfloor \bmod N. \tag{4.31}$$

Let z^{near} be a nearest integer to Mr/N, choosing either one in a tie. It is an allowed output label: $0 \le Mr/N \le M - M/N < M - 1$.

Consecutive integers are one unit apart, so a real number is within $1/2$ of a nearest integer. Applying this to Mr/N and dividing the distance by M gives

$$\left|z^{\text{near}} - \frac{Mr}{N}\right| \le \frac{1}{2} \quad \Longrightarrow \quad \left|\frac{z^{\text{near}}}{M} - \frac{r}{N}\right| = \frac{1}{M}\left|z^{\text{near}} - \frac{Mr}{N}\right| \le \frac{1}{2M}. \tag{4.32}$$

Multiplying this phase error by N shows exactly why the rounded numerator is correct:

$$\left|\frac{Nz^{\text{near}}}{M} - r\right| = N\left|\frac{z^{\text{near}}}{M} - \frac{r}{N}\right| \le \frac{N}{2M} < \frac{1}{2}. \tag{4.33}$$

Consequently $\text{Round}_N(z^{\text{near}}) = r$: a nearest phase estimate recovers the numerator exactly. The final reduction modulo N simply expresses the candidate as a residue. This is the accuracy we need; the remaining question is how often the circuit achieves it.

For a concrete Q4 instance, take $p = 7$, so $N = 6$, and $m = n = 3$, so $M = 8$. If $s = 5$, then the true first eigenphase is $5/6$, and its nearest measurement label is $z_g^{\text{near}} = 7$. The phase estimate is inexact, but the rescaled numerator is recovered exactly:

$$\frac{z_g^{\text{near}}}{M} = \frac{7}{8} \approx \frac{5}{6}, \qquad \text{Round}_6(7) = \text{round}\left(\frac{6 \cdot 7}{8}\right) = 5 = s. \tag{4.34}$$

This example describes the nearest-label event; it does not assert that every measurement returns 7. The next calculation bounds the probability of the event that makes rounding succeed.

Probability of a sufficiently accurate readout. Fix a target eigenstate with the phase just considered:

$$U\,|\psi\rangle = e^{2\pi i\theta}\,|\psi\rangle\,, \qquad \theta = r/N. \tag{4.35}$$

The finite circuit has three stages: create coherent branch labels, encode the phase on those branches, and decode them into possible readout labels. For three phase qubits, its structure is

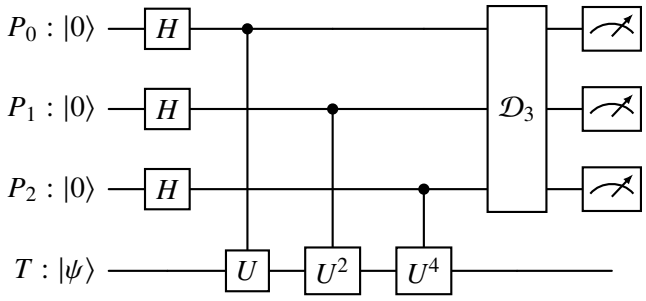


Figure 8. Three-bit phase estimation on a fixed target eigenstate. The controlled powers encode the phase, and the inverse Fourier transform $\mathcal{D}_3$ produces amplitudes on the eight measurable labels.

The Hadamards first create M equally weighted coherent branches $|j\rangle$. Writing $j = \sum_{\ell=0}^{m-1} q_\ell 2^\ell$ with $q_\ell \in \{0, 1\}$ makes the controlled-power step explicit: a control bit equal to zero skips its power, while a bit equal to one applies it. Their combined action is

$$\prod_{\ell=0}^{m-1} (U^{2^\ell})^{q_\ell} = U^{\sum_{\ell=0}^{m-1} q_\ell 2^\ell} = U^j. \tag{4.36}$$

Since the target is an eigenstate,

$$U^j\,|\psi\rangle = (e^{2\pi i\theta})^j\,|\psi\rangle = e^{2\pi ij\theta}\,|\psi\rangle\,. \tag{4.37}$$

The target therefore factors out unchanged. The acquired phase is relative between different control branches, so it can affect their later interference. Finally, the inverse Fourier transform $\mathcal{D}_m$ combines those branches into the possible readout labels.

The complete unitary evolution is

$$\begin{aligned} |0^m\rangle_P |\psi\rangle_T &\xrightarrow{H^{\otimes m}} \frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} |j\rangle_P |\psi\rangle_T \\ &\xrightarrow{\text{controlled powers of } U} \left(\frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} e^{2\pi i j\theta} |j\rangle_P \right) |\psi\rangle_T \xrightarrow{\mathcal{D}_m} \left(\sum_{z=0}^{M-1} \alpha_\theta(z) |z\rangle_P \right) |\psi\rangle_T . \end{aligned} \tag{4.38}$$

To calculate the probability of the desired readout, we need its amplitude. Every incoming branch j contributes to the same output label z, because the decoder acts by

$$\mathcal{D}_m |j\rangle = \frac{1}{\sqrt{M}} \sum_{z=0}^{M-1} e^{-2\pi i j z/M} |z\rangle . \tag{4.39}$$

These contributions can reinforce or cancel, so their amplitudes must be added before taking the squared magnitude. Multiplying each incoming amplitude by its decoder factor and summing over j gives

$$\alpha_\theta(z) = \sum_{j=0}^{M-1} \underbrace{\frac{e^{2\pi i j\theta}}{\sqrt{M}}}_{\text{incoming amplitude}} \underbrace{\frac{e^{-2\pi i j z/M}}{\sqrt{M}}}_{\text{decoder contribution}} = \frac{1}{M} \sum_{j=0}^{M-1} e^{2\pi i j(\theta - z/M)}. \tag{4.40}$$

The two normalization factors give $1/M$.

The same calculation has a useful geometric interpretation. The acquired phase-register wave and the finite-grid waves are

$$|\Phi_\theta\rangle = \frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} e^{2\pi i j\theta} |j\rangle , \qquad |w_z\rangle = \frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} e^{2\pi i j z/M} |j\rangle . \tag{4.41}$$

The M grid waves form an orthonormal basis, and $\mathcal{D}_m |w_z\rangle = |z\rangle$. Thus the amplitude just derived is the component of the acquired wave along the candidate grid wave:

$$\alpha_\theta(z) = \langle w_z \mid \Phi_\theta\rangle, \qquad |\Phi_\theta\rangle = \sum_z \alpha_\theta(z) |w_z\rangle \longmapsto \sum_z \alpha_\theta(z) |z\rangle . \tag{4.42}$$

Unless the acquired wave equals one grid wave exactly, it has components along several of them. The unitary decoder preserves this expansion; it does not collapse the state or randomly select one component.

Measuring the phase register returns one integer $\widetilde{z}$, with

$$\Pr[\widetilde{z} = z \mid \theta] = |\alpha_\theta(z)|^2 . \tag{4.43}$$

This formula makes the earlier cancellation picture precise. If $\theta = k/M$, it gives $\alpha_\theta(k) = 1$ and zero amplitude at every other label. Otherwise the readout is spread over several labels. I now bound the probability of the particular nearest label that makes rounding succeed.

To see why nearby labels have large amplitudes, put $\delta_z := \theta - z/M$. Then

$$\alpha_\theta(z) = \frac{1}{M} \sum_{j=0}^{M-1} e^{2\pi i j \delta_z}. \tag{4.44}$$

Each summand is a unit arrow in the complex plane. Advancing j by one rotates the next arrow by $2\pi\delta_z$. When the candidate is close to the true phase modulo one turn, the mismatch is small, so the arrows reinforce one another. For distant candidates they rotate through many directions and largely cancel. The factor j accumulates the phase mismatch across the branches and makes the comparison sensitive.

For $\delta_z \notin \mathbb{Z}$, summing the geometric progression and taking the squared magnitude gives

$$\alpha_\theta(z) = \frac{1}{M} \frac{1 - e^{2\pi i M \delta_z}}{1 - e^{2\pi i \delta_z}}, \qquad |\alpha_\theta(z)|^2 = \frac{1}{M^2} \frac{\sin^2(\pi M \delta_z)}{\sin^2(\pi \delta_z)}. \tag{4.45}$$

The sine ratio follows from $|1 - e^{iu}| = 2|\sin(u/2)|$. This converts the interference picture into the quantitative bound below.

Set $\varepsilon := M\theta - z^{\text{near}}$, so $|\varepsilon| \le 1/2$. For $\varepsilon \neq 0$, substituting $\delta_{z^{\text{near}}} = \varepsilon/M$ into that formula gives

$$\Pr[\widetilde{z} = z^{\text{near}} \mid \theta] = \frac{1}{M^2} \frac{\sin^2(\pi\varepsilon)}{\sin^2(\pi\varepsilon/M)}. \tag{4.46}$$

Here $0 < \pi|\varepsilon| \le \pi/2$. Concavity of sine on $[0, \pi/2]$ places its graph above the chord joining $(0,0)$ and $(\pi/2, 1)$, so $\sin(\pi|\varepsilon|) \ge 2|\varepsilon|$. The inequality $\sin u \le u$ for $u \ge 0$ gives $\sin(\pi|\varepsilon|/M) \le \pi|\varepsilon|/M$. Using the first inequality in the numerator and the second in the denominator yields

$$\Pr[\widetilde{z} = z^{\text{near}} \mid \theta] \ge \frac{1}{M^2} \left(\frac{2\,|\varepsilon|}{\pi\,|\varepsilon|\,/M} \right)^2 = \frac{4}{\pi^2}. \tag{4.47}$$

When $\varepsilon = 0$, this probability is 1. Thus one finite readout recovers its integer numerator by rounding with probability at least $4/\pi^2$. Appendix C.1 retains a standalone, stage-by-stage refresher and develops the circular nearest-label convention for generic phases; the Q4 phases above require no wraparound.

For the two Q4 readouts on a fixed $|\chi_s\rangle$, the numerators are s and $t_s := sx \bmod N$. Write their measured labels as z_g, z_y and their amplitudes as

$$\alpha_s(z_g) := \alpha_{s/N}(z_g), \qquad \beta_s(z_y) := \alpha_{t_s/N}(z_y). \tag{4.48}$$

Writing the two fixed-target transformations separately makes their roles visible. With G and Y denoting the respective phase workspaces,

$$\begin{aligned} |0^m\rangle_G\,|\chi_s\rangle_T &\xmapsto{\text{PEA}(U_g)} \left(\sum_{z_g=0}^{M-1} \alpha_s(z_g)\,\big|z_g\big\rangle_G \right) |\chi_s\rangle_T\,, \\ |0^m\rangle_Y\,|\chi_s\rangle_T &\xmapsto{\text{PEA}(U_y)} \left(\sum_{z_y=0}^{M-1} \beta_s(z_y)\,\big|z_y\big\rangle_Y \right) |\chi_s\rangle_T\,. \end{aligned} \tag{4.49}$$

For each fixed s, the squared amplitudes in each row sum to one. The first distribution is peaked near Ms/N and the second near Mt_s/N. These are distributions of possible readouts of the same unchanged eigenstate; they are not new eigenstates or new eigenphases. Accordingly, rescaling a sufficiently accurate first readout estimates s, while rescaling the second estimates t_s:

$$\frac{Nz_g}{M} \approx s, \qquad \frac{Nz_y}{M} \approx t_s. \tag{4.50}$$

Let $z_g^{\text{near}}(s)$ and $z_y^{\text{near}}(s)$ denote nearest integers to Ms/N and Mt_s/N, respectively. Each designated output has probability at least $4/\pi^2$, and the rounding result gives

$$\text{Round}_N\big(z_g^{\text{near}}(s)\big) = s, \qquad \text{Round}_N\big(z_y^{\text{near}}(s)\big) = t_s. \tag{4.51}$$

This completes the calculation for a fixed eigenstate. It remains to show what the two measurements produce when the target starts in $|1\rangle$.

4.II.ii Correlation of the finite readouts

The single-eigenstate calculation shows how an accurate readout recovers its numerator. I now establish why the two readouts can recover numerators with the required relation when the prepared target is a superposition. Because the first finite measurement need not isolate one $|\chi_s\rangle$, this requires following the retained target through both estimations. With two logical phase registers G, Y and the target T, the initial state is

$$|0^m\rangle_G\,|0^m\rangle_Y\,|1\rangle_T = \frac{1}{\sqrt{N}} \sum_{s=0}^{N-1} |0^m\rangle_G\,|0^m\rangle_Y\,|\chi_s\rangle_T\,. \tag{4.52}$$

First fix one term of this superposition. The first PEA acts on G and T, while the identity acts on the untouched register Y. The fixed-s rule established above therefore gives

$$|0^m\rangle_G\,|0^m\rangle_Y\,|\chi_s\rangle_T \xmapsto{\text{PEA}(U_g)} \sum_{z_g=0}^{M-1} \alpha_s(z_g)\,\big|z_g\big\rangle_G\,|0^m\rangle_Y\,|\chi_s\rangle_T\,. \tag{4.53}$$

The same linear transformation is applied to every s-term already present in $|1\rangle$. Substituting the fixed-branch rule before combining the sums makes the step explicit:

$$\begin{aligned}\frac{1}{\sqrt{N}}\sum_{s=0}^{N-1}|0^m\rangle_G\,|0^m\rangle_Y\,|\chi_s\rangle_T &\xmapsto{\mathrm{PEA}(U_g)} \frac{1}{\sqrt{N}}\sum_{s=0}^{N-1}\left(\sum_{z_g=0}^{M-1}\alpha_s(z_g)\,\big|z_g\big\rangle_G\,|0^m\rangle_Y\,|\chi_s\rangle_T\right)\\ &= \frac{1}{\sqrt{N}}\sum_{s=0}^{N-1}\sum_{z_g=0}^{M-1}\alpha_s(z_g)\,\big|z_g\big\rangle_G\,|0^m\rangle_Y\,|\chi_s\rangle_T\,.\end{aligned} \tag{4.54}$$

No value of s or z_g has been selected at this stage. Each $|\chi_s\rangle$ branch remains present and is now correlated with its own finite-PEA amplitudes over z_g. Now suppose measuring G returns the particular label $\widetilde{z}_g$. This measurement keeps every term whose G-register equals $\big|\widetilde{z}_g\big\rangle_G$ and removes every term with $z_g \neq \widetilde{z}_g$. Applying that rule to the state above gives the unnormalized post-measurement state

$$\frac{1}{\sqrt{N}}\sum_{s=0}^{N-1}\sum_{z_g=0}^{M-1}\alpha_s(z_g)\,\big|z_g\big\rangle_G\,|0^m\rangle_Y\,|\chi_s\rangle_T \xrightarrow{\text{measure } G:\,\widetilde{z}_g} \frac{1}{\sqrt{N}}\big|\widetilde{z}_g\big\rangle_G\,|0^m\rangle_Y\sum_{s=0}^{N-1}\alpha_s(\widetilde{z}_g)\,|\chi_s\rangle_T\,. \tag{4.55}$$

The whole joint state has collapsed: all other measured labels are gone. But the surviving label need not identify one value of s. Its probability is the squared norm of the surviving vector. Since the $|\chi_s\rangle$ are orthonormal,

$$\Pr[\widetilde{z}_g] = \frac{1}{N}\left\|\sum_{s=0}^{N-1}\alpha_s(\widetilde{z}_g)\,|\chi_s\rangle\right\|^2 = \frac{1}{N}\sum_{s=0}^{N-1}\big|\alpha_s(\widetilde{z}_g)\big|^2\,. \tag{4.56}$$

Dividing the surviving state by the square root of this probability cancels the factor $1/\sqrt{N}$. The normalized joint state is therefore

$$\big|\widetilde{z}_g\big\rangle_G\,|0^m\rangle_Y\,\frac{\displaystyle\sum_{s=0}^{N-1}\alpha_s(\widetilde{z}_g)\,|\chi_s\rangle_T}{\sqrt{\displaystyle\sum_{s=0}^{N-1}\big|\alpha_s(\widetilde{z}_g)\big|^2}}. \tag{4.57}$$

The reason several s-branches can remain is label overlap. Finite PEA gives each eigenphase a peak over several possible z_g values, so the same observed label $\widetilde{z}_g$ may have nonzero amplitude on several different $|\chi_s\rangle$ branches. If both $s = a$ and $s = b$ can produce this label, the surviving target factor contains both $\alpha_a(\widetilde{z}_g)\,|\chi_a\rangle$ and $\alpha_b(\widetilde{z}_g)\,|\chi_b\rangle$.

A numerical two-label model makes the filtering concrete. This is an illustrative measurement model, not a claim about the exact PEA amplitudes of a particular choice of p, g, y. Start with $(|\chi_1\rangle + |\chi_2\rangle)/\sqrt{2}$ and suppose the two branches acquire the readout amplitudes

$$\begin{aligned}|0^m\rangle_G\,|\chi_1\rangle_T &\longmapsto \left(\frac{\sqrt{3}}{2}\,|5\rangle_G + \frac{1}{2}\,|6\rangle_G\right)|\chi_1\rangle_T\,,\\ |0^m\rangle_G\,|\chi_2\rangle_T &\longmapsto \left(\frac{1}{2}\,|5\rangle_G + \frac{\sqrt{3}}{2}\,|6\rangle_G\right)|\chi_2\rangle_T\,.\end{aligned} \tag{4.58}$$

Either eigenstate branch can produce either label. Applying these rules by linearity and grouping the result by the measurable label gives

$$\frac{1}{\sqrt{2}}\left[|5\rangle_G\left(\frac{\sqrt{3}}{2}\,|\chi_1\rangle + \frac{1}{2}\,|\chi_2\rangle\right)_T + |6\rangle_G\left(\frac{1}{2}\,|\chi_1\rangle + \frac{\sqrt{3}}{2}\,|\chi_2\rangle\right)_T\right]. \tag{4.59}$$

Observing $G = 5$ removes the entire label-6 component. The probability of that observation is $\frac{1}{2}(\frac{3}{4} + \frac{1}{4}) = \frac{1}{2}$, so normalizing the surviving vector leaves

$$\frac{\sqrt{3}}{2}\,|\chi_1\rangle + \frac{1}{2}\,|\chi_2\rangle\,. \tag{4.60}$$

The whole joint state has collapsed, but the target still contains both eigenstate branches with unequal weights. Measuring G answered which finite label was observed; it did not identify which branch produced it.

This is exactly where the finite situation differs from the earlier ideal picture. There the phase labels attached to different s values were treated as perfectly distinct, so observing one label removed every other $|\chi_s\rangle$ branch. Here overlapping finite-resolution peaks leave the generally nonuniform coherent superposition above. It is not yet a classical mixture: no measurement has distinguished the surviving s-branches.

The second PEA acts on this retained target. On each surviving $|\chi_s\rangle$ branch it uses the U_y eigenphase t_s/N, so it attaches the amplitudes $\beta_s(z_y)$ without changing the branch label s. Conditional on the observed first label $\widetilde{z}_g$, the resulting state is proportional to

$$\sum_{s=0}^{N-1}\sum_{z_y=0}^{M-1}\alpha_s(\widetilde{z}_g)\beta_s(z_y)\left|\widetilde{z}_g\right\rangle_G\left|z_y\right\rangle_Y|\chi_s\rangle_T\,. \tag{4.61}$$

Equivalently, if both measurements are postponed until the end, the complete two-PEA state is

$$\frac{1}{\sqrt{N}}\sum_{s=0}^{N-1}\sum_{z_g=0}^{M-1}\sum_{z_y=0}^{M-1}\alpha_s(z_g)\beta_s(z_y)\left|z_g\right\rangle_G\left|z_y\right\rangle_Y|\chi_s\rangle_T\,. \tag{4.62}$$

The orthogonality needed to read the resulting probabilities follows from the same root-of-unity sum used earlier:

$$\langle\chi_{s'}|\chi_s\rangle=\frac{1}{N}\sum_{j=0}^{N-1}\omega^{(s'-s)j}=\delta_{s,s'}. \tag{4.63}$$

Therefore the probability of a classical pair of measured labels is

$$\Pr[z_g,z_y]=\frac{1}{N}\sum_{s=0}^{N-1}\left|\alpha_s(z_g)\right|^2\left|\beta_s(z_y)\right|^2. \tag{4.64}$$

Equivalently, conditional on one latent label s, the two readout distributions factor as

$$\Pr[z_g,z_y\mid s]=\left|\alpha_s(z_g)\right|^2\left|\beta_s(z_y)\right|^2. \tag{4.65}$$

Thus the *classical outcome distribution* is exactly as if one first chose a uniform hidden label s and then sampled both finite-PEA distributions for that same s. This is only an interpretation of the measured statistics: the circuit never measures such a hidden label, and the conditional target above remains a coherent superposition.

Hence, conditional on that same latent s, the two nearest-label events have joint probability at least

$$q:=\left(\frac{4}{\pi^2}\right)^2=\frac{16}{\pi^4}. \tag{4.66}$$

On this sufficient event, the rounding calculation gives the exact correlated pair (s,t_s) with $t_s=sx \bmod N$. Outputs outside this event are not automatically wrong: a non-nearest label may still round correctly, and in any case the next step checks the proposed relation exactly.

4.II.iii Candidate verification and discrete-log recovery

One preparation of $|1\rangle$[5], followed by the sequential PEA of U_g and U_y on the same retained target, will be called a *pair run*. Every pair run first produces only the candidate pair

$$\widehat{s}:=\mathrm{Round}_N(z_g),\qquad \widehat{t}:=\mathrm{Round}_N(z_y). \tag{4.67}$$

The nearest-label events need not have occurred. Instead, the candidate is tested by the entirely classical, exact congruence

$$g^{\widehat{t}}\stackrel{?}{\equiv}y^{\widehat{s}}\pmod p. \tag{4.68}$$

All quantities in this test are known. Moreover, because $y=g^x \bmod p$ and g has order N, it is equivalent to the relation we need:

$$g^{\widehat{t}}\equiv y^{\widehat{s}}\pmod p\iff g^{\widehat{t}}\equiv g^{x\widehat{s}}\pmod p\iff \widehat{t}\equiv x\widehat{s}\pmod N. \tag{4.69}$$

A failing candidate is discarded. A passing candidate is an exact modular equation involving the true x, even if it did not arise from the designated nearest labels. Only after this validation are the hats dropped and the pair

[5]This is an ordinary computational-basis preparation: starting from the standard all-zero target $|0\rangle^{\otimes n}$, apply an X gate to the qubit carrying the 2^0 bit. No separate state-preparation algorithm or nontrivial asymptotic cost is therefore hidden here.

stored as (s_i, t_i) with

$$t_i \equiv s_i x \pmod{N}. \tag{4.70}$$

I use two verified pairs for the recovery step, matching the trial analyzed below. If

$$\gcd(s_1, s_2, N) = 1, \tag{4.71}$$

the extended Euclidean algorithm supplies integers a, b, c such that $as_1 + bs_2 + cN = 1$. Multiplying by x and substituting the verified relations gives

$$x \equiv (as_1 + bs_2 + cN)x \pmod{N} \equiv a(s_1 x) + b(s_2 x) \pmod{N} \equiv at_1 + bt_2 \pmod{N}. \tag{4.72}$$

The cN term vanishes modulo N. Thus neither s_1 nor s_2 needs to be individually invertible: the two coefficients need only have no common prime factor together with N. The resulting representative $\widetilde{x} \in \{0, \ldots, N-1\}$ is accepted only after the final check $g^{\widetilde{x}} \equiv y \pmod{p}$.

Optional use of additional verified relations. One verified pair already suffices when $\gcd(s_1, N) = 1$, giving $x \equiv s_1^{-1} t_1 \pmod{N}$. Alternatively, verified pairs may be retained until $\gcd(N, s_1, \ldots, s_k) = 1$. Generalized Bézout coefficients then give

$$a_1 s_1 + \cdots + a_k s_k + bN = 1 \quad \Longrightarrow \quad x \equiv a_1 t_1 + \cdots + a_k t_k \pmod{N}. \tag{4.73}$$

With $d_k := \gcd(N, s_1, \ldots, s_k)$, the identity $d_{k+1} = \gcd(d_k, s_{k+1}) \mid d_k$ shows that an additional verified relation can only leave the remaining gcd unchanged or decrease it. The two-pair algorithm below already suffices for the claimed bounds.

4.II.iv Success probability and repetitions

For the stated probability and resource guarantees, one *recovery trial* consists of two independent pair runs. Validate both pairs and attempt the two-pair Bézout recovery; if anything fails, discard that trial and start another. Fixing the two latent branch labels σ_1, σ_2, the four nearest-label events—two PEAs in each pair run—have probability at least $(4/\pi^2)^4$. Therefore, without assuming that validation leaves the labels uniform,

$$\Pr[\text{recovery trial succeeds}] \geq \frac{1}{N^2} \sum_{\substack{0 \leq \sigma_1, \sigma_2 < N \\ \gcd(\sigma_1, \sigma_2, N) = 1}} \left(\frac{4}{\pi^2}\right)^4 = \left(\frac{4}{\pi^2}\right)^4 \Pr_{\sigma_1, \sigma_2 \text{ uniform}}[\gcd(\sigma_1, \sigma_2, N) = 1]. \tag{4.74}$$

For independent uniform σ_1, σ_2, the required counting can be made explicit. Fix a prime $\ell \mid N$. Exactly N/ℓ of the N residue classes modulo N are divisible by ℓ, so $\Pr[\ell \mid \sigma_i] = 1/\ell$ and, by independence of the two samples, $\Pr[\ell \mid \sigma_1 \text{ and } \ell \mid \sigma_2] = 1/\ell^2$. Moreover, the Chinese remainder theorem makes these divisibility conditions independent for distinct prime divisors of N. The gcd equals 1 precisely when no prime $\ell \mid N$ divides both samples. Therefore

$$\Pr[\gcd(\sigma_1, \sigma_2, N) = 1] = \prod_{\substack{\ell \mid N \\ \ell \text{ prime}}} (1 - \ell^{-2}). \tag{4.75}$$

Every factor lies in $(0, 1)$, so extending the finite product to all primes can only decrease it. Euler's product for the Riemann zeta function, together with $\zeta(2) = \pi^2/6$, then gives [13]

$$\prod_{\substack{\ell \mid N \\ \ell \text{ prime}}} (1 - \ell^{-2}) \geq \prod_{\ell \text{ prime}} (1 - \ell^{-2}) = \frac{1}{\zeta(2)} = \frac{6}{\pi^2}. \tag{4.76}$$

Consequently every recovery trial succeeds with probability at least the positive constant

$$\rho_0 := \left(\frac{4}{\pi^2}\right)^4 \frac{6}{\pi^2} = \frac{1536}{\pi^{10}} > 0.0164. \tag{4.77}$$

With exact phase rotations, the expected number of recovery trials is at most $1/\rho_0 < 61$, hence the expected number of pair runs is at most $2/\rho_0 < 122$; these are conservative constant bounds.

To guarantee a prescribed failure probability $\eta \in (0, 1)$ in this exact-rotation model, consider R independent recovery trials. Each trial fails with probability at most $1 - \rho_0$, so the probability that all R trials fail is at most $(1 - \rho_0)^R$. The elementary inequality $1 - u \leq e^{-u}$ gives the simpler upper bound $e^{-\rho_0 R}$. Requiring this bound to

be at most η yields the complete chain

$$e^{-\rho_0 R} \le \eta \quad \Longleftrightarrow \quad -\rho_0 R \le \ln \eta \quad \Longleftrightarrow \quad R \ge \frac{\ln(1/\eta)}{\rho_0}. \tag{4.78}$$

Since the number of trials must be an integer, choose

$$R := \left\lceil \frac{\ln(1/\eta)}{\rho_0} \right\rceil. \tag{4.79}$$

Uniformly for every $\eta \in (0, 1)$, this exact choice satisfies $R = O(1 + \log(1/\eta))$. In the usual high-confidence regime $\eta \to 0$, the shorter notation $O(\log(1/\eta))$ is equivalent. For this choice, the joint failure probability indeed obeys

$$(1 - \rho_0)^R \le e^{-\rho_0 R} \le \eta. \tag{4.80}$$

The pair-level test and the final check prevent an incorrect x from ever being accepted. Thus the indefinite repeat-until-verified version is a *Las Vegas algorithm*: its stopping time is random, but every answer it returns is guaranteed to be correct; randomness affects the runtime, not the correctness of the output. The capped version instead returns either the correct x or an explicit failure, with failure probability at most η. For example, choosing $\eta = 0.01$ gives the conservative cap

$$R = \left\lceil \frac{\ln 100}{\rho_0} \right\rceil = 281, \tag{4.81}$$

so the probability of obtaining no verified solution by the cap is at most 1%. In other words, one complete capped execution returns a verified solution with probability at least 99% and reports explicit failure with probability at most 1%; it never returns an incorrect solution. Across 100 independent complete executions, the expected number of reported failures is therefore at most one, although the actual number need not equal one. This numerical guarantee uses only the worst-case lower bound ρ_0; it is not a prediction that a typical instance requires 281 trials.

The implementation discussion below also gives a fixed finite gate set with approximated feedback rotations. Its error budget preserves trial success probability at least $\rho_0/2$. In that model, the corresponding cap is

$$R_{\text{fixed}} := \left\lceil \frac{2\ln(1/\eta)}{\rho_0} \right\rceil, \qquad (1 - \rho_0/2)^{R_{\text{fixed}}} \le \eta. \tag{4.82}$$

Exact classical verification still prevents an incorrect output; the change affects only the success constant and the required number of trials.

4.III Efficient implementation and resource bounds

4.III.i Efficient controlled powers

The preceding analysis used the controlled powers required by phase estimation abstractly. I now show why they remain efficient. Fix $c \in \{g, y\}$. An n-bit phase estimate logically requires the n controlled operations $U_c^{2^j}$, one associated with each binary phase position j, where $0 \le j < n$. This requirement belongs to ordinary PEA and does not yet assume an iterative implementation; the later one-qubit version merely performs the same logical operations sequentially while reusing their control qubit. Repeated application of the multiplier gives, on a valid group label,

$$U_c^r \, |a\rangle = |c^r a \bmod p\rangle \, . \tag{4.83}$$

Both sides fix every outside label under our full-space definition of U_c, so the identity holds on the entire n-qubit space. In particular, defining the known classical constant

$$c_j := c^{2^j} \bmod p \tag{4.84}$$

gives the operator identity

$$U_c^{2^j} = U_{c_j}. \tag{4.85}$$

Thus the power 2^j does *not* mean applying U_c exponentially many times. Repeated squaring computes the constants

$$c_0 = c, \qquad c_{j+1} = c_j^2 \bmod p \tag{4.86}$$

classically; one controlled multiplier by the known constant c_j is then compiled for that binary phase position.

Gate counts use the standard reversible-circuit model, in which constant-size one- and two-qubit gates and Toffoli gates have unit cost; the classically specified one-qubit feedback rotations in iterative PEA are counted as one-qubit gates. The fixed-set refinement below quantifies the cost and success probability when those rotations are approximated. A clean reversible multiplier by the hardwired constant c_j can be built by writing $a = \sum_{k=0}^{n-1} a_k 2^k$, precomputing $d_{j,k} = 2^k c_j \bmod p$, and conditionally adding $d_{j,k}$ to an n-qubit accumulator whenever $a_k = 1$. The forward accumulation uses n controlled modular additions. After a swap, cleaning the original register uses n controlled modular subtractions compiled from c_j^{-1}, which exists because $c_j \in \mathbb{Z}_p^*$. A reversible range flag conditions both halves on $1 \le a \le p-1$, thereby fixing every outside label; it can be uncomputed from the output because multiplication by nonzero c_j preserves membership in $\{1, \dots, p-1\}$. Thus one clean controlled multiplier uses $2n = O(n)$ modular-addition primitives, each costing $O(n)$ gates with ripple-carry arithmetic, and therefore costs $O(n^2)$ gates with $A(n) = O(n)$ reusable arithmetic ancillas [12, §II, Eqs. (4)–(6); §III.B–C, Figs. 4–5; §IV]. The complete accumulator invariant and uncomputation are developed in Appendix C.2.

For each base $c \in \{g, y\}$, repeated squaring, modular inversion, and doubling precompute the n constants c_j, their inverses, the forward addends $d_{j,k}$, and the cleanup addends $e_{j,k} := 2^k c_j^{-1} \bmod p$. These are $O(n^2)$ hardwired n-bit constants for each base. With schoolbook arithmetic, this one-time classical compilation costs $O(n^3)$ bit operations; the resulting controlled multipliers are then reused in every trial.

Qubit-saving refinement. The solution above is already complete with ordinary $m = n$ qubit phase estimation, whose peak memory remains $O(n)$. To reduce the physical phase workspace as far as possible, I invoke standard iterative phase estimation: it applies the same n controlled powers sequentially and, by measuring, resetting, and classically feeding forward one control qubit, reproduces the same n-bit outcome distribution and the same target state conditioned on that outcome [5, arXiv pp. 3–4, Eqs. (10)–(11), Fig. 2]. This optional implementation refinement replaces n simultaneous phase qubits by one reusable qubit; it changes no part of the preceding correctness proof and no asymptotic gate bound. Indeed, ordinary PEA would use $n + n + A(n) = O(n)$ qubits, while the refined implementation uses $n + 1 + A(n)$.

4.III.ii Quantum and classical resource counts

Quantum memory means the largest number of qubits present at one time rather than a sum over sequential stages. Within each PEA the controlled multipliers are applied one after another. When one multiplier finishes, its accumulator, carry qubits, and range flags have returned to $|0\rangle$ and are reused by the next multiplier. The two PEAs in a pair run and the two pair runs in a recovery trial are also sequential, while their measured bits are stored classically. Therefore the peak occurs during one controlled modular multiplication and is

$$Q(n) = \underbrace{n}_{\text{target}} + \underbrace{1}_{\text{reused phase qubit}} + \underbrace{A(n)}_{\text{arithmetic ancillas}} = O(n), \tag{4.87}$$

where $A(n) = O(n)$. This is asymptotically optimal: representing residues of an n-bit prime already requires the n-qubit target register. Here $A(n)$ includes the n-qubit product accumulator, carry and comparison workspace, the range flag, and a constant number of control ancillas.

For the gate count, one n-bit PEA has n controlled-power stages, hence n controlled modular multipliers. One pair run contains a PEA for U_g followed by a PEA for U_y, hence $2n$ multipliers. One recovery trial uses two independent pair runs, hence

$$2 \text{ pair runs} \times 2 \text{ PEAs per pair} \times n \text{ stages per PEA} = 4n \tag{4.88}$$

controlled multipliers. Since each costs $O(n^2)$ gates,

$$G_{\text{trial}}(n) = 4n \cdot O(n^2) + O(n) = O(n^3). \tag{4.89}$$

The lower-order $O(n)$ term consists of a constant number of one-qubit operations, measurements, and resets for each of the $O(n)$ sequential phase bits, together with preparing the target as $|1\rangle$. Because the expected number of recovery trials is below the constant 61, the indefinite verified algorithm has expected quantum gate complexity

$$\mathbb{E}[G(n)] = O(n^3). \tag{4.90}$$

This is total gate work, not circuit depth.

Feedback rotations over a fixed finite gate set. For a finite-set implementation, choose the inverse-closed universal set $\{H, T, T^\dagger, \mathrm{CNOT}\}$, where $T = \mathrm{diag}(1, e^{i\pi/4})$. It implements X and Toffoli exactly with constant-size circuits [2, §4.3, Fig. 4.9; §4.5.3], so the arithmetic incurs only constant overhead. Combine the classical phase corrections for each measured bit into one feedback rotation; at most $4n$ such rotations per recovery trial need approximation. Give each rotation operator-norm error at most $\delta = \rho_0/(8n)$, with an irrelevant global phase ignored. Applying the same telescoping error estimate as in Question 1(b) to the intervening measurement and classical-control channels bounds the total variation distance between the exact and synthesized trial distributions by $4n\delta = \rho_0/2$. Consequently, the synthesized trial still succeeds with probability at least $\rho_0/2$. Solovay–Kitaev synthesis [4, §2, Theorem 1; §3] uses $O(\mathrm{polylog}\, n)$ gates and classical synthesis time per rotation, so the additional $O(n\,\mathrm{polylog}\, n)$ work is lower order than $O(n^3)$. The expected trial count is now at most $2/\rho_0 < 122$, and the cap in (4.82) gives failure probability at most η. All synthesized rotations act on the reused phase qubit, so the peak quantum space is unchanged.

The classical count has the same leading order with schoolbook arithmetic. The one-time constant and circuit precomputation costs $O(n^3)$ bit operations. Within a trial, rounding the measured labels costs at most $O(n^2)$, while the pair checks $g^{\widehat{t}} \equiv y^{\widehat{s}} \pmod p$ and the final check $g^{\widetilde{x}} \equiv y \pmod p$ use modular exponentiation. Repeated squaring performs $O(n)$ modular multiplications of n-bit integers, each costing $O(n^2)$, so each such check costs $O(n^3)$. The gcd and extended Euclidean calculations cost $O(n^2)$ and are lower order. A constant number of checks per trial and a constant expected number of trials therefore give

$$\mathbb{E}[C_{\mathrm{classical}}(n)] = O(n^3). \tag{4.91}$$

Finally, imposing the corresponding cap R or R_{fixed}, both $O(1 + \log(1/\eta))$, multiplies the quantum and classical running times but does not increase the peak workspace:

	qubits	quantum gates	classical bit operations
repeat until verified, expected	$O(n)$	$O(n^3)$	$O(n^3)$
cap with failure at most η	$O(n)$	$O\left(n^3\left(1+\log\frac{1}{\eta}\right)\right)$	$O\left(n^3\left(1+\log\frac{1}{\eta}\right)\right)$

(4.92)

In the exact-rotation model, the concrete 99% guarantee above uses $R = 281$ and permits at most 562 pair runs, 1124 phase-estimation blocks, and therefore $1124n$ controlled modular-multiplier calls. These are conservative worst-case caps; for fixed $\eta = 0.01$ they are constants with respect to n, so the capped quantum and classical costs are still $O(n^3)$. In all cases classical verification ensures that the algorithm never reports an incorrect discrete logarithm.

4.III.iii Final algorithm and guarantees

With $N = p - 1$ and $m = n$, one recovery trial proceeds as follows.

1. Perform two independent pair runs. In each run, prepare the target in $|1\rangle$, run iterative PEA for U_g, and then run iterative PEA for U_y on that same retained target. Reuse the phase qubit and store its readouts classically; prepare a fresh $|1\rangle$ target at the start of the next pair run.
2. Round each run's readouts using Round_N to obtain $(\widehat{s}_i, \widehat{t}_i)$ for $i = 1, 2$. Verify $g^{\widehat{t}_i} \equiv y^{\widehat{s}_i} \pmod p$ for both pairs. If either check fails, end this trial without an answer.
3. For the two verified pairs, test $\gcd(s_1, s_2, N) = 1$. If it fails, end this trial without an answer. Otherwise, find a, b, c with $as_1 + bs_2 + cN = 1$ and compute the representative $\widetilde{x} = (at_1 + bt_2) \bmod N$.
4. Return $\widetilde{x}$ only if $g^{\widetilde{x}} \equiv y \pmod p$. After any unsuccessful trial, start a fresh independent trial. The capped version stops after R trials with exact rotations, or R_{fixed} trials with synthesized rotations, and reports explicit failure if no trial has returned an answer.

Final result. For the given n-bit prime p, generator g, and $y \in \mathbb{Z}_p^*$, the repeat-until-verified algorithm returns the unique $x \in \{0, \ldots, p-2\}$ satisfying $y = g^x \bmod p$. With iterative phase estimation and linear-workspace reversible multiplication, it uses $n + 1 + O(n) = O(n)$ qubits and has expected quantum and classical complexity $O(n^3)$. Trial success is at least $\rho_0 = 1536/\pi^{10}$ with exact rotations and $\rho_0/2$ over the specified finite gate set. Using the corresponding cap R or R_{fixed} gives quantum and classical cost $O(n^3(1+\log(1/\eta)))$, with explicit failure probability at most η and incorrect-answer probability zero.

QUESTION 5

Quantum Fingerprinting

P5a The swap test

PROBLEM

For $|\phi_x\rangle, |\phi_y\rangle$ in any Hilbert space, show that the circuit below outputs 0 with probability $(1 + |\langle\phi_x|\phi_y\rangle|^2)/2$, where $\text{SWAP}\,|\phi_x\rangle\,|\phi_y\rangle = |\phi_y\rangle\,|\phi_x\rangle$.

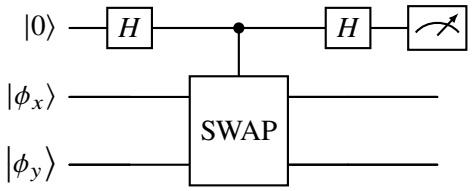


Figure 9. The **swap test**: a single controlled SWAP acts on the two registers, which need not be single qubits.

$\Pr[0] = \frac{1}{2}(1 + |\langle\phi_x|\phi_y\rangle|^2)$: equal to 1 iff the states agree up to phase, and to $\frac{1}{2}$ iff they are orthogonal.

P5a.i State evolution through the swap test

Denote the control register by A and the two state registers by X and Y. The circuit begins in the product state

$$|\Psi_0\rangle = |0\rangle_A\,|\phi_x\rangle_X\,|\phi_y\rangle_Y\,. \tag{P5a.1}$$

The first Hadamard acts only on A. Since $H\,|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, while the identity acts on X and Y, the state becomes

$$\begin{aligned}|\Psi_1\rangle &= (H_A \otimes I_{XY})\,|0\rangle_A\,|\phi_x\rangle_X\,|\phi_y\rangle_Y = \left(\frac{|0\rangle_A + |1\rangle_A}{\sqrt{2}}\right)|\phi_x\rangle_X\,|\phi_y\rangle_Y \\ &= \frac{1}{\sqrt{2}}\left(|0\rangle_A\,|\phi_x\rangle_X\,|\phi_y\rangle_Y + |1\rangle_A\,|\phi_x\rangle_X\,|\phi_y\rangle_Y\right).\end{aligned} \tag{P5a.2}$$

The controlled swap acts separately on the two control branches. For arbitrary $|u\rangle, |v\rangle \in \mathcal{H}$,

$$|0\rangle_A\,|u\rangle_X\,|v\rangle_Y \longmapsto |0\rangle_A\,|u\rangle_X\,|v\rangle_Y\,, \qquad |1\rangle_A\,|u\rangle_X\,|v\rangle_Y \longmapsto |1\rangle_A\,|v\rangle_X\,|u\rangle_Y\,. \tag{P5a.3}$$

Thus the control-0 term in (P5a.2) is unchanged, while the X and Y registers are exchanged in the control-1 term. By linearity,

$$|\Psi_2\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle_A\,|\phi_x\rangle_X\,|\phi_y\rangle_Y + |1\rangle_A\,|\phi_y\rangle_X\,|\phi_x\rangle_Y\right). \tag{P5a.4}$$

For readability, abbreviate the two possible assignments of the states to the registers:

$$|\Phi_{xy}\rangle := |\phi_x\rangle_X \otimes |\phi_y\rangle_Y\,, \qquad |\Phi_{yx}\rangle := |\phi_y\rangle_X \otimes |\phi_x\rangle_Y\,. \tag{P5a.5}$$

The order of the subscripts records which state occupies register X and which state occupies register Y: xy is the aligned assignment and yx is the swapped assignment. Thus (P5a.4) is

$$|\Psi_2\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle_A\,|\Phi_{xy}\rangle + |1\rangle_A\,|\Phi_{yx}\rangle\right). \tag{P5a.6}$$

The second Hadamard again acts only on A. Using $H\,|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $H\,|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, we obtain

$$\begin{aligned}|\Psi_f\rangle &= \frac{1}{\sqrt{2}}\left(H\,|0\rangle_A\,|\Phi_{xy}\rangle + H\,|1\rangle_A\,|\Phi_{yx}\rangle\right) = \frac{1}{2}\left[(|0\rangle_A + |1\rangle_A)\,|\Phi_{xy}\rangle + (|0\rangle_A - |1\rangle_A)\,|\Phi_{yx}\rangle\right] \\ &= \frac{1}{2}\left(|0\rangle_A\,|\Phi_{xy}\rangle + |1\rangle_A\,|\Phi_{xy}\rangle + |0\rangle_A\,|\Phi_{yx}\rangle - |1\rangle_A\,|\Phi_{yx}\rangle\right)\end{aligned}$$

$$= \frac{1}{2}|0\rangle_A \left(|\Phi_{xy}\rangle + |\Phi_{yx}\rangle\right) + \frac{1}{2}|1\rangle_A \left(|\Phi_{xy}\rangle - |\Phi_{yx}\rangle\right). \tag{P5a.7}$$

The minus sign therefore occurs only in the $|1\rangle_A |\Phi_{yx}\rangle$ component; it comes from the identity $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$.

P5a.ii Acceptance probability from the symmetric branch

Define the symmetric and antisymmetric branch vectors

$$|\Phi_+\rangle := \frac{1}{2}\left(|\Phi_{xy}\rangle + |\Phi_{yx}\rangle\right), \qquad |\Phi_-\rangle := \frac{1}{2}\left(|\Phi_{xy}\rangle - |\Phi_{yx}\rangle\right). \tag{P5a.8}$$

Equation (P5a.7) can therefore be written as

$$|\Psi_f\rangle = |0\rangle_A |\Phi_+\rangle_{XY} + |1\rangle_A |\Phi_-\rangle_{XY}. \tag{P5a.9}$$

Thus $|\Phi_+\rangle$ and $|\Phi_-\rangle$ are the two components of the joint state associated with the possible control outcomes; their squared norms give the corresponding probabilities.

For comparison, the familiar Born rule says that if a single qubit is in the state $\alpha|0\rangle + \beta|1\rangle$, then measuring it in the computational basis gives outcome 0 with probability $|\alpha|^2$. In (P5a.9), the control qubit may be entangled with X and Y, so the quantity attached to $|0\rangle_A$ is the vector $|\Phi_+\rangle$ rather than a scalar amplitude. The projector form of the Born rule makes the corresponding probability precise.

Measuring only the control qubit and obtaining 0 corresponds to the projector

$$P_0 = |0\rangle\langle 0|_A \otimes I_{XY}. \tag{P5a.10}$$

Here I_{XY} appears because the two state registers are not being measured. Applying the projector removes the control-1 branch:

$$\begin{aligned} P_0|\Psi_f\rangle &= (|0\rangle\langle 0|_A \otimes I_{XY})(|0\rangle_A |\Phi_+\rangle + |1\rangle_A |\Phi_-\rangle) \\ &= |0\rangle_A \langle 0|0\rangle |\Phi_+\rangle + |0\rangle_A \langle 0|1\rangle |\Phi_-\rangle = |0\rangle_A |\Phi_+\rangle, \end{aligned} \tag{P5a.11}$$

because $\langle 0|0\rangle = 1$ and $\langle 0|1\rangle = 0$.

The Born rule states that the probability of this event is the squared norm of the projected component. Since P_0 is an orthogonal projector, $P_0^\dagger P_0 = P_0$, and therefore

$$\begin{aligned} \Pr[A=0] = \left\||P_0|\Psi_f\rangle\right\|^2 = \langle\Psi_f| P_0^\dagger P_0 |\Psi_f\rangle = \langle\Psi_f| P_0 |\Psi_f\rangle = (\langle 0|_A \langle\Phi_+| + \langle 1|_A \langle\Phi_-|)\,|0\rangle_A |\Phi_+\rangle \\ = \langle 0|0\rangle \langle\Phi_+|\Phi_+\rangle + \langle 1|0\rangle \langle\Phi_-|\Phi_+\rangle = \langle\Phi_+|\Phi_+\rangle = \||\Phi_+\rangle\|^2 = \frac{1}{4}\left\||\Phi_{xy}\rangle + |\Phi_{yx}\rangle\right\|^2. \end{aligned} \tag{P5a.12}$$

Thus this is exactly the familiar rule $\Pr[0] = |\alpha|^2$, with the scalar amplitude α replaced by the complete branch vector $|\Phi_+\rangle$.

We now evaluate the remaining squared norm from its definition:

$$\begin{aligned} \left\||\Phi_{xy}\rangle + |\Phi_{yx}\rangle\right\|^2 &= \left(\langle\Phi_{xy}| + \langle\Phi_{yx}|\right)\left(|\Phi_{xy}\rangle + |\Phi_{yx}\rangle\right) \\ &= \langle\Phi_{xy}|\Phi_{xy}\rangle + \langle\Phi_{xy}|\Phi_{yx}\rangle + \langle\Phi_{yx}|\Phi_{xy}\rangle + \langle\Phi_{yx}|\Phi_{yx}\rangle. \end{aligned} \tag{P5a.13}$$

Substituting the definitions from (P5a.5), the squared norms of the aligned and swapped assignments are

$$\begin{aligned} \langle\Phi_{xy}|\Phi_{xy}\rangle &= \left(\langle\phi_x|_X \otimes \langle\phi_y|_Y\right)\left(|\phi_x\rangle_X \otimes |\phi_y\rangle_Y\right) = \langle\phi_x|\phi_x\rangle \langle\phi_y|\phi_y\rangle = 1\cdot 1 = 1, \\ \langle\Phi_{yx}|\Phi_{yx}\rangle &= \left(\langle\phi_y|_X \otimes \langle\phi_x|_Y\right)\left(|\phi_y\rangle_X \otimes |\phi_x\rangle_Y\right) = \langle\phi_y|\phi_y\rangle \langle\phi_x|\phi_x\rangle = 1\cdot 1 = 1. \end{aligned} \tag{P5a.14}$$

The final equality in each line uses the normalization of the corresponding input state. The inner product on a tensor-product space factorizes. Consequently,

$$\begin{aligned} \langle\Phi_{xy}|\Phi_{yx}\rangle &= \left(\langle\phi_x|_X \otimes \langle\phi_y|_Y\right)\left(|\phi_y\rangle_X \otimes |\phi_x\rangle_Y\right) \\ &= \langle\phi_x|\phi_y\rangle \langle\phi_y|\phi_x\rangle = \langle\phi_x|\phi_y\rangle \overline{\langle\phi_x|\phi_y\rangle} = \left|\langle\phi_x|\phi_y\rangle\right|^2. \end{aligned} \tag{P5a.15}$$

The same calculation with the two assignments interchanged gives $\langle\Phi_{yx}|\Phi_{xy}\rangle = |\langle\phi_x|\phi_y\rangle|^2$. Hence

$$\big\||\Phi_{xy}\rangle + |\Phi_{yx}\rangle\big\|^2 = 2 + 2\,|\langle\phi_x|\phi_y\rangle|^2 \tag{P5a.16}$$

$$\Pr[0] = \frac{1}{4}\left(2 + 2\,|\langle\phi_x|\phi_y\rangle|^2\right) = \boxed{\frac{1 + |\langle\phi_x|\phi_y\rangle|^2}{2}}. \tag{P5a.17}$$

This is the desired probability.

P5b Polynomial fingerprint states

PROBLEM

Let $q = p^a$ and $\mathbb{F}_q$ the field with q elements. For f of degree $\le n$ over $\mathbb{F}_q$ set $|\phi_f\rangle = q^{-1/2}\sum_{x\in\mathbb{F}_q}|x, f(x)\rangle$. Show that distinct f, g of degree $\le n$ satisfy $0 \le \langle\phi_f|\phi_g\rangle \le n/q$.

We interpret the state as a uniform quantum encoding of the polynomial's input–output table. Its overlap with another such state is therefore the fraction of inputs on which the two polynomials agree.

P5b.i Fingerprint states and agreement points

Let $\mathbb{F}_q$ be the finite field with q elements. The condition $q = p^a$, where p is prime, guarantees that such a field exists. For this proof, the important facts are that $\mathbb{F}_q$ contains exactly q possible inputs and that ordinary polynomial algebra works over it.

A polynomial f defines a table of q input–output pairs,

$$\{(x, f(x)) : x \in \mathbb{F}_q\}. \tag{P5b.1}$$

The state

$$|\phi_f\rangle = \frac{1}{\sqrt{q}}\sum_{x\in\mathbb{F}_q}|x, f(x)\rangle \tag{P5b.2}$$

is the uniform quantum encoding of this table, or equivalently of the graph of f. More explicitly,

$$|x, f(x)\rangle = |x\rangle_X \otimes |f(x)\rangle_V\,, \tag{P5b.3}$$

where the first register stores the input x and the second stores the corresponding output $f(x)$.

There are q mutually orthogonal basis states in the superposition, each with amplitude $1/\sqrt{q}$. Therefore,

$$q\left|\frac{1}{\sqrt{q}}\right|^2 = 1, \tag{P5b.4}$$

so $|\phi_f\rangle$ is normalized.

Now let $f \neq g$, where both polynomials have degree at most n. We want to bound

$$\langle\phi_f \mid \phi_g\rangle, \tag{P5b.5}$$

the overlap between their two fingerprint states. Since the states encode the input–output tables of f and g, their overlap should come exactly from the inputs at which the two polynomials produce the same output. We now show this formally.

Expanding the inner product gives

$$\langle\phi_f \mid \phi_g\rangle = \left(\frac{1}{\sqrt{q}}\sum_{x\in\mathbb{F}_q}\langle x, f(x)|\right)\left(\frac{1}{\sqrt{q}}\sum_{y\in\mathbb{F}_q}|y, g(y)\rangle\right) = \frac{1}{q}\sum_{x,y\in\mathbb{F}_q}\langle x, f(x) \mid y, g(y)\rangle. \tag{P5b.6}$$

The states $|x, f(x)\rangle$ and $|y, g(y)\rangle$ are computational-basis states labelled by ordered pairs. Consequently,

$$\langle x, f(x) \mid y, g(y)\rangle = \begin{cases} 1, & x = y \text{ and } f(x) = g(y), \\ 0, & \text{otherwise.} \end{cases} \tag{P5b.7}$$

The condition $x = y$ eliminates all terms with different inputs. After setting $y = x$, a surviving term contributes 1 exactly when $f(x) = g(x)$. Therefore,

$$\langle \phi_f \mid \phi_g \rangle = \frac{1}{q} \sum_{x \in \mathbb{F}_q} \mathbf{1}[f(x) = g(x)] = \frac{\left|\{x \in \mathbb{F}_q : f(x) = g(x)\}\right|}{q}. \tag{P5b.8}$$

Thus the overlap is exactly the fraction of field elements on which f and g agree. This immediately shows that

$$0 \leq \langle \phi_f \mid \phi_g \rangle. \tag{P5b.9}$$

P5b.ii Bounding agreement points by polynomial degree

It remains to bound the number of agreement points. Define

$$h := f - g. \tag{P5b.10}$$

Because $f \neq g$, the polynomial h is not the zero polynomial. Furthermore,

$$\deg h = \deg(f - g) \leq \max\{\deg f, \deg g\} \leq n. \tag{P5b.11}$$

For every $x \in \mathbb{F}_q$,

$$f(x) = g(x) \iff (f - g)(x) = 0 \iff h(x) = 0. \tag{P5b.12}$$

Hence the inputs on which f and g agree are precisely the roots of h. We now invoke the standard polynomial root theorem, familiar from Linear Algebra 2: a nonzero polynomial of degree d over a field has at most d distinct roots. Since $\deg h \leq n$, it follows that

$$\left|\{x \in \mathbb{F}_q : f(x) = g(x)\}\right| \leq n. \tag{P5b.13}$$

Substituting this into the overlap formula gives

$$\boxed{0 \leq \langle \phi_f \mid \phi_g \rangle = \frac{\left|\{x \in \mathbb{F}_q : f(x) = g(x)\}\right|}{q} \leq \frac{n}{q}}. \tag{P5b.14}$$

Thus two different degree-$\leq n$ polynomials produce fingerprint states with overlap at most n/q.

5.a An $O(\log n)$-qubit simultaneous-message protocol

> **PROBLEM**
>
> Alice holds x and Bob holds y in $\{0,1\}^n$; they share no entanglement and no randomness, cannot communicate, and each send one message to a referee R who must decide whether $x = y$. Give a quantum protocol in which each message is $O(\log n)$ qubits and R errs with probability at most $1/100$. (Classically $\Theta(\sqrt{n})$ bits are necessary and sufficient.)

We combine the two preliminary results. P5b turns each input into a polynomial fingerprint and bounds the overlap of unequal fingerprints. P5a then converts that overlap into the probability of an outcome that the referee can observe by a swap test. Repeating the comparison on independently prepared pairs reduces the false-accept probability.

5.a.i Polynomial fingerprints and the message protocol

More precisely, P5b associates a polynomial f with the state

$$\left|\phi_f\right\rangle = \frac{1}{\sqrt{q}} \sum_{u \in \mathbb{F}_q} \left|u, f(u)\right\rangle. \tag{5.a.1}$$

Its inner product with another fingerprint is exactly the fraction of field inputs on which the two polynomials agree. Thus, when q is sufficiently large relative to the polynomial degree, distinct polynomials have fingerprints with small overlap. P5a supplies the operational meaning of this bound: a swap test on $\left|\phi_f\right\rangle$ and $\left|\phi_g\right\rangle$ outputs 0 with probability

$$\frac{1}{2}\left(1 + \left|\langle \phi_f \mid \phi_g \rangle\right|^2\right). \tag{5.a.2}$$

Identical fingerprints therefore always produce 0, whereas a small overlap makes the probability of output 0 bounded away from 1. This is the bridge between the algebraic bound in P5b and the referee's decision procedure.

The input length n is a public parameter: the problem states that $x, y \in \{0,1\}^n$. Hence Alice, Bob, and the referee may use the same field and the same encoding convention. Let

$$q = 2^a, \tag{5.a.3}$$

where a, and therefore q, will be chosen after the error analysis. The prime subfield of $\mathbb{F}_q$ is $\mathbb{F}_2$, so each input bit may be regarded as an element of $\mathbb{F}_q$. Write

$$x = (x_0, \ldots, x_{n-1}), \qquad y = (y_0, \ldots, y_{n-1}), \tag{5.a.4}$$

and define

$$f_x(T) = \sum_{j=0}^{n-1} x_j T^j, \qquad f_y(T) = \sum_{j=0}^{n-1} y_j T^j. \tag{5.a.5}$$

If $x = y$, all corresponding coefficients agree, so $f_x = f_y$. If $x \neq y$, at least one coefficient differs, so $f_x - f_y$ is a nonzero polynomial. The largest power used in either polynomial is T^{n-1}; therefore,

$$\deg(f_x - f_y) \leq n - 1. \tag{5.a.6}$$

P5b consequently gives, for $x \neq y$,

$$\left|\langle \phi_{f_x} \mid \phi_{f_y} \rangle\right| \leq \frac{n-1}{q}. \tag{5.a.7}$$

We now describe the messages and the referee's procedure. Let r be a positive integer, also to be chosen after the error analysis. Alice independently prepares r copies of $\left|\phi_{f_x}\right\rangle$, and Bob independently prepares r copies of $\left|\phi_{f_y}\right\rangle$. These are fresh preparations, not clones of an unknown state: each sender knows a classical description of the input and runs the same preparation circuit r separate times. As usual in the ideal quantum communication-complexity model, local state preparation and computation are noiseless and free; the protocol's error comes from measurement randomness, not from physical preparation or transmission noise.

Each basis state in one fingerprint has the form

$$|u, f_x(u)\rangle = |u\rangle \otimes |f_x(u)\rangle\,. \tag{5.a.8}$$

Because $q = 2^a$, one field element has 2^a possible values and is represented by an a-qubit register. The ordered pair in (5.a.8) therefore consists of two a-qubit registers, so one fingerprint contains $2a$ qubits. Alice bundles her r fingerprints into her single message to the referee, and Bob does the same; each complete message thus contains

$$2ar \tag{5.a.9}$$

qubits.

The referee pairs Alice's ith fingerprint with Bob's ith fingerprint and performs one swap test on each of the r fresh pairs. The referee declares $x = y$ if and only if all r swap tests output 0. This comparison is basis-independent and does not require the referee to know x, y, f_x, or f_y; its acceptance probability is exactly the expression supplied by P5a.

5.a.ii Correctness and error amplification

We first prove correctness when $x = y$. In this case $f_x = f_y$, and hence

$$\left|\phi_{f_x}\right\rangle = \left|\phi_{f_y}\right\rangle. \tag{5.a.10}$$

Every swap test receives two identical states and outputs 0 with probability 1. Thus all r tests output 0, and

$$\Pr[\text{error} \mid x = y] = 0. \tag{5.a.11}$$

Now suppose $x \neq y$, and set

$$s := \left|\langle \phi_{f_x} \mid \phi_{f_y} \rangle\right|. \tag{5.a.12}$$

By (5.a.7), $s \leq (n-1)/q$. P5a therefore bounds the probability that one swap test falsely outputs 0 by

$$p = \frac{1}{2}(1 + s^2) \leq \frac{1}{2}\left(1 + \left(\frac{n-1}{q}\right)^2\right). \tag{5.a.13}$$

The r pairs were prepared independently, and a separate swap test is applied to each pair. Consequently, the outcomes are independent. The referee incorrectly declares equality only if every test outputs 0, so

$$\Pr[\text{error} \mid x \neq y] \leq \left[\frac{1}{2}\left(1 + \left(\frac{n-1}{q}\right)^2\right)\right]^r. \tag{5.a.14}$$

5.a.iii Parameter choice and communication cost

It remains to choose the parameters. The problem statement requires the error probability to be at most $1/100$. A swap test outputs 0 with probability at least $1/2$ for every pair of states, because its acceptance probability is $\frac{1}{2}(1 + s^2)$ with $s^2 \geq 0$. Thus even perfectly orthogonal unequal fingerprints would have r-test false-accept probability at least 2^{-r}. Since

$$2^{-6} = \frac{1}{64} > \frac{1}{100}, \qquad 2^{-7} = \frac{1}{128} < \frac{1}{100}, \tag{5.a.15}$$

six repetitions cannot suffice within this repeated-swap-test protocol. We therefore choose the smallest potentially sufficient value,

$$r = 7. \tag{5.a.16}$$

We now choose the field size. The overlap bound in (5.a.7) contains the ratio $(n-1)/q$. To make this overlap bounded by a constant independent of the input length, we choose q to be at least a constant multiple of n. Thus, let $q \geq cn$ for a constant $c > 1$. The factor of n then cancels:

$$\frac{n-1}{q} < \frac{n}{q} \leq \frac{n}{cn} = \frac{1}{c}. \tag{5.a.17}$$

With $r = 7$, the false-accept probability is therefore at most

$$\left[\frac{1}{2}\left(1 + \frac{1}{c^2}\right)\right]^7. \tag{5.a.18}$$

Requiring this bound to be at most $1/100$, and using $c > 1$, we take seventh roots and isolate c:

$$\begin{aligned} \left[\frac{1}{2}\left(1 + \frac{1}{c^2}\right)\right]^7 \leq \frac{1}{100} &\iff \frac{1}{2}\left(1 + \frac{1}{c^2}\right) \leq 100^{-1/7} \\ &\iff \frac{1}{c^2} \leq 2 \cdot 100^{-1/7} - 1 \iff c \geq \frac{1}{\sqrt{2 \cdot 100^{-1/7} - 1}} \approx 5.28. \end{aligned} \tag{5.a.19}$$

We take the convenient integer value $c = 6$. We also require q to be a power of 2, so we choose the smallest such power satisfying $q \geq 6n$. Since $q = 2^a$, the condition $q \geq 6n$ is equivalent to

$$2^a \geq 6n \iff a \geq \log_2(6n). \tag{5.a.20}$$

The exponent a must be an integer, both because it specifies the field $\mathbb{F}_{2^a}$ and because it is the number of qubits in one field-element register. Hence the smallest valid choice is

$$a = \lceil \log_2(6n) \rceil, \qquad q = 2^a. \tag{5.a.21}$$

This explains the ceiling: if $6n$ is not itself a power of 2, we move to the next integer exponent, equivalently the next power of 2. By minimality,

$$6n \leq q < 12n. \tag{5.a.22}$$

In particular, $(n-1)/q < 1/6$. Therefore, for $x \neq y$, one test falsely accepts with probability at most

$$p < \frac{1}{2}\left(1 + \frac{1}{36}\right) = \frac{37}{72}, \tag{5.a.23}$$

and the seven independent tests falsely accept with probability at most

$$\Pr[\text{error} \mid x \neq y] < \left(\frac{37}{72}\right)^7 < 0.0095 < \frac{1}{100}. \tag{5.a.24}$$

Finally, the problem statement requires each complete message to contain only $O(\log n)$ qubits. By (5.a.9), each sender transmits

$$2ar = 14a \tag{5.a.25}$$

qubits. From $a = \lceil \log_2(6n) \rceil$ we have

$$a < \log_2(6n) + 1 = \log_2(12n) = O(\log n). \tag{5.a.26}$$

Consequently, $14a = O(\log n)$. The protocol uses one $O(\log n)$-qubit message from each sender, has no error when $x = y$, and has error probability below $1/100$ when $x \neq y$, as required. ■

5.a.iv Interpretation and limitations of quantum fingerprinting

Although the question asks only for the protocol, its correctness, and its communication cost, I found it helpful to pause after the proof and consider what this abstract result means in practice, and why the supplied swap test is a natural comparison tool for the referee. The following two brief observations record that perspective; they are offered as context rather than as part of the proof.

Interpretation as a quantum fingerprint. The closest classical analogy to this protocol is hashing for equality testing. The map $x \mapsto |\phi_{f_x}\rangle$ compresses an n-bit input into an $O(\log n)$-qubit state, but the state is not intended to reveal or reconstruct x. Instead, different inputs produce nonorthogonal states whose overlap is guaranteed to be small. This small overlap plays a role analogous to a low collision probability in classical hashing: using the swap test, the referee can test whether two senders supplied the same fingerprint without learning the strings themselves.

The analogy should not be taken too far. A quantum fingerprint is not a classical or cryptographic hash: it cannot be read as a reusable digest, it cannot be copied freely, and the construction does not establish collision resistance, preimage resistance, or authentication. Its main significance is therefore communication-theoretic. In the simultaneous-message model without shared randomness or entanglement, it reduces each message from the optimal classical size $\Theta(\sqrt{n})$ to $O(\log n)$ qubits [7]. If shared randomness and ordinary hashing are available, the corresponding classical equality test is already inexpensive, so this protocol should be understood primarily as a separation between communication models rather than as a practical replacement for conventional hashing.

Role and limitations of the swap test. The referee receives quantum states without knowing their classical descriptions, so a universal, basis-independent comparison is natural. The swap test accepts identical fingerprints with certainty, and its acceptance probability depends exactly on the overlap bounded by P5b; this makes it the direct bridge from the algebraic fingerprint construction to an observable decision. Its limitations also explain the protocol's design: even orthogonal states are accepted with probability $1/2$, and the measurement generally disturbs its inputs, so amplification requires fresh independently prepared pairs. The controlled-swap circuit is not necessarily the easiest physical implementation.

5.b The circuits and their complexity

> **PROBLEM**
>
> Describe the circuits for Alice, Bob and R, and state their complexity in qubits and gates. If a quantum version of a known classical algorithm is used, its details may be omitted, but the algorithm must be the right one — note that for non-prime q, arithmetic in $\mathbb{F}_q$ is not arithmetic modulo q. Bounds should be as tight as possible.

Part (a) requires Alice and Bob to prepare seven copies of their polynomial fingerprints and requires the referee to compare the seven corresponding pairs. We now realize these operations as circuits and count their resources. We count standard one-, two-, and three-qubit gates. Toffoli and controlled-SWAP gates both have constant-size exact decompositions into one- and two-qubit gates, so this convention changes the asymptotic counts only by constant factors.

5.b.i Sender preparation circuits

Recall that $q = 2^a$. Fix a one-to-one labeling of the a-qubit computational basis by the elements of $\mathbb{F}_q$, and write $|u\rangle$ for the basis state labeled by $u \in \mathbb{F}_q$. This labeling only determines how field elements are stored. All values $f_z(u)$ below are computed using addition and multiplication in the extension field $\mathbb{F}_q$, not integer arithmetic modulo q.

It is enough to describe one sender. Write $z = x$ for Alice and $z = y$ for Bob, and let

$$f_z(T) = \sum_{j=0}^{n-1} z_j T^j. \tag{5.b.1}$$

The sender begins with an a-qubit argument register arg and an a-qubit value register val in $|0^a\rangle_{\text{arg}} |0^a\rangle_{\text{val}}$. Hadamard gates act on all a qubits of arg and on no qubit of val. Their action is

$$\begin{aligned}(H_{\text{arg}}^{\otimes a} \otimes I_{\text{val}}) |0^a\rangle_{\text{arg}} |0^a\rangle_{\text{val}} &= \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)_{\text{arg}}^{\otimes a} |0^a\rangle_{\text{val}} \\ &= \frac{1}{(\sqrt{2})^a} \sum_{b_1,\ldots,b_a \in \{0,1\}} |b_1 \cdots b_a\rangle_{\text{arg}} |0^a\rangle_{\text{val}} = \frac{1}{\sqrt{q}} \sum_{u \in \mathbb{F}_q} |u\rangle_{\text{arg}} |0^a\rangle_{\text{val}} .\end{aligned} \tag{5.b.2}$$

The middle expression distributes the a tensor factors over all a-bit strings. The last equality merely relabels those same $2^a = q$ basis strings by the elements of $\mathbb{F}_q$. Thus the Hadamards create the required coherent uniform superposition without measuring the argument register.

Because z is classical and local classical computation is not charged in the simultaneous-message model, the preparation circuit may depend on z. The sender classically computes the table $u \mapsto f_z(u)$ using the actual field operations of $\mathbb{F}_q$. For each u, let $t_u \in \{0,1\}^a$ be the fixed basis label of $f_z(u)$. The executed quantum circuit then uses the clean hard-wired lookup

$$L_z : \quad |u\rangle_{\text{arg}} |v\rangle_{\text{val}} \left|0^{O(a)}\right\rangle_{\text{work}} \longmapsto |u\rangle_{\text{arg}} |v \oplus t_u\rangle_{\text{val}} \left|0^{O(a)}\right\rangle_{\text{work}}, \tag{5.b.3}$$

where $\oplus$ is bitwise XOR on the stored labels. A standard reversible implementation processes one hard-wired row s at a time: it computes the flag $[u = s]$, conditionally XORs t_s into val, and uncomputes the flag. Each row costs $O(a)$ gates, so all q rows cost $O(qa)$ gates and use $O(a)$ reusable workspace. This is the a-bit-output analogue of the coherent hard-wired lookup used in Lecture 13.[6] Appendix D.1 gives the gates for one row and explains why only the matching row acts on each address branch.

Applying L_z to the superposition in (5.b.2) gives, by linearity,

$$\begin{aligned}\frac{1}{\sqrt{q}} \sum_{u \in \mathbb{F}_q} |u\rangle_{\text{arg}} |0^a\rangle_{\text{val}} \left|0^{O(a)}\right\rangle_{\text{work}} &\xrightarrow{L_z} \frac{1}{\sqrt{q}} \sum_{u \in \mathbb{F}_q} |u\rangle_{\text{arg}} |f_z(u)\rangle_{\text{val}} \left|0^{O(a)}\right\rangle_{\text{work}} \\ &= \left|\phi_{f_z}\right\rangle_{\text{arg,val}} \left|0^{O(a)}\right\rangle_{\text{work}} .\end{aligned} \tag{5.b.4}$$

No further uncomputation is needed: $f_z(u)$ is the desired output, while the lookup has already returned every scratch qubit to zero.

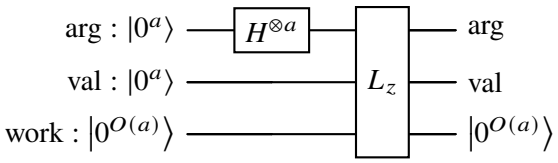


Figure 10. One sender's fingerprint-preparation circuit. Here $z = x$ for Alice and $z = y$ for Bob. The first two output registers jointly contain $\left|\phi_{f_z}\right\rangle$, and the lookup workspace is clean.

Each sender runs this preparation seven times, retaining the completed $2a$-qubit fingerprints and reusing the $O(a)$ lookup workspace. It therefore sends $7(2a) = 14a$ qubits, has peak quantum space $14a + O(a)$, and uses $7\big(a + O(qa)\big) = O(qa)$ quantum gates. Part (a) gives $6n \le q < 12n$, so $q = \Theta(n)$ and $a = \log_2 q = \Theta(\log n)$. Hence each sender communicates $O(\log n)$ qubits, has $O(\log n)$ peak quantum space, and uses $O(n \log n)$ quantum gates. The classical lookup description contains q words of a bits, hence qa bits; its synthesis is the uncharged local preprocessing specified above.

[6] Course Lecture 13, communication-complexity discussion, paragraph "Move 1 — Alice writes in her bit, by a coherent lookup."

5.b.ii The referee's circuit

For each $i = 1, \dots, 7$, the referee applies the swap-test circuit proved in P5a to Alice's and Bob's ith fingerprints. Because one fingerprint contains $2a$ qubits, the controlled swap in one test is implemented by $2a$ three-qubit controlled-SWAP gates, all controlled by the test ancilla and each exchanging one corresponding message-qubit pair. Thus the seven tests use

$$28a + 7 \text{ qubits}, \qquad 14a \text{ controlled-SWAP gates}, \qquad 14 \text{ Hadamard gates}, \tag{5.b.5}$$

together with seven measurements. Consequently, the referee's quantum gate count is $O(a) = O(\log n)$.

Let $b_1, \dots, b_7$ be the seven measured ancilla bits. The referee declares "equal" exactly when $b_1 = b_2 = \cdots = b_7 = 0$. This is an ordinary classical all-zero test on seven fixed bits, so it requires only $O(1)$ classical gates. By the correctness proof in part (a), the resulting decision has no error when $x = y$ and error probability below $1/100$ when $x \neq y$.

5.b.iii Overall resource counts

Party	Message sent	Peak qubits	Quantum gates	Measurements
Alice	$14a$	$14a + O(a)$	$O(qa)$	none
Bob	$14a$	$14a + O(a)$	$O(qa)$	none
Referee	none	$28a + 7$	$O(a)$	7
Protocol	$28a = O(\log n)$	$O(\log n)$	$O(qa) = O(n \log n)$	7

Classically, each sender synthesizes a qa-bit lookup description, while the referee performs only $O(1)$ Boolean postprocessing. These counts realize the protocol of part (a) with the required communication and error bounds. ∎

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